SA-AI (Spalart–Allmaras with Autogenous Inception): Technical Summary

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1 Model

1.1 Transport equation and blended sources

$$\frac{D\tilde{\nu}}{Dt} = P - D + \frac{1}{\sigma}\nabla\cdot\left[(c_{\nu,\mathrm{ai}}\,\nu + \tilde{\nu})\,\nabla\tilde{\nu}\right] + \frac{c_{b2}}{\sigma}\,|\nabla\tilde{\nu}|^2, \tag{1}$$

$$P = \max\left[\,(1-\sigma_P)\,a\,S\,\omega\,\tilde{\nu},\ \ \sigma_P\,P_{\mathrm{SA}}\,\right], \qquad D = \sigma_D\,D_{\mathrm{SA}}, \tag{2}$$

$\sigma_P \equiv 1$ (and with it $\sigma_D = 1$), $c_{\nu,\mathrm{ai}} = 1$, and $b \equiv 0$ recover baseline SA exactly.

$$P_{\mathrm{SA}} = c_{b1}\tilde{S}\,\tilde{\nu}, \qquad D_{\mathrm{SA}} = c_{w1} f_w \left(\frac{\tilde{\nu}}{d}\right)^2, \qquad \tilde{S} = \omega + \frac{\tilde{\nu}}{\kappa^2 d^2}\,f_{v2}, \qquad f_{v2} = 1 - \frac{\chi}{1+\chi f_{v1}},$$
$$f_{v1} = \frac{\chi^3}{\chi^3 + c_{v1}^3}, \qquad f_w = g\left(\frac{1+c_{w3}^6}{g^6+c_{w3}^6}\right)^{1/6}, \qquad g = r + c_{w2}(r^6 - r), \qquad r = \min\left(\frac{\tilde{\nu}}{\tilde{S}\,\kappa^2 d^2},\,10\right), \tag{3}$$

The non-negative $\tilde{S}$ modification and the negative-$\tilde{\nu}$ safeguards of Allmaras et al. (2012) are retained, with one consistency requirement: the negative-branch diffusion factor f_n is formed with the scaled coefficient $c_{\nu,\mathrm{ai}}\,\nu$, which keeps the total diffusivity positive, $c_{\nu,\mathrm{ai}}\nu + f_n\tilde{\nu} > 0$.

$$\sigma_P = \sigma_t(\chi;\tau), \qquad \sigma_t(\chi;\tau) = \max\left[1 - e^{-(\chi-1)/\tau},\ 0\right], \qquad \sigma_D = 1 - \frac{c_{b1}}{\kappa^2 c_{w1}}\,(1-\sigma_P). \tag{4}$$

1.2 Local indicators

In a thin quasi-two-dimensional layer the profile is captured to second order about any wall-normal point by three velocity scales formed from the wall distance d: the velocity u, the shear $d\,u'$, and the curvature $\frac{1}{2}d^2u''$. Normalized by their common magnitude $R = \sqrt{u^2 + (du')^2 + (\frac{1}{2}d^2u'')^2}$, they place the local state on the unit sphere of directions

$$(X, Y, Z) = \left(u,\ d\,u',\ \tfrac{1}{2}d^2u''\right)/R, \tag{5}$$

$$\hat{\Omega} = \frac{Y}{\sqrt{X^2+Y^2}} \quad \text{(vorticity fraction)}, \qquad \hat{I} = Y - X - Z \quad \text{(accumulated inflection)}. \tag{6}$$

A near-wall parabola $u = By + Cy^2$ satisfies $\hat{I} = 0$ identically.

$$I(d) \equiv R\,\hat{I} = d\,u' - u - \tfrac{1}{2}d^2u'' = -\int_0^d \tfrac{1}{2}\,s^2\,u'''(s)\,\mathrm{d}s, \tag{7}$$

Every profile leaves the wall with $\hat{I}$ third-order small ($u_w''' = 0$ on a steady wall). The vorticity Reynolds number $Re_\Omega = d^2\omega/\nu$ is reserved for the onset gate. Fig. 1.

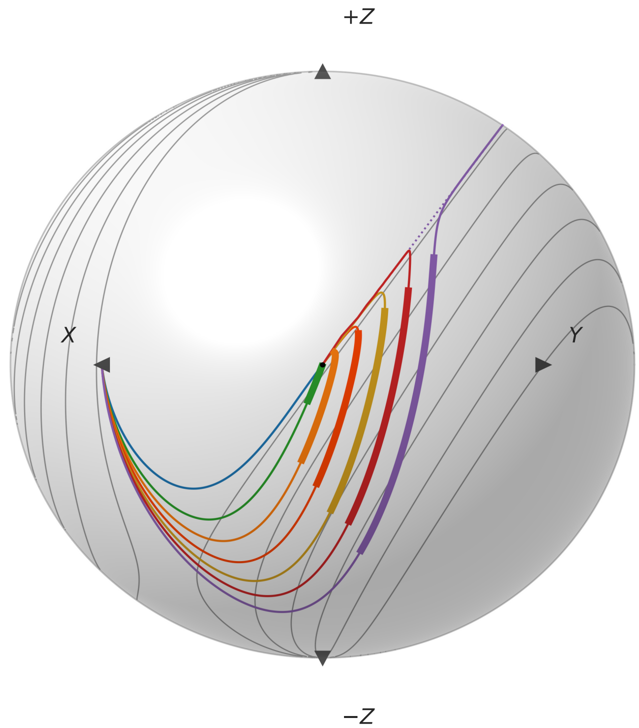


Figure 1: Indicator sphere. Falkner–Skan profiles $\beta = +0.30,\ 0,\ -0.10,\ -0.16,\ -0.19,\ -0.1988$; separated Stewartson $\beta = -0.19$, $H \approx 4.9$. Attached profiles at $Re_\theta = 500$, reversed-flow at $Re_\theta = 200$; $\hat{\Omega}\hat{I}$ contours $\{0.025, 0.2, 0.4, 0.6, 0.8, 1.0\}$.

In the solver the triple is assembled from four local vector-calculus objects: the velocity $\mathbf{u}$, the vorticity $\boldsymbol{\omega}$, the wall-distance vector $\mathbf{d} = d\,\nabla d$, and the curvature vector $\boldsymbol{\ell} = \frac{1}{2}\langle \mathbf{d}, \mathbf{d}\rangle\, \nabla^2\mathbf{u}$, with $\nabla^2\mathbf{u} = -\nabla\times\boldsymbol{\omega}$ supplying the second derivative.

$$(X,\,Y,\,Z) \;=\; \Big(\langle \mathbf{u},\mathbf{u}\rangle,\ \sqrt{\langle \mathbf{u},\mathbf{u}\rangle\,\langle \mathbf{d},\mathbf{d}\rangle\,\langle \boldsymbol{\omega},\boldsymbol{\omega}\rangle},\ \langle \boldsymbol{\ell},\mathbf{u}\rangle\Big), \qquad \hat{\Omega} = \frac{Y}{\sqrt{X^2+Y^2}}, \quad \hat{I} = \frac{Y-X-Z}{\sqrt{X^2+Y^2+Z^2}}, \tag{8}$$

The solver's equivalent ratio realization $(|\mathbf{u}|,\ d|\boldsymbol{\omega}|,\ \frac{1}{2}d^2(\nabla^2\mathbf{u})\cdot\hat{\mathbf{u}})$ coincides with it wherever $\mathbf{u} \neq 0$.

1.3 Amplification rate and onset threshold

The complete amplification source is

$$P_{\mathrm{AI}} = a\,S\,\omega\,\tilde{\nu}, \qquad a = a_{\max}\ \mathrm{clip}\big\langle\,\hat{\Omega}\,\hat{I}\,\big\rangle_0^1, \qquad S = S\big(Re_\Omega/Re_\Omega^{\mathrm{c}}(\hat{\Omega}\hat{I})\big), \qquad S(\zeta) = \tfrac{1}{2}\big[1+\tanh\big((\zeta-1)/w\big)\big], \tag{9}$$

$$Re_\Omega^{\mathrm{c}}(\hat{\Omega}\hat{I}) = k\ \mathrm{softmin}_2\Big(C,\ A + B\,\langle\hat{\Omega}\hat{I}\rangle_+^{-2}\Big), \qquad \mathrm{softmin}_2(x_1,x_2) = \frac{x_1x_2}{\sqrt{x_1^2+x_2^2}}, \qquad \langle x\rangle_+ = \max(x,0). \tag{10}$$

Fig. 2: graze ratios 0.93–1.14, root-mean-square $\sim 6\%$, from incipient separation through $\beta = +0.15$. The marched Blasius $N = 9$ crossing lands at $Re_\theta = 1181$, 7% past Drela's 1108; across the attached family the $N = 1$ stations land on the Drela–Giles stations to 6% root-mean-square (adverse wedges within $\pm 4\%$, the favorable pair 11–12% early). By $\beta = +0.35$ the rate is $\approx 0.1\times$ Drela and the onset $\approx 3\times$ its critical Re_θ; by $\beta \approx 0.45$ transition has retreated beyond any station of practical relevance, and stronger favorable wedges are suppressed outright (Figs. 3–4).

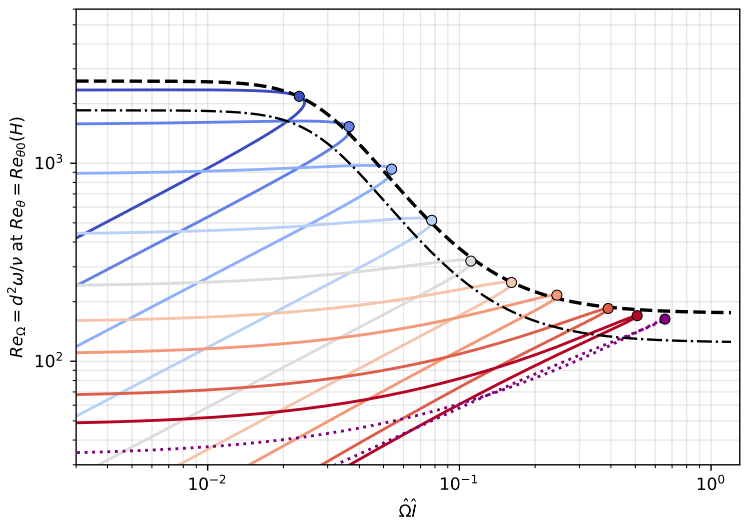


Figure 2: Onset-threshold graze. Attached family $\beta = +0.15$, $+0.10$, $+0.05$, 0, -0.05, -0.10, -0.15, -0.19, -0.1988, each at its Drela–Giles $Re_{\theta 0}(H)$; separated Stewartson $\beta = -0.19$ ($H \approx 4.9$, $Re_{\theta 0} \approx 26$). $\max_y \hat{\Omega}\hat{I} = 0.162/0.078/0.037$ at $\beta = -0.10/0/+0.10$. Envelope $k = 1$; model threshold $k = 0.712$.

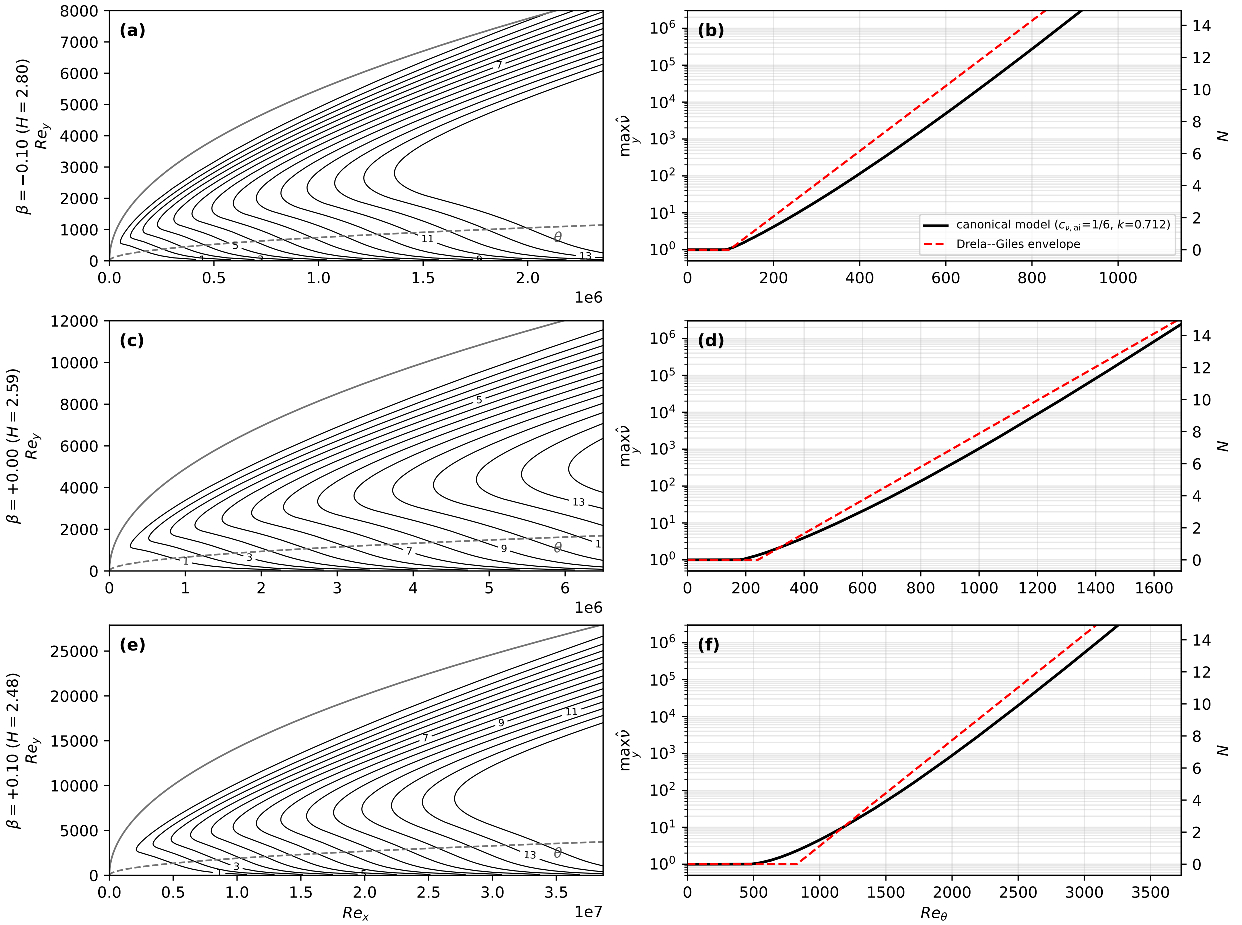


Figure 3: Disturbance transport, three wedges. Adverse $\beta = -0.10$ ($\max_y \hat{\Omega}\hat{I} = 0.162$), Blasius $\beta = 0$ (0.078), favorable $\beta = +0.10$ (0.037); seeded $\hat{\nu}_\infty = 1$, $a_{\max} = 0.19$, $(c_{\nu,\mathrm{ai}}, k) = (\frac{1}{6}, 0.712)$. Anchor $N = 9$ crossing 7% past Drela; favorable $N = 1$ station $\sim$11% early.

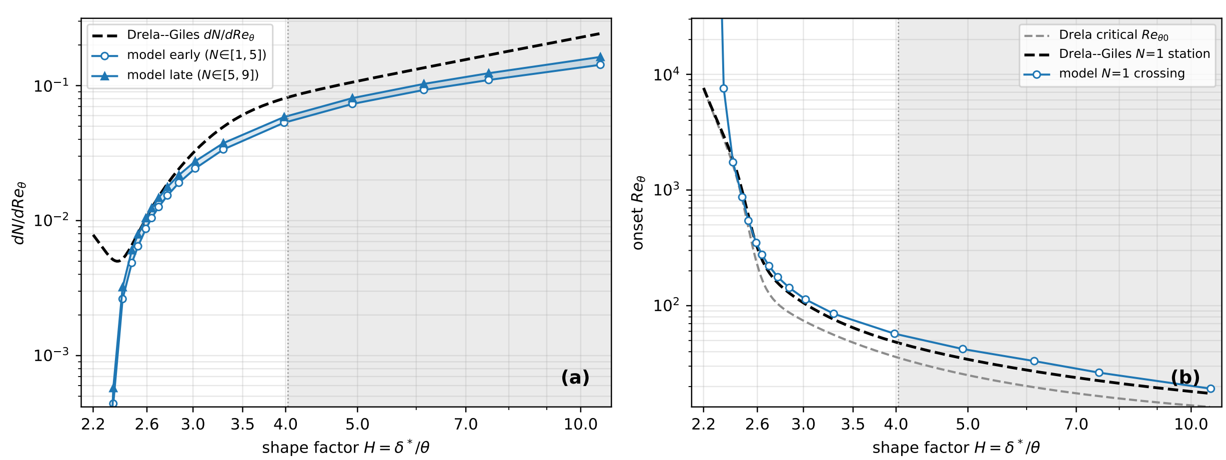


Figure 4: Calibrated model vs Drela–Giles. Stagnation $H \approx 2.2$, Blasius $H \approx 2.59$, separation limit $H = 3.98$, reversed-flow Stewartson $H \approx 10.6$; $a_{\max} = 0.19$, $c_{\nu,\mathrm{ai}} = 1/6$. (b) onset tracks the Drela–Giles $N = 1$ stations within $\sim \pm 12\%$ from $H = 2.44$ ($\beta = +0.15$) to $H = 3.98$.

1.4 Reduced laminar diffusion

The molecular coefficient in the diffusion term of Eq. (1) only is scaled by $c_{\nu,\mathrm{ai}} < 1$, leaving the mean-flow viscosity, the SA production and destruction, and $\chi = \tilde{\nu}/\nu$ (formed with the true ν) untouched. The frozen-profile growth eigenvalue on Blasius at $Re_\theta = 400$ delivers only 0.49 of the Drela–Giles rate, and the marched inception fails outright (the Blasius $N = 1$ crossing sits at $Re_\theta = 392$ even with the threshold removed entirely). Down the ladder $c_{\nu,\mathrm{ai}} = 1, \frac{1}{2}, \frac{1}{3}, \frac{1}{6}, \frac{1}{12}$ that station's eigenvalue ratio climbs 0.49, 0.75, 0.87, 1.02, 1.13: the value

$$c_{\nu,\mathrm{ai}} = \tfrac{1}{6}$$

puts the just-past-critical stations on the correlation (the separation branch holds 0.82–0.91). The equilibrium disturbance-band thickness scales as $\sqrt{c_{\nu,\mathrm{ai}}}$, so every halving thins the band the grid must resolve by $\sqrt{2}$.

1.5 Handover blend and destruction tie

The destruction is not gated off but tied: σ_D tracks σ_P with slope $c_{b1}/(\kappa^2 c_{w1}) \approx 0.249$ from the floor $(1 + c_{b2})/(\sigma c_{w1}) \approx 0.751$, the two summing to one by the definition of c_{w1}. The tie is derived from the constant-stress wall layer, on which SA's solution is the linear profile $\tilde{\nu} = \kappa u_\tau y$ with height-independent terms $P_{\mathrm{SA}} = c_{b1} u_\tau^2$, $D_{\mathrm{SA}} = c_{w1}\kappa^2 u_\tau^2$, and self-diffusion $(1 + c_{b2})\kappa^2 u_\tau^2/\sigma$; requiring the gated balance to hold pointwise gives Eq. (4), under which the linear profile is an exact solution of the gated equation at every height and for any τ—continuously and discretely (verified to below 10^{-10} in profile and log-law intercept with two unrelated discretizations). In the laminar range the destruction floor is inert at the disturbance's quadratic scale: its ratio to the amplification production stays below 10^{-3} up to the $\chi = 1$ crossing for every wedge and seed tested, and at $\chi = c_{v1}$ is at most 1.4% of the σ_P-weighted SA production it is tied against; marching the calibration instrument with the floor included moves the anchor and prediction stations by less than 1%.

1.6 Eddy-viscosity assembly in lifted transitional layers

The near-wall damping f_{v1} enforces the attached-wall-layer identity $\chi = \kappa y^+$. Writing $y^+ := \chi/\kappa$, the reconstruction

$$q = \frac{|\mathbf{u}|\, d/\nu}{y^+\, U^+(y^+)}, \qquad U^+(y^+) = \tfrac{1}{\kappa}\ln(1 + \kappa y^+) + 7.8\big(1 - e^{-y^+/11} - \tfrac{y^+}{11} e^{-y^+/3}\big) \text{ (Reichardt)}, \tag{11}$$

The assembly is blended,

$$\mu_t = \big[(1 - b)\, f_{v1}(\chi) + b\big]\, \rho\tilde{\nu}, \qquad b = s(\chi)\, G(q), \qquad s = \mathrm{clip}(\chi - 1, 0, 1), \qquad G = \mathrm{clip}((q - 2)/2, 0, 1), \tag{12}$$

inert for $\chi < 1$ (negligible μ_t either way), for $\chi \gtrsim 30$ ($f_{v1} \to 1$), and in the attached inner layer ($q = 1$, G shut). Controlled on/off batteries bound the net effect on attached-layer outer edges below 1.5 drag counts (flat plate and NLF at their cold-budget protocol).

1.7 Freestream seed

A case seeded with χ_∞ reaches the handover where its envelope crosses $N_{\mathrm{crit}} = \ln(c_{v1}/\chi_\infty)$.

$$N_{\mathrm{crit}}(Tu) = -8.43 - 2.4\, \ln(Tu_{\mathrm{frac}}), \qquad \chi_\infty = c_{v1}\, e^{-N_{\mathrm{crit}}}, \tag{13}$$

One e-fold in χ_∞ is one unit of N_{crit}, which on the Blasius envelope ($dN/dRe_\theta \approx 10^{-2}$) shifts onset by $\Delta Re_\theta \approx 100$. Strongly bypass-dominated conditions ($Tu \gtrsim 1\%$) are out of scope: at $Tu = 1\%$ the map leaves only $N_{\mathrm{crit}} \approx 2.6$, and by $Tu \approx 3\%$ the seed exceeds c_{v1} outright.

1.8 Constants

Baseline SA, unchanged:

$$c_{b1} = 0.1355, \quad \sigma = \tfrac{2}{3}, \quad c_{b2} = 0.622, \quad \kappa = 0.41, \quad c_{w1} = \frac{c_{b1}}{\kappa^2} + \frac{1 + c_{b2}}{\sigma}, \quad c_{w2} = 0.3, \quad c_{w3} = 2, \quad c_{v1} = 7.1.$$

SA-AI:

constant	value	determination
$a_{\max}$	0.19	tanh-layer Kelvin–Helmholtz temporal eigenvalue, $\omega_{i,\max}/\omega_{\text{peak}} = 0.1897$
(C, A, B)	(2600, 175, 2)	graze envelope of the attached Falkner–Skan family at its critical Re_θ (rms $\sim 6\%$)
k	0.712	Blasius march crosses $N = 1$ at the Drela–Giles station $Re_\theta = 338$ at $c_{\nu,\text{ai}} = \frac{1}{6}$
w	0.35	onset ramp width; $N = 1$ stations move $< 1\%$ halved, $< 2.5\%$ doubled
$c_{\nu,\text{ai}}$	1/6	frozen-profile growth eigenvalue reaches the Drela–Giles rate just past critical (1.02 at $Re_\theta = 400$)
τ	4	numerics bracket: handover 78% complete at $\chi = c_{v1}$, 97% at $\chi \approx 14.8$
σ_D tie	Eq. (4)	derived: constant-stress wall-layer balance; linear wall solution exact
bypass (s, G)	Eq. (12)	parameter-free law-of-the-wall discriminator; G ramps over $q \in (2, 4)$
$(A, B)_{\text{Mack}}$	(−8.43, 2.4)	Mack 1977 e^N correlation, wholesale

1.9 Running the model: branch selection and seed handling

On the frozen Hiemenz stagnation field with the freestream seed exactly zero, standard SA sustains the wedge above a critical nose Reynolds number bisected to $Re_r = 4.66–4.74 \times 10^5$ (Fig. 5).

Table 1: Attachment-anchored branch: steady max χ versus nose Reynolds number $Re_r = L^2$ on the frozen Hiemenz wedge, standard SA, $\chi_\infty = 0$. Below the bisected critical band the layer collapses to $\chi = 0$.

$L = \sqrt{Re_r}$	Re_r	steady max χ
≤ 683	$\leq 4.66 \times 10^5$	$< 10^{-3}$ (collapses to 0)
683–689	$4.66–4.74 \times 10^5$	critical band (bisected)
689	4.74×10^5	2.19
700	4.90×10^5	6.15
1000	1.00×10^6	24.1
3000	9.00×10^6	75.7

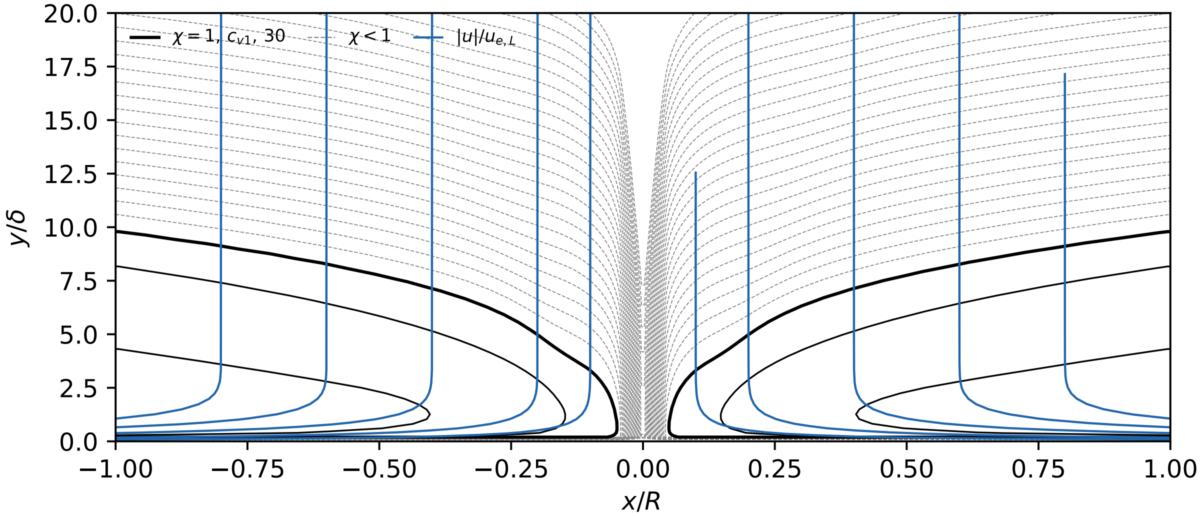


Figure 5: Attachment-anchored turbulent branch, $L = 3000$ ($Re_r = 9 \times 10^6$), $\max \chi = 76$. Standard SA, frozen Hiemenz, $\chi_\infty = 0$. Abscissa x/R (arc length in nose radii, $R/\delta = L$); ordinate y/δ. χ contours dashed per decade for $\chi < 1$ (10^{-24}–10^{-1}; the zero freestream leaves no floor, so χ decays smoothly over decades), solid at 1, c_{v1}, 30; velocity magnitude $|u|/u_{e,L}$ (blue, normalized by the $x = L$ edge speed) at 0.1, 0.2, 0.4, 0.6, 0.8. Branch data in Table 1.

The protocol used for every result in this paper: the $\tilde{\nu}$-equation residual is scaled by $f^{\,1-g}$, where g rises smoothly from 0 (laminar) to 1 (turbulent) with χ and $f = 10^{-2}$ (the solver uses $g = 1 - e^{-(\chi - c_{v1})/4}$ above $\chi = c_{v1}$, zero below). Because the freestream boundary condition flows through the same scaling, the freestream viscosity ratio supplied to the solver is $\chi_\infty f$; the seeds quoted throughout are the physical χ_∞, recovered at the boundary-layer edge. (In the reference implementation the scaling is `AI_LAMINAR_SLOWDOWN` $= f$, and the case files carry the pre-multiplied boundary seed $\chi_\infty f$; running a committed case at $f = 1$ without rescaling the seed mis-seeds it by $1/f$.)

2 Results

Coupled RANS: Flow360, unstructured node-centered second-order finite volume, Roe flux, $M = 0.1$. Airfoil cases: $\chi_\infty = c_{v1} e^{-9} \approx 8.76 \times 10^{-4}$ (the $N_{\text{crit}} = 9$ anchor; via Mack's map, $Tu \approx 0.07\%$); two independently generated mesh families—all-triangular unstructured cavity meshes, constrained-Delaunay through the boundary layer, and all-quadrilateral structured O-grids—at three refinement levels, 1.5×10^4 to 2.6×10^5 nodes. Transition locations are the solver's near-wall $\chi = 1$ crossing (the $\chi = c_{v1}$ crossing sits up to $0.05\,c$ aft where the front grows slowly); bubble stations are signed-C_f zero crossings. The cross-solver replication is a cell-centered, pressure-based, incompressible OpenFOAM implementation of the model on the same flat-plate grid specification and the same structured airfoil grids (the port omits the f_{v1} bypass); thirty airfoil cases plus the flat-plate sweep.

2.1 Flat plate, $Tu = 0.04$–0.60%

Structured 320×80 grid, $Re_x \le 6 \times 10^6$, $y^+ \lesssim 0.3$; five seeds from Mack's map ($\chi_\infty \approx 2.3 \times 10^{-4}$, 1.2×10^{-3}, 6.3×10^{-3}, 2.9×10^{-2}, 1.5×10^{-1}). Figure 6: at the $\chi = c_{v1}$ crossing the onset brackets the Abu-Ghannam & Shaw correlation within 10% across the range. Figure 7: $\chi = 1$ crossings within 7% in Re_θ. Table 2: both solvers' crossings.

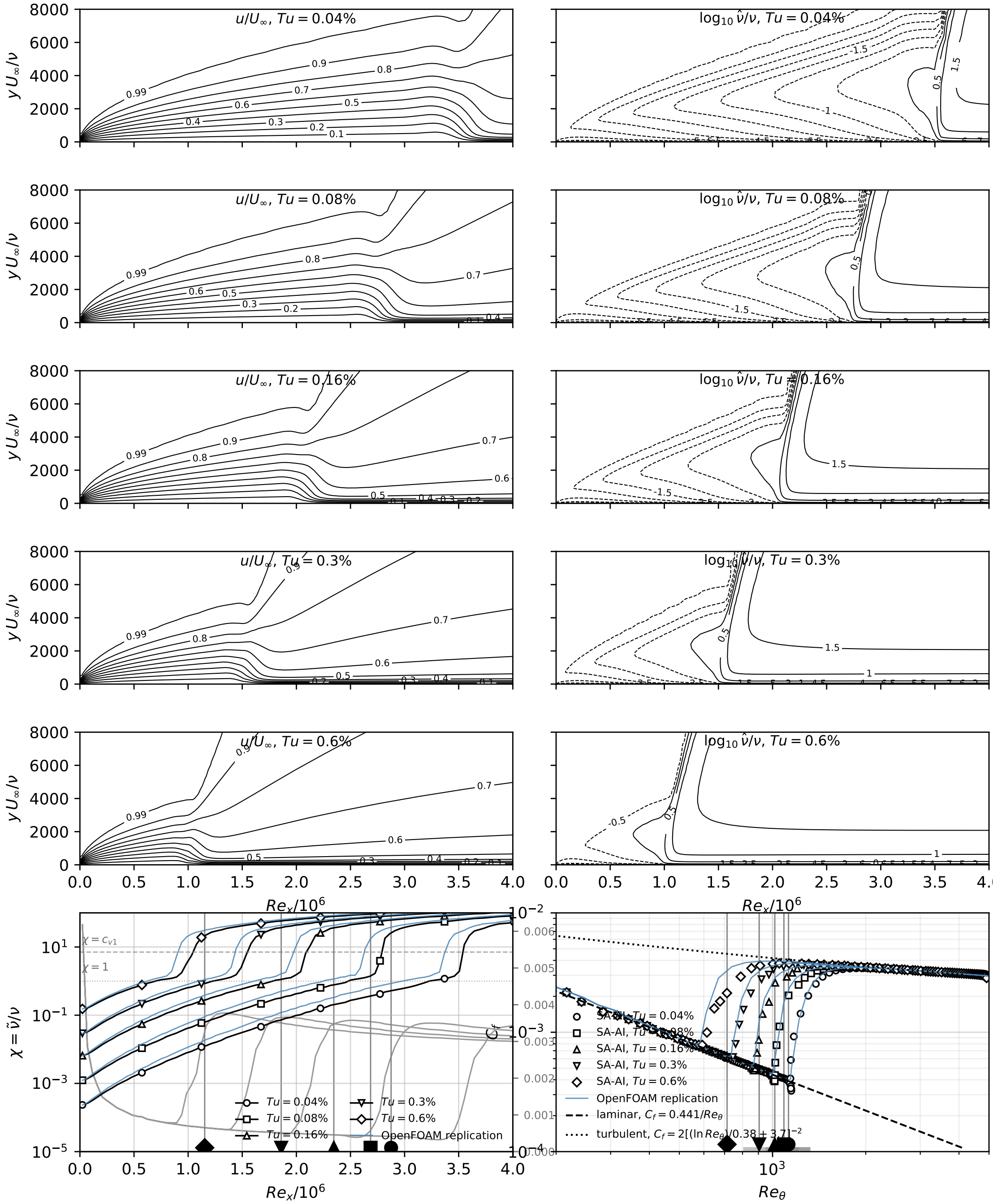


Figure 6: Flat plate, Flow360. Five levels $Tu = 0.04$–0.60%; references $\chi = 1$ and $\chi = c_{v1} = 7.1$; laminar $C_f = 0.441/Re_\theta$, turbulent $C_f = 2[(\ln Re_\theta)/0.38 + 3.7]^{-2}$; $\chi = c_{v1}$ onset brackets Abu-Ghannam & Shaw within 10%.

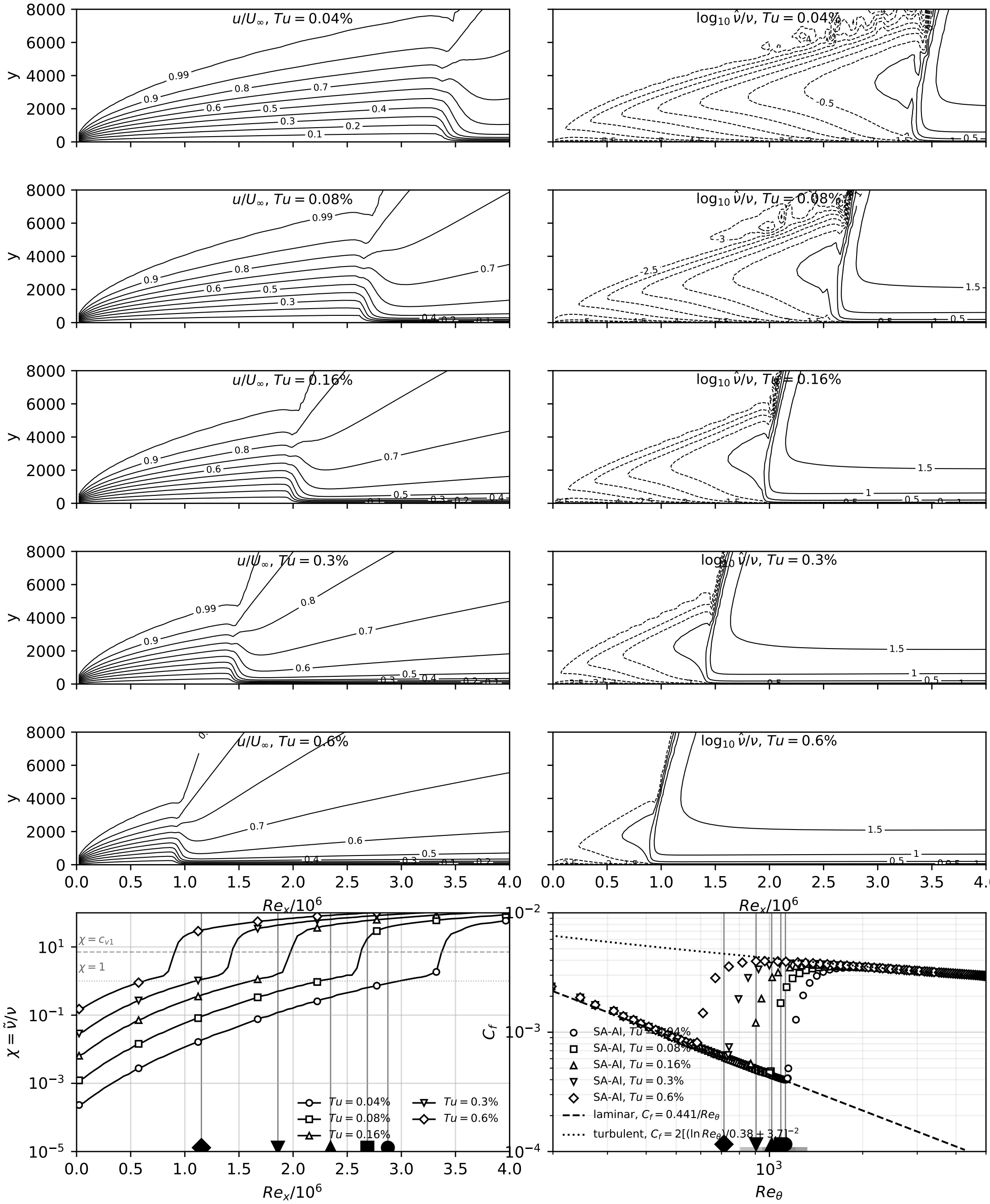


Figure 7: Flat plate, OpenFOAM. Same sweep as Fig. 6, five levels $Tu = 0.04$–0.60%.

Table 2: Flat-plate transition-onset Re_θ (integrated from the computed profiles) at the $\chi=1$ and $\chi=c_{v1}$ crossings, Flow360 and the OpenFOAM replication (OF), against the Abu-Ghannam & Shaw correlation.

Tu [%]	AGS Re_θ	$\chi=1$	$\chi=c_{v1}$	OF $\chi=1$	OF $\chi=c_{v1}$
0.04	1126	1149	1203	1098	1154
0.08	1088	1002	1069	949	1004
0.16	1017	845	934	798	887
0.3	905	700	815	656	739
0.6	713	524	685	487	589

2.2 NLF(1)-0416 at $Re = 4 \times 10^6$

$\alpha \in \{0°, 4°, 9°, 15°\}$ on both families at L0–L2 plus $\alpha = -8°, -4°$ on all six grids (thirty-six cases). Figure 8: transition fronts against the LTPT microphone measurements, Coder's AFT (OVERFLOW) at $N_{\text{crit}} = 10.07$ and recalibrated 7.18, the fourteen First AIAA Transition Modeling Workshop submittals, SA-LM2015, Langtry–Menter, and XFOIL. Figure 9: drag polar against Somers TP-1861, the e^9 reference, and fully turbulent SA. Figures 10–12: surface distributions. Table 3: forces and fronts. Table 4: at L2 every front within $0.016\,c$ and the lift within 0.9% of the SA-AI values.

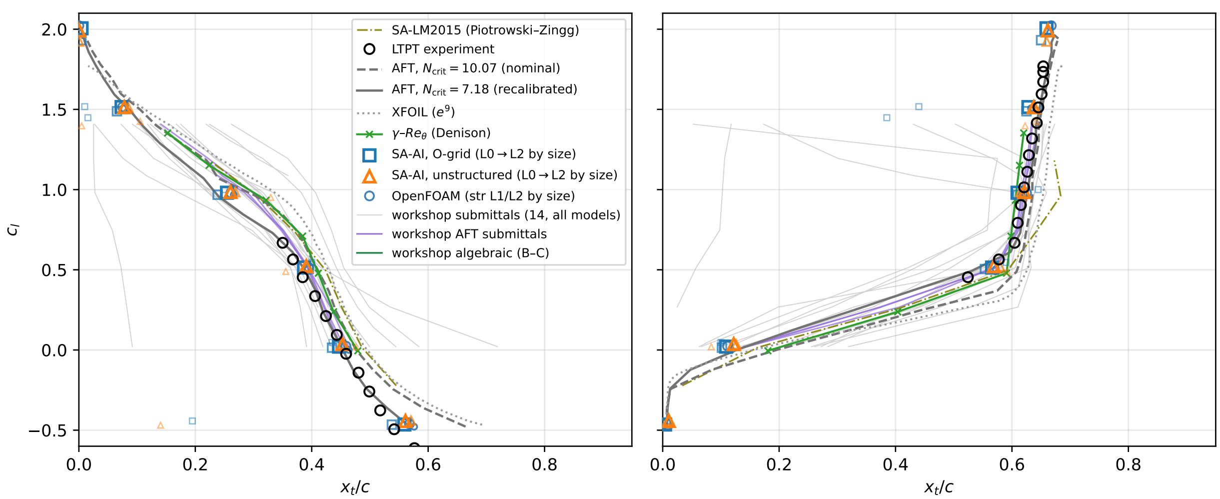


Figure 8: NLF(1)-0416 transition fronts, $Re = 4 \times 10^6$; near-wall $\chi = 1$ convention. SA-AI at $\alpha = 0°, 4°, 9°, 15°, -8°, -4°$; workshop sweep $\alpha = -4°$–$8°$.

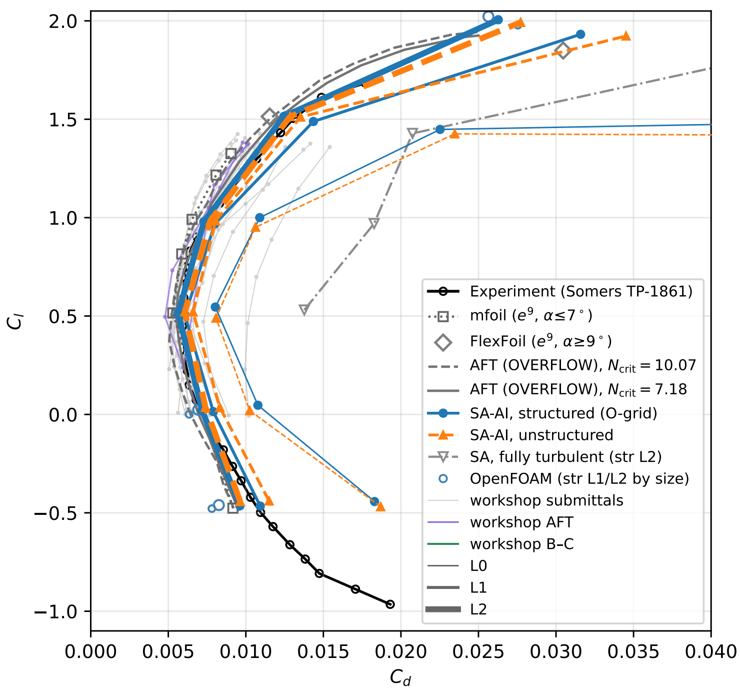


Figure 9: NLF(1)-0416 drag polar, $Re = 4 \times 10^6$. Fully turbulent SA ($\chi_\infty = 3$): 1.6–2.5× above the SA-AI drag, +144/+150% in the bucket at $\alpha = 0°/4°$, +67/+74% at 9°/15°.

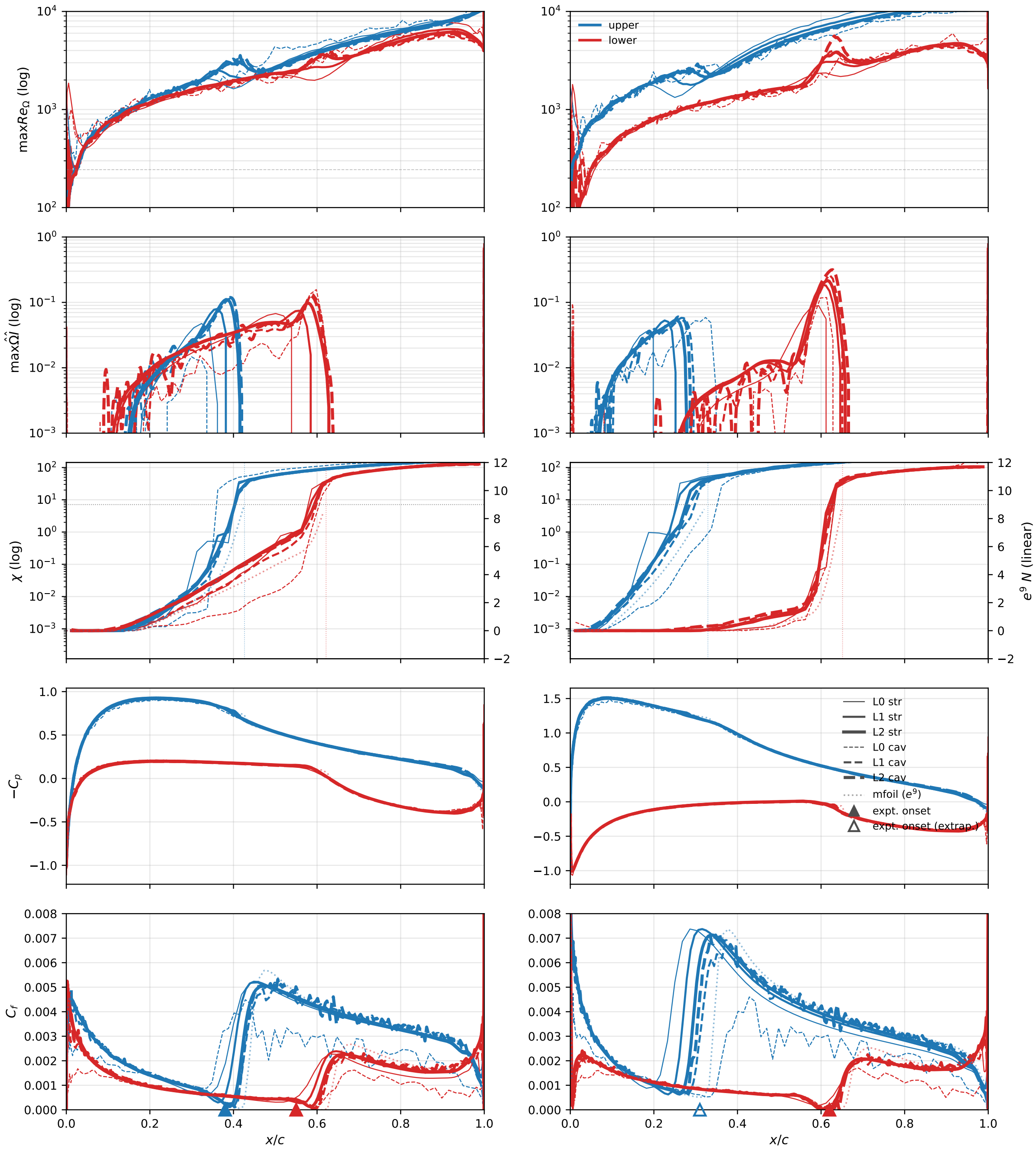


Figure 10: NLF(1)-0416, $\alpha = 0°, 4°$, $Re = 4 \times 10^6$.

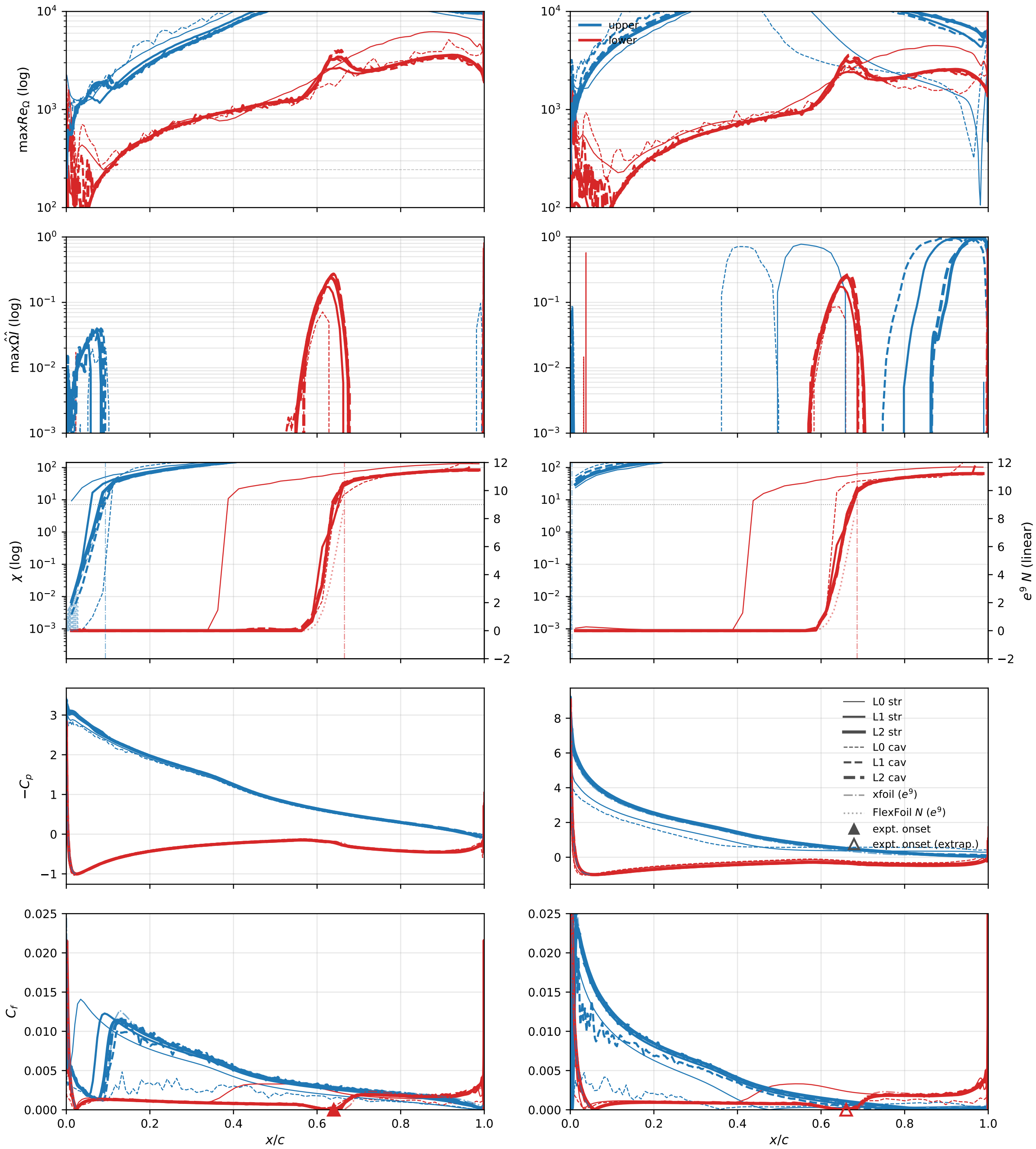


Figure 11: NLF(1)-0416, $\alpha = 9^\circ, 15^\circ$, $Re = 4 \times 10^6$.

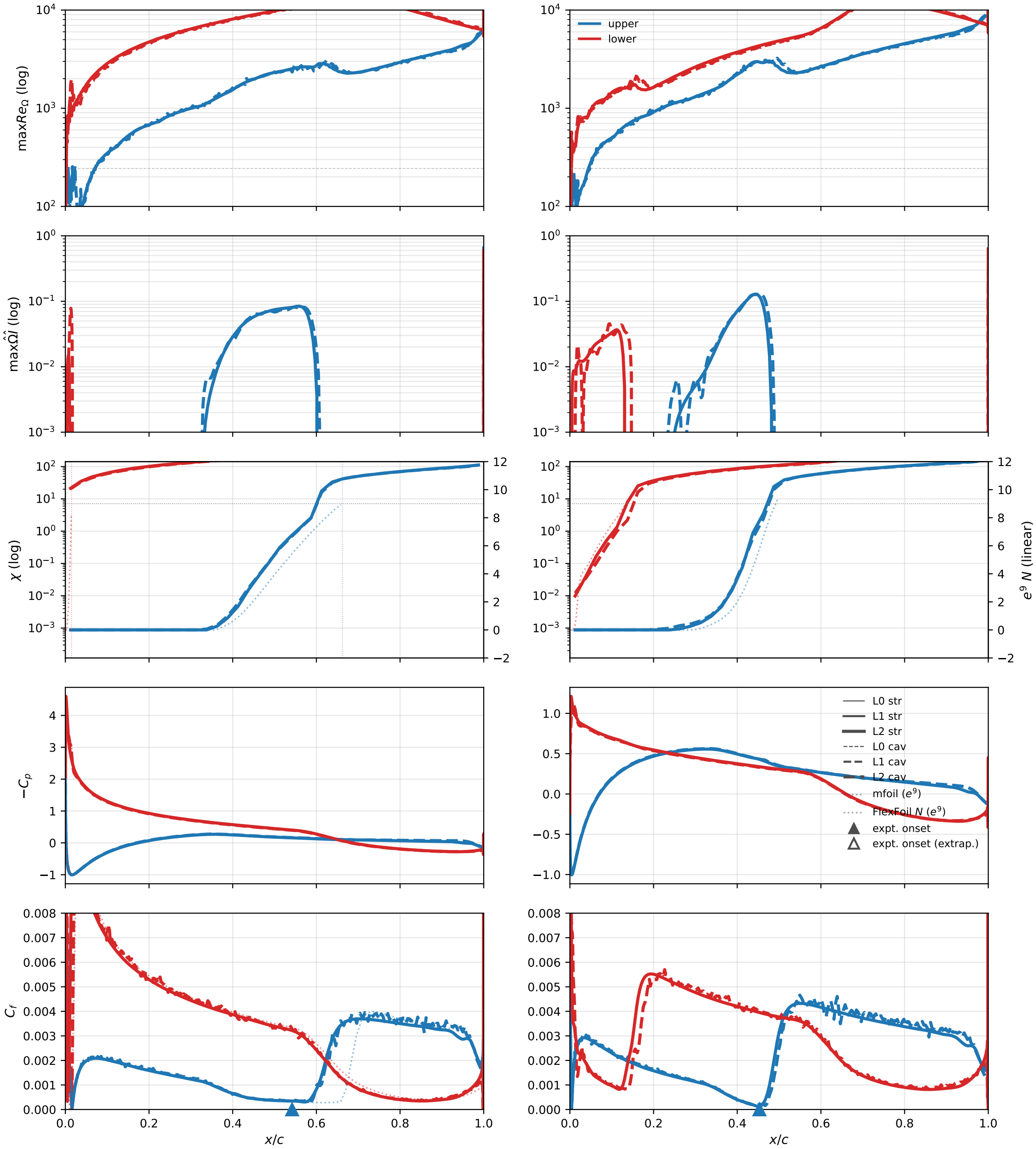


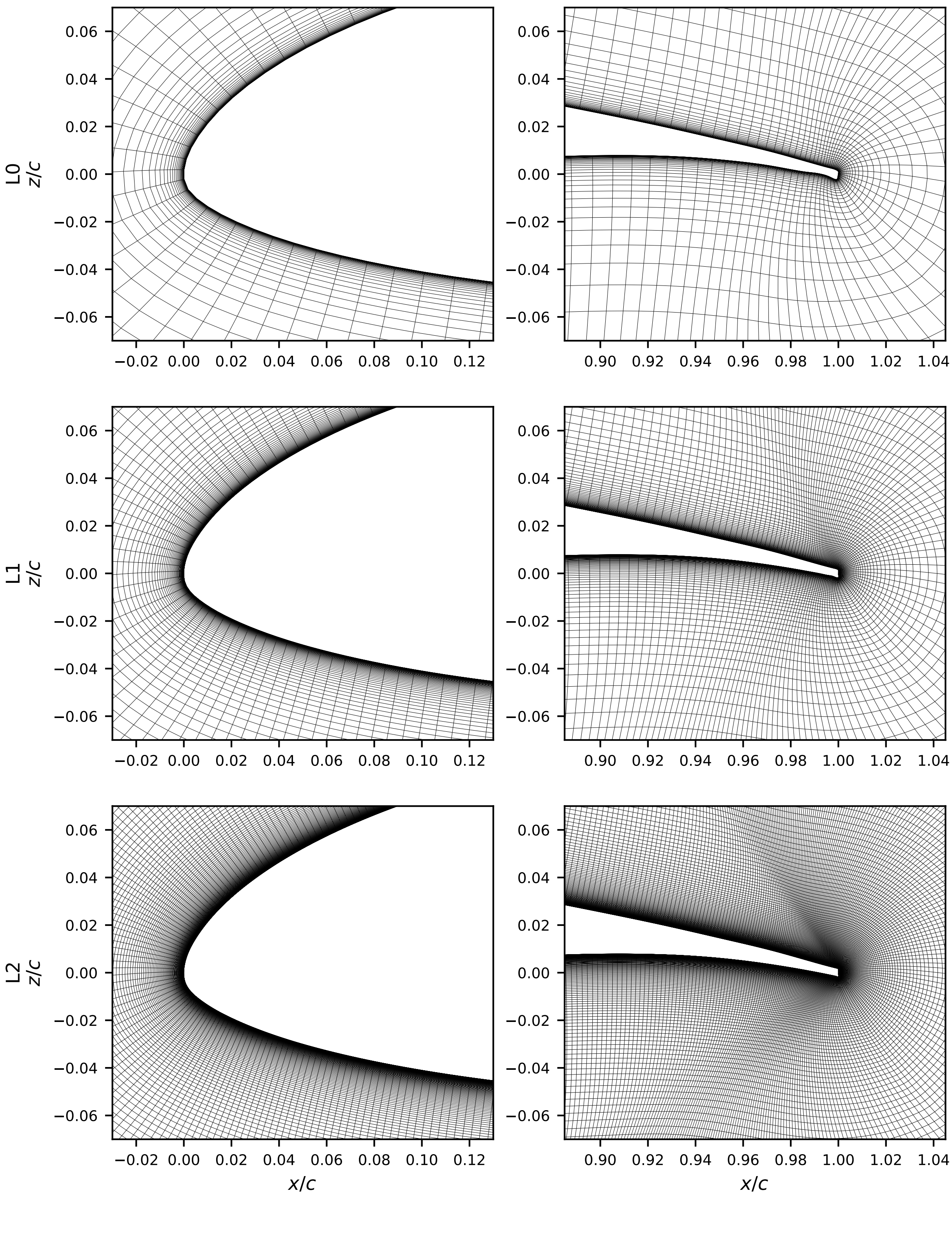
Figure 12: NLF(1)-0416, $\alpha = -8°, -4°$, $Re = 4 \times 10^6$.

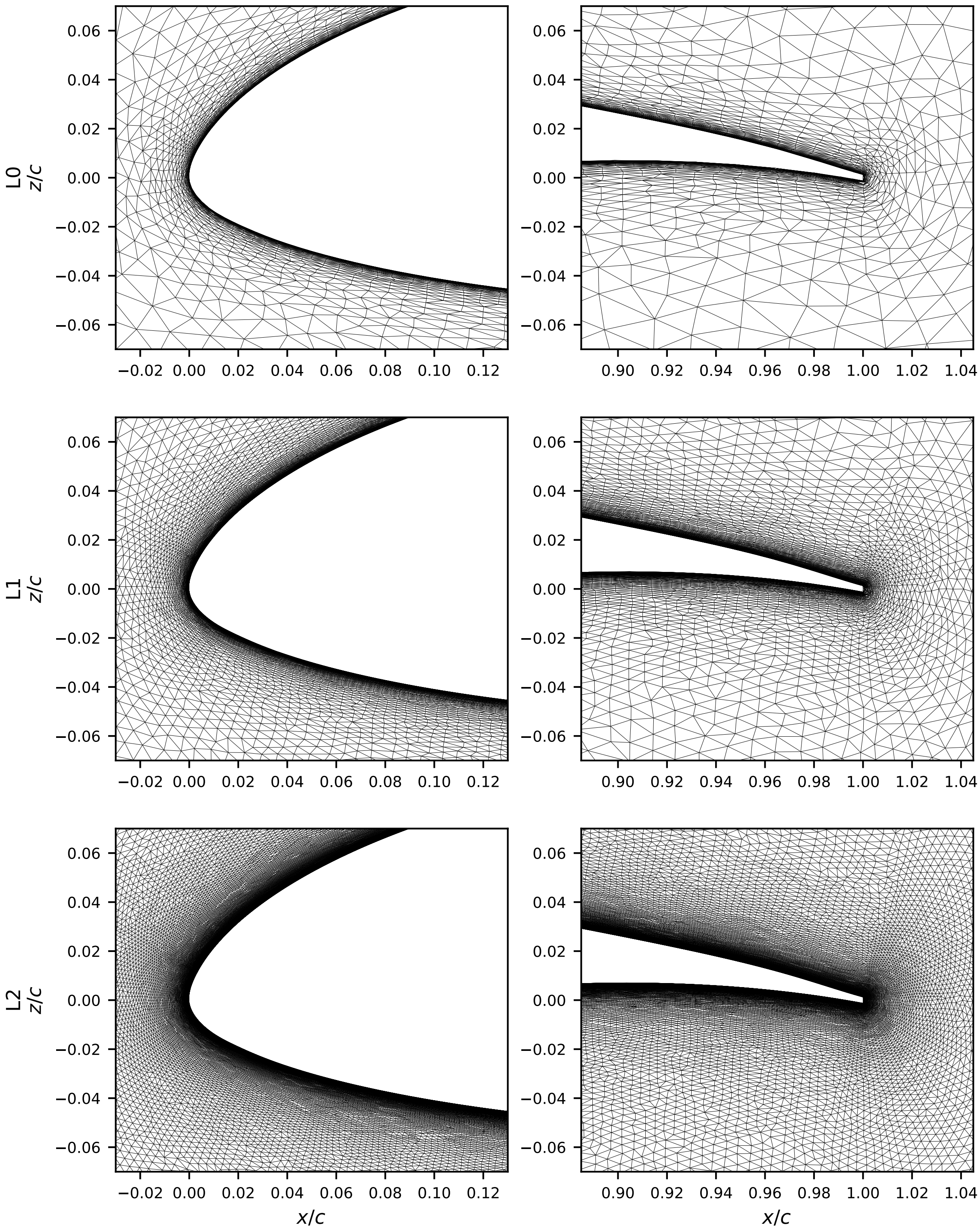
Table 3: NLF(1)-0416, $Re = 4\times10^6$: computed forces and transition locations (near-wall $\chi = 1$ front stations).

α	grid	structured O-grid				unstructured cavity			
		C_L	C_D	x_{tr}^{up}	x_{tr}^{lo}	C_L	C_D	x_{tr}^{up}	x_{tr}^{lo}
-8	L0	-0.4426	0.01830	0.195	0.011	-0.4689	0.01869	0.140	0.012
-8	L1	-0.4656	0.01093	0.538	0.006	-0.4397	0.01150	0.571	0.011
-8	L2	-0.4646	0.00962	0.559	0.004	-0.4426	0.00964	0.561	0.010
-4	L0	0.0469	0.01078	0.435	0.100	0.0197	0.01025	0.476	0.083
-4	L1	0.0142	0.00790	0.435	0.102	0.0357	0.00832	0.452	0.126
-4	L2	0.0208	0.00718	0.447	0.109	0.0363	0.00739	0.453	0.122
0	L0	0.5458	0.00803	0.400	0.590	0.4896	0.00808	0.355	0.592
0	L1	0.5048	0.00607	0.395	0.553	0.5209	0.00660	0.392	0.579
0	L2	0.5139	0.00564	0.387	0.566	0.5219	0.00606	0.391	0.569
4	L0	1.0001	0.01090	0.250	0.645	0.9513	0.01062	0.330	0.621
4	L1	0.9678	0.00800	0.238	0.610	0.9839	0.00805	0.269	0.614
4	L2	0.9805	0.00730	0.255	0.610	0.9848	0.00772	0.260	0.621
9	L0	1.4483	0.02252	0.015	0.385	1.4251	0.02345	0.105	0.641
9	L1	1.4886	0.01436	0.065	0.625	1.5112	0.01353	0.085	0.635
9	L2	1.5146	0.01245	0.073	0.628	1.5139	0.01286	0.078	0.638
15	L0	1.5172	0.07109	0.010	0.440	1.3976	0.11602	0.005	0.623
15	L1	1.9317	0.03159	0.004	0.650	1.9232	0.03454	0.001	0.659
15	L2	2.0057	0.02628	0.005	0.659	1.9924	0.02772	0.000	0.662

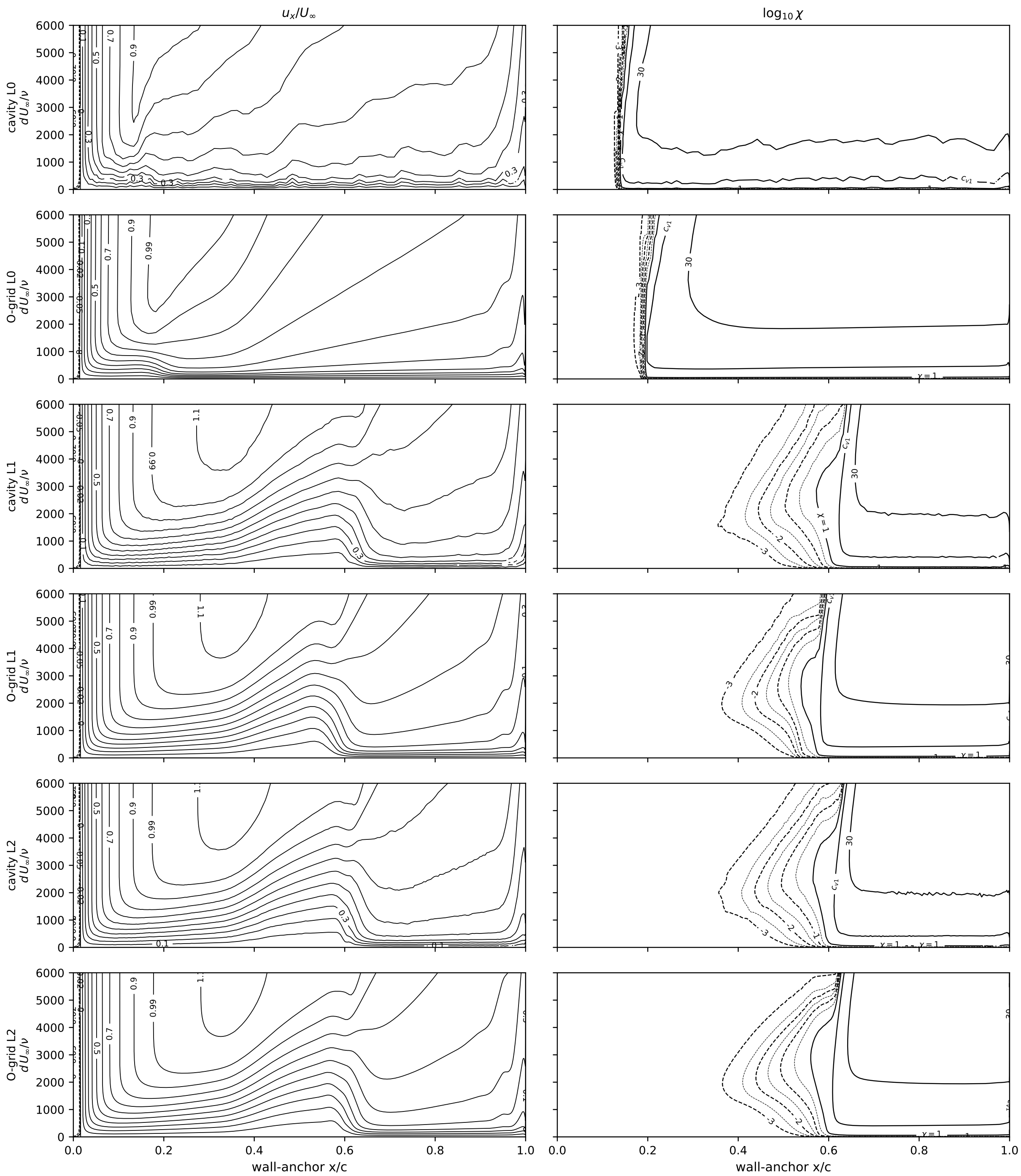
Table 4: NLF(1)-0416: the independent OpenFOAM implementation on the structured L0, L1 and L2 grids, fronts at the same near-wall $\chi = 1$ convention (the port omits the f_{v1} bypass).

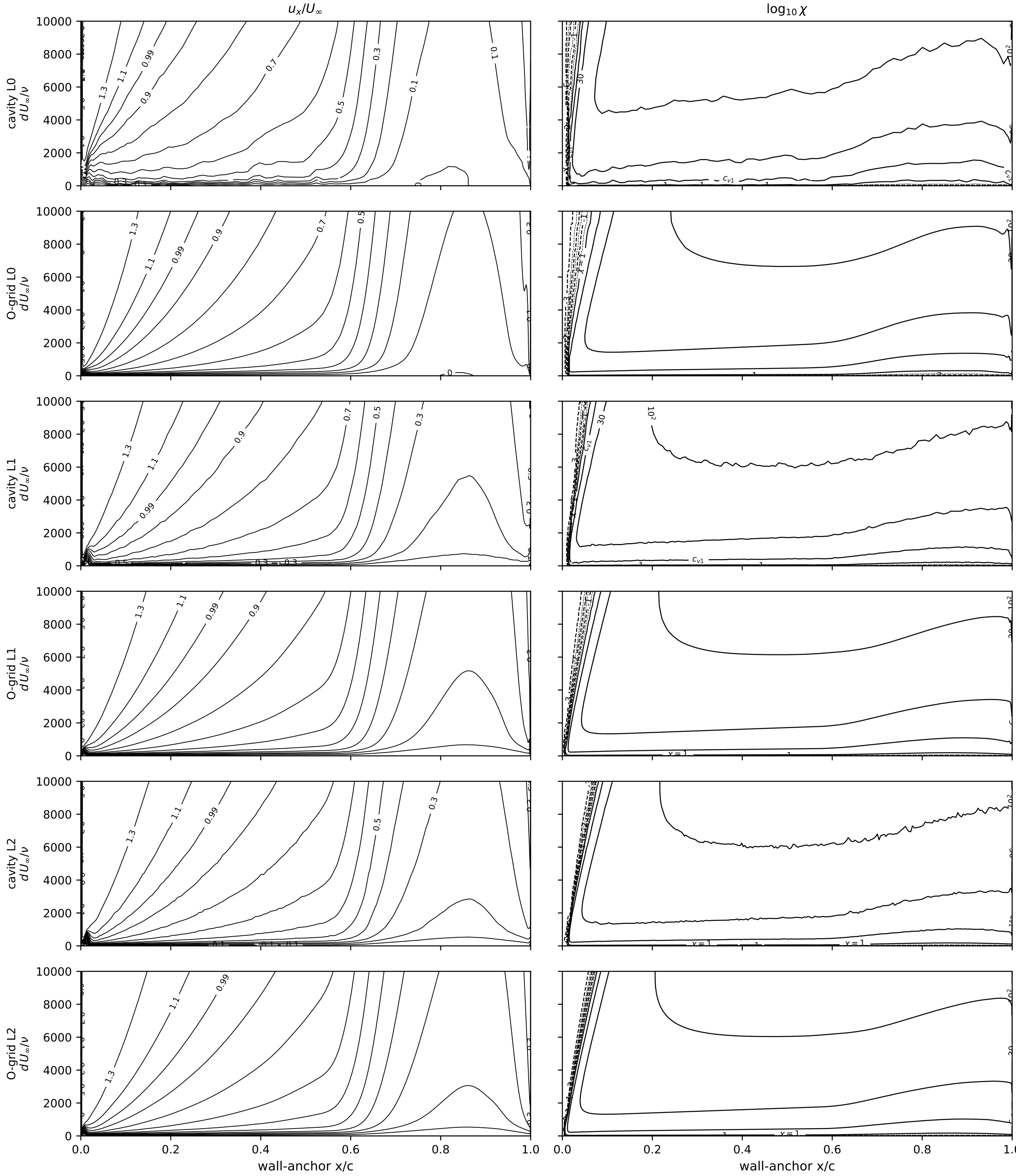
α	grid	C_L	C_D	x_{tr}^{up}	x_{tr}^{lo}
-8	L0	-0.4460	0.01241	0.556	0.007
-8	L1	-0.4785	0.00781	0.576	0.000
-8	L2	-0.4606	0.00826	0.568	0.000
-4	L0	0.0388	0.00748	0.424	0.018
-4	L1	0.0005	0.00634	0.464	0.117
-4	L2	0.0239	0.00693	0.459	0.111
0	L0	0.5473	0.00669	0.387	0.564
0	L1	0.5034	0.00603	0.393	0.572
0	L2	0.5153	0.00573	0.395	0.568
4	L0	1.0114	0.00955	0.271	0.618
4	L1	0.9670	0.00778	0.266	0.616
4	L2	0.9793	0.00749	0.261	0.625
9	L0	1.5277	0.01496	0.063	0.618
9	L1	1.5066	0.01273	0.076	0.633
9	L2	1.5151	0.01270	0.075	0.644
15	L0	1.7210	0.05409	0.000	0.634
15	L1	1.9780	0.02754	0.000	0.665
15	L2	2.0221	0.02564	0.000	0.667

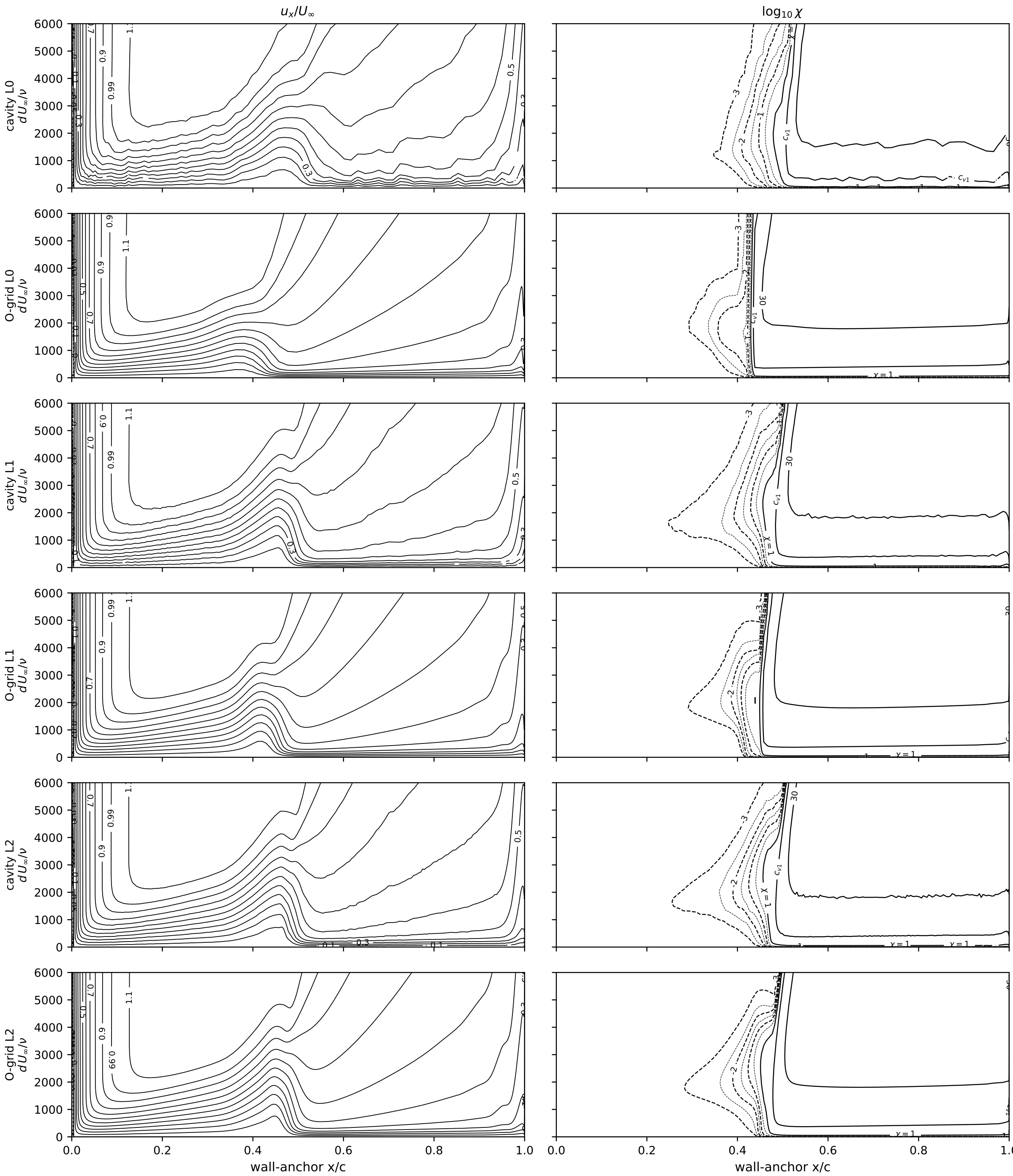

L0
L1
L2
z/c
x/c
0.06
0.04
0.02
0.00
−0.02
−0.04
−0.06
−0.02 0.00 0.02 0.04 0.06 0.08 0.10 0.12
0.90 0.92 0.94 0.96 0.98 1.00 1.02 1.04



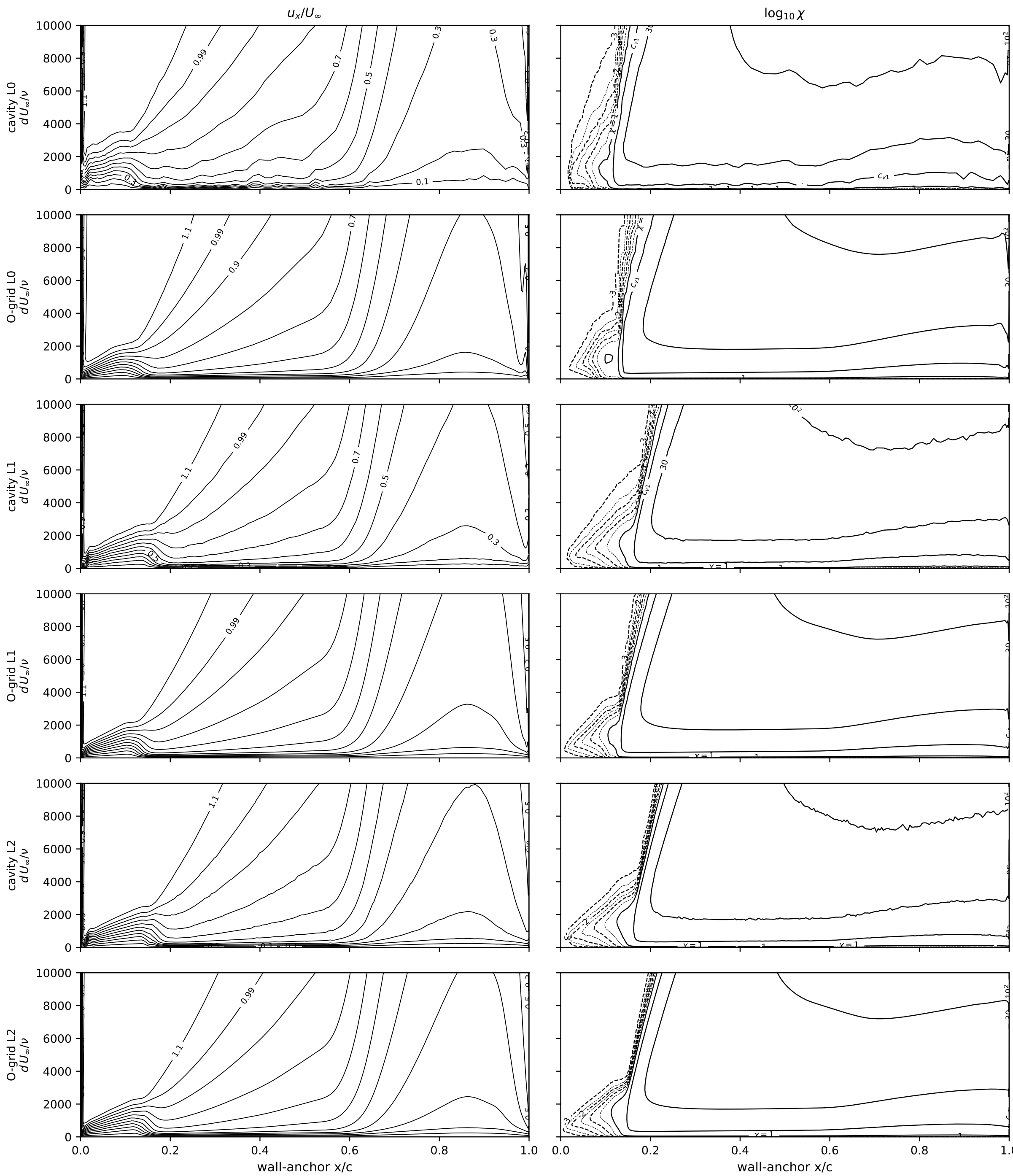
L0
L1
L2
z/c
x/c


u_x/U_∞
$\log_{10}\chi$
cavity L0 dU_∞/ν
O-grid L0 dU_∞/ν
cavity L1 dU_∞/ν
O-grid L1 dU_∞/ν
cavity L2 dU_∞/ν
O-grid L2 dU_∞/ν
wall-anchor x/c
wall-anchor x/c

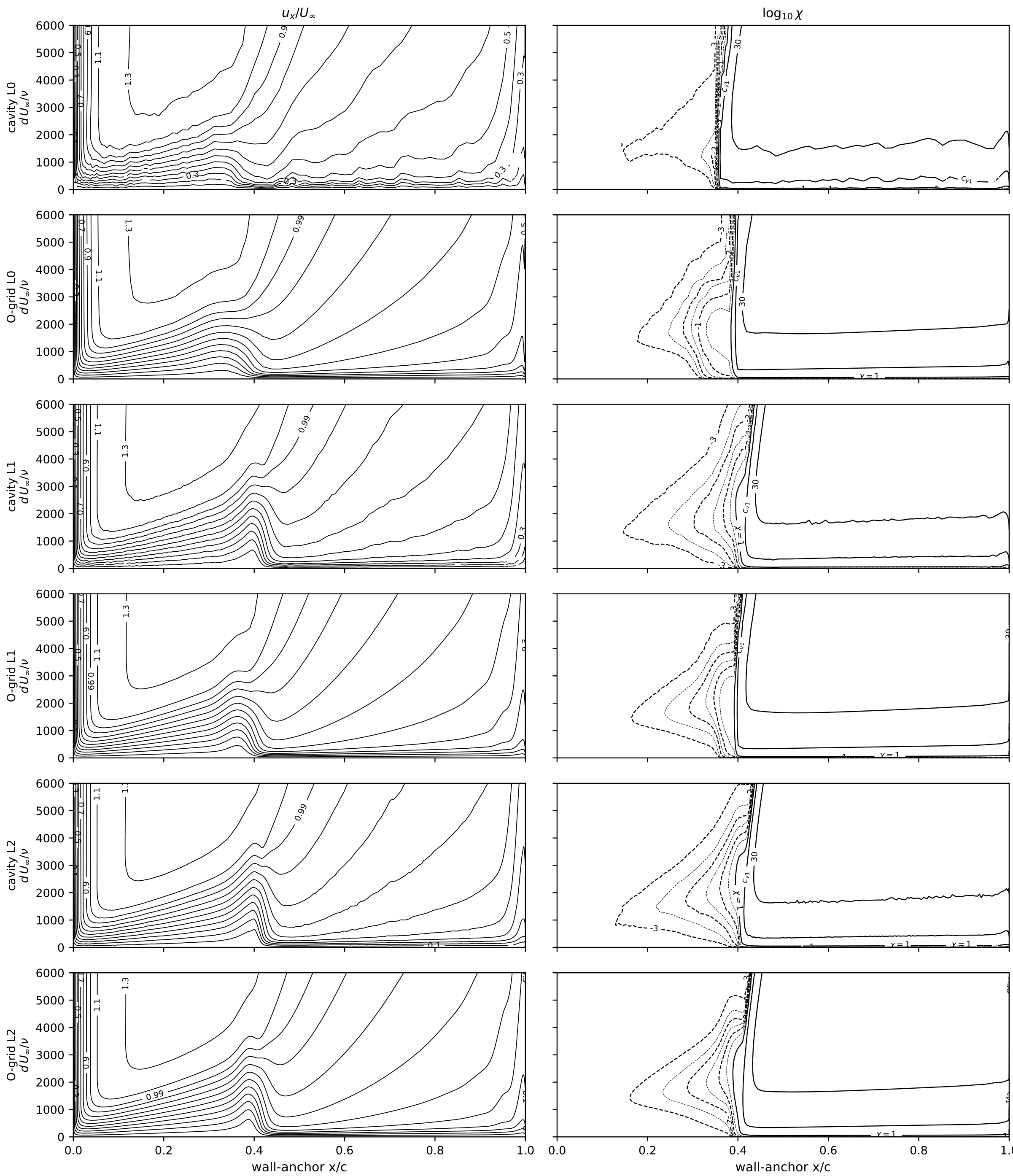

u_x/U_∞
$\log_{10}\chi$
cavity L0
O-grid L0
cavity L1
O-grid L1
cavity L2
O-grid L2
dU_∞/ν
wall-anchor x/c
wall-anchor x/c



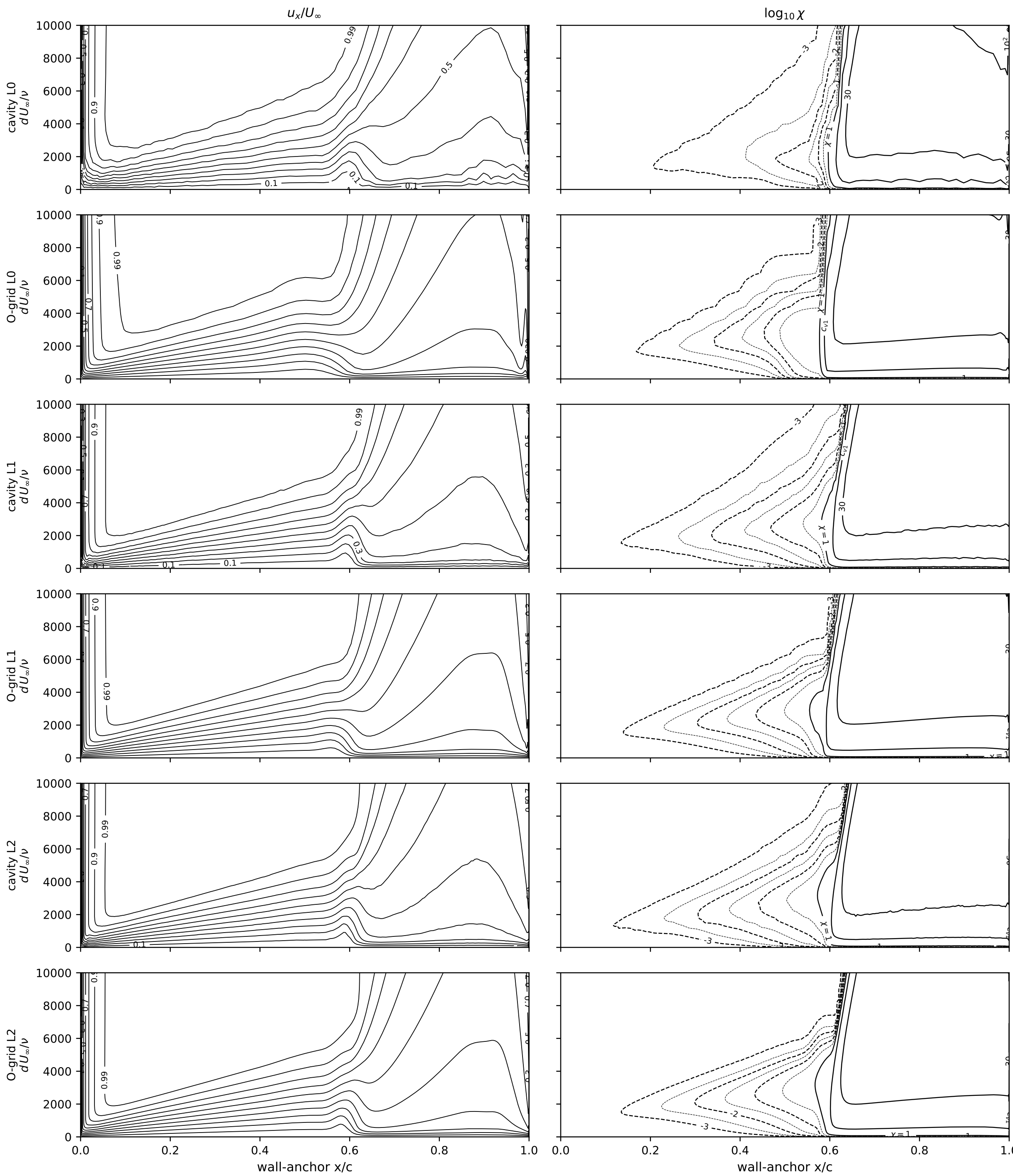
u_x/U_∞
$\log_{10}\chi$
cavity L0
O-grid L0
cavity L1
O-grid L1
cavity L2
O-grid L2
dU_∞/ν
wall-anchor x/c
wall-anchor x/c



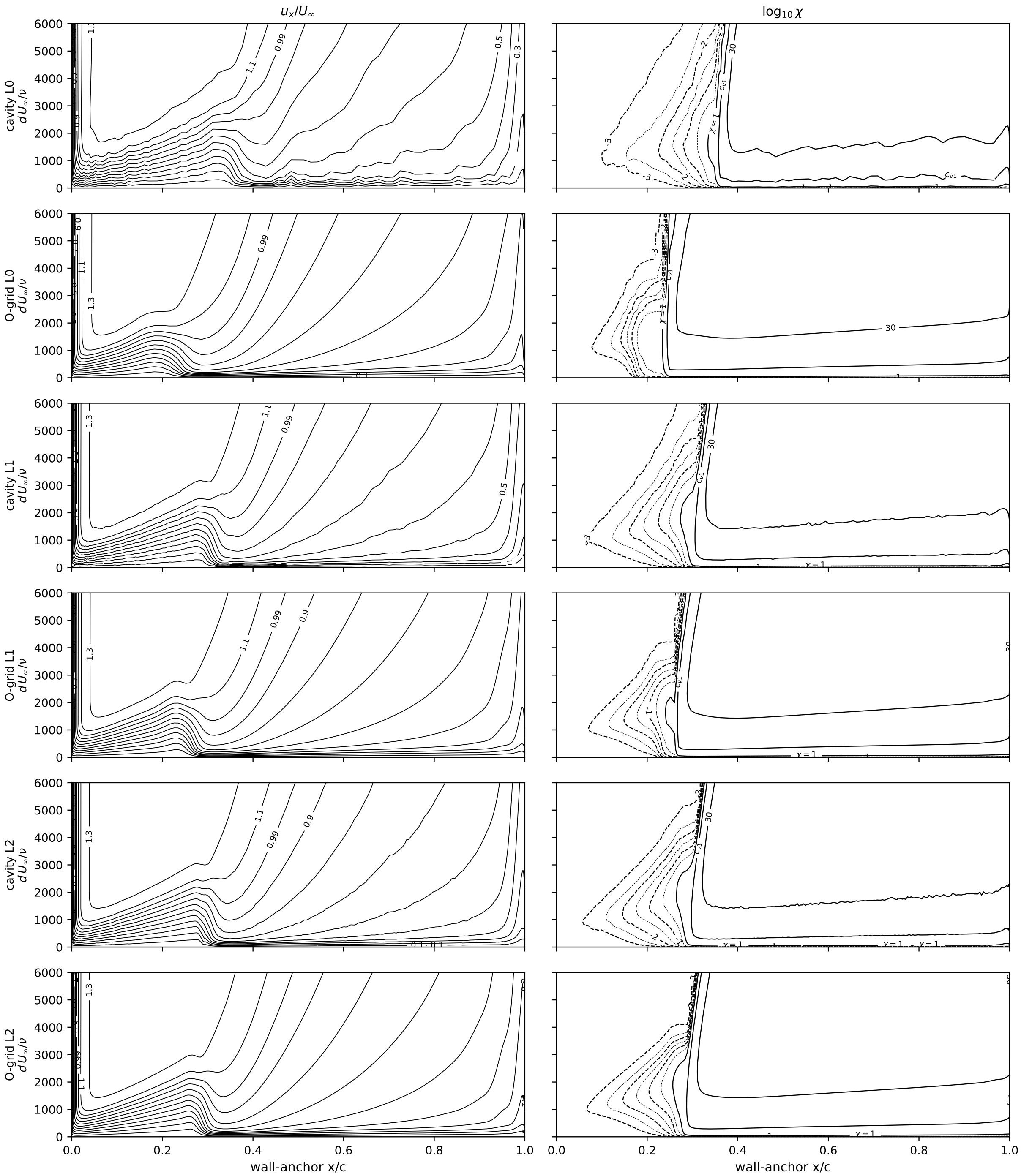
u_x/U_∞
$\log_{10}\chi$
cavity L0
O-grid L0
cavity L1
O-grid L1
cavity L2
O-grid L2
$d\,U_\infty/\nu$
wall-anchor x/c

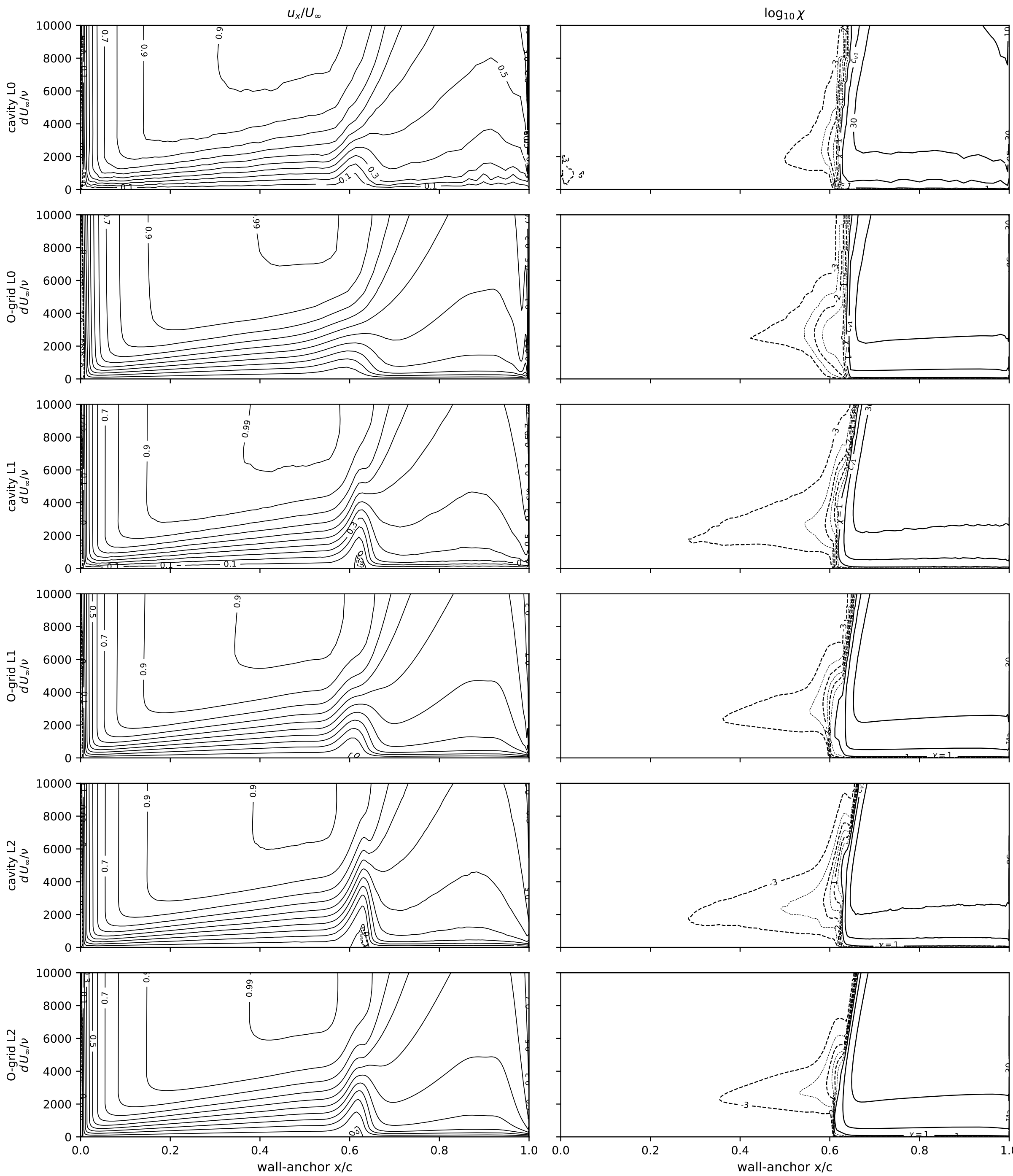


u_x/U_∞
$\log_{10}\chi$
cavity L0
O-grid L0
cavity L1
O-grid L1
cavity L2
O-grid L2
dU_∞/ν
wall-anchor x/c

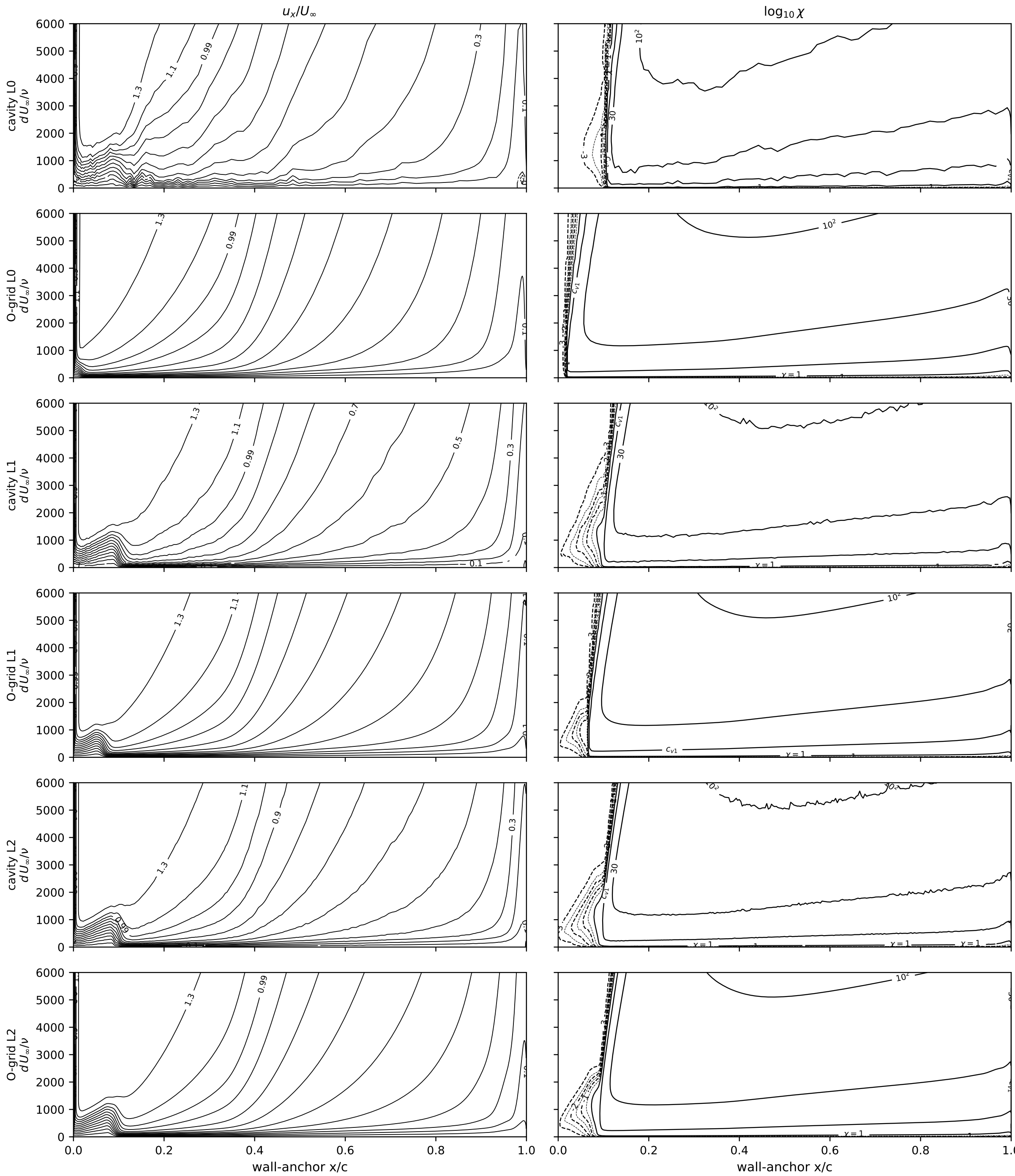


u_x/U_∞
$\log_{10}\chi$
cavity L0 $d\,U_\infty/\nu$
O-grid L0 $d\,U_\infty/\nu$
cavity L1 $d\,U_\infty/\nu$
O-grid L1 $d\,U_\infty/\nu$
cavity L2 $d\,U_\infty/\nu$
O-grid L2 $d\,U_\infty/\nu$
wall-anchor x/c
wall-anchor x/c

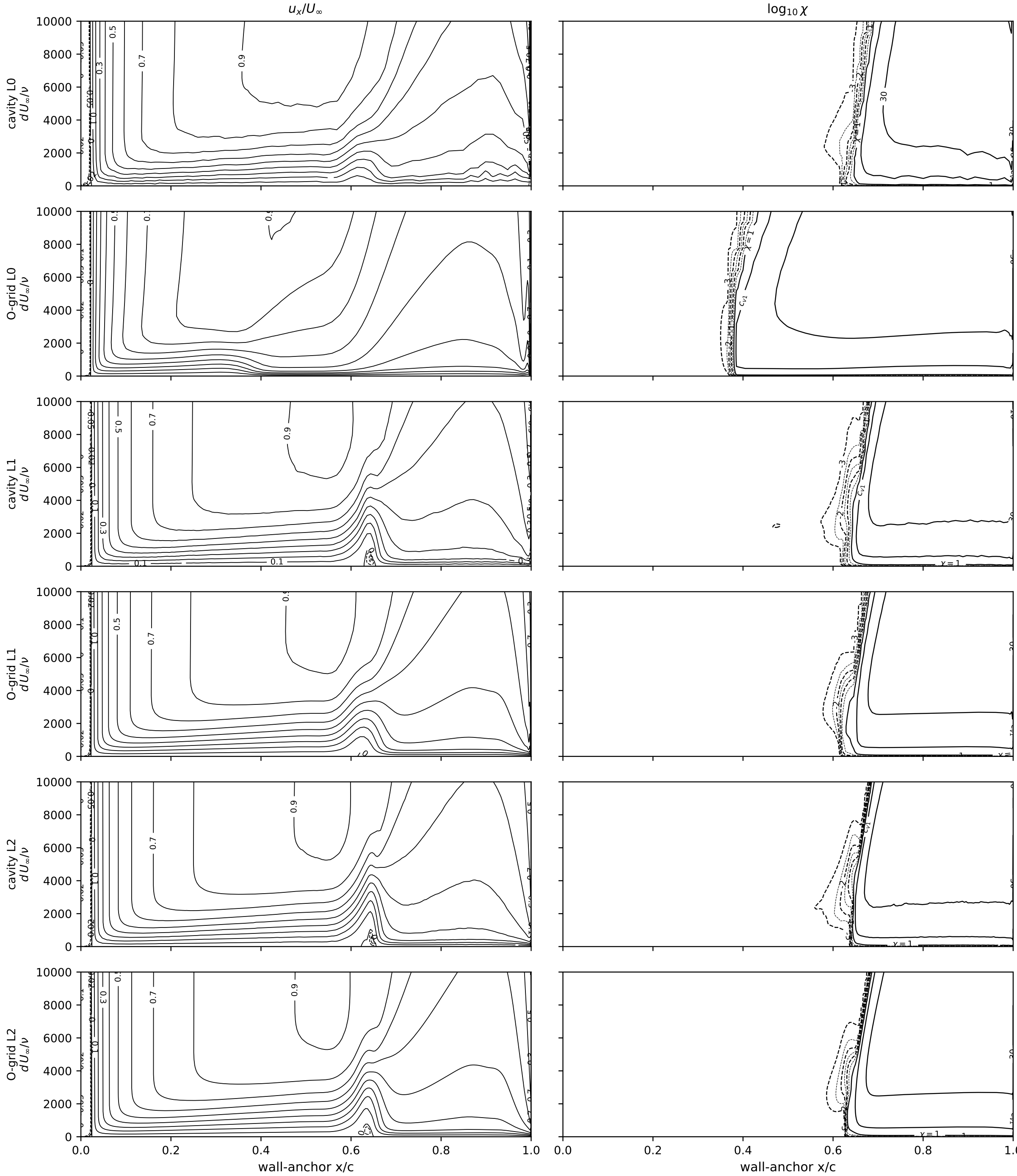

u_x/U_∞
$\log_{10}\chi$
cavity L0 dU_∞/ν
O-grid L0 dU_∞/ν
cavity L1 dU_∞/ν
O-grid L1 dU_∞/ν
cavity L2 dU_∞/ν
O-grid L2 dU_∞/ν
wall-anchor x/c
wall-anchor x/c



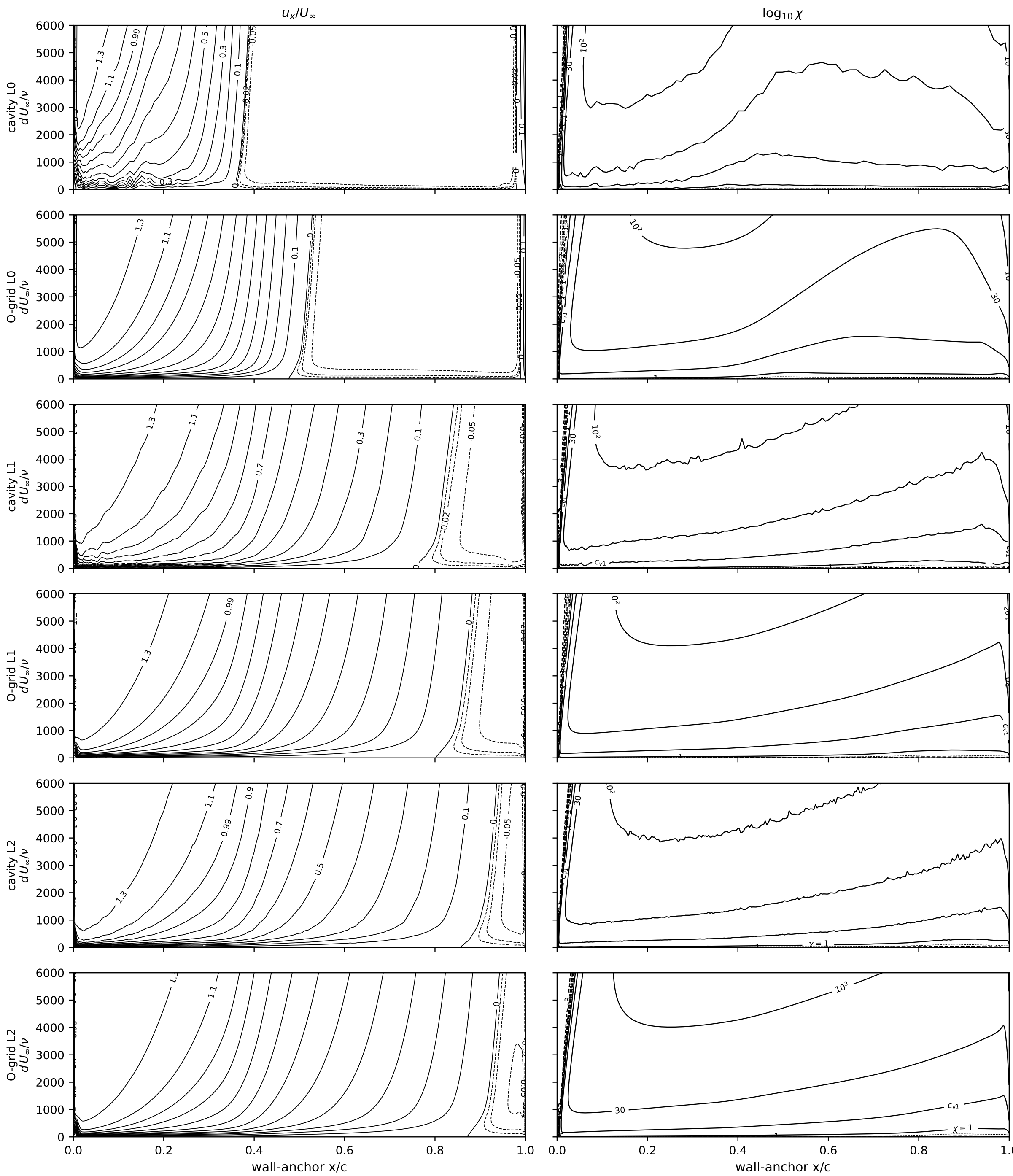
u_x/U_∞
$\log_{10}\chi$
cavity L0
O-grid L0
cavity L1
O-grid L1
cavity L2
O-grid L2
dU_∞/ν
wall-anchor x/c
wall-anchor x/c



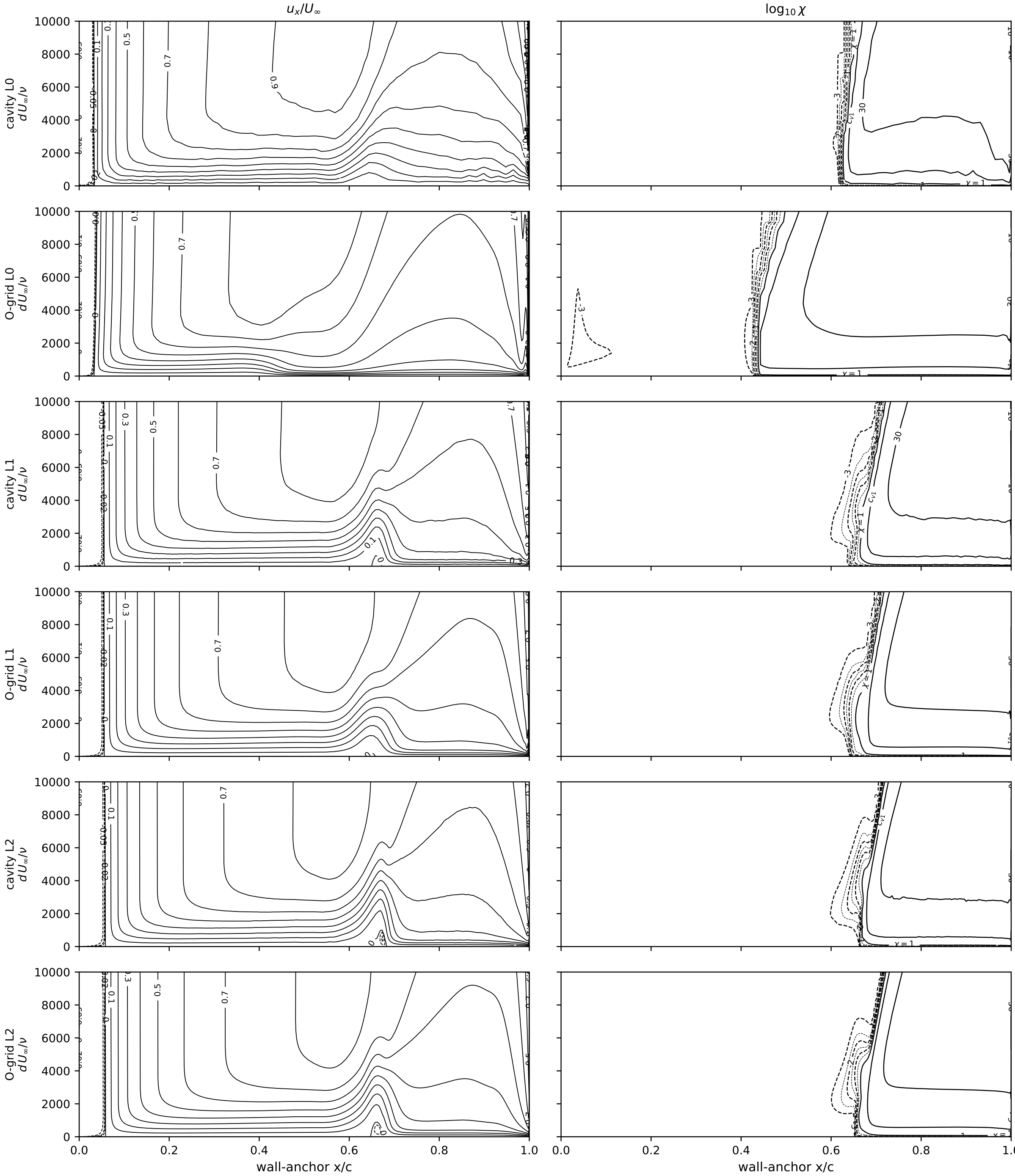
u_x/U_∞
$\log_{10}\chi$
cavity L0
O-grid L0
cavity L1
O-grid L1
cavity L2
O-grid L2
$d\,U_\infty/\nu$
wall-anchor x/c



u_x/U_∞
$\log_{10}\chi$
cavity L0
O-grid L0
cavity L1
O-grid L1
cavity L2
O-grid L2
dU_∞/ν
wall-anchor x/c



u_x/U_∞
$\log_{10}\chi$
cavity L0 $d\,U_\infty/\nu$
O-grid L0 $d\,U_\infty/\nu$
cavity L1 $d\,U_\infty/\nu$
O-grid L1 $d\,U_\infty/\nu$
cavity L2 $d\,U_\infty/\nu$
O-grid L2 $d\,U_\infty/\nu$
wall-anchor x/c
wall-anchor x/c



2.3 Eppler 387 at $Re = 2 \times 10^5$

$\alpha \in \{0°, 2°, 5°, 7°\}$ on both families at L0–L2 plus six extension incidences ($-2°$, $1°$, $3°$, $4°$, $6°$, $8.5°$) on the finest pair, and the two bounding incidences ($-2°$, $8.5°$) on L0 and L1 as well, so all six grids span the measured polar end to end (forty-four solutions, one solver build); same seed. Figure 13: drag polar against the LTPT measurement (McGhee et al., TM-4062), the e^9 reference, fully turbulent SA, and the digitized γ–Re_θ and SA-BC curves (-6.4% to $+3.7\%$ over $\alpha = 0°$–$7°$). Figure 14: bubble stations against the oil flow, Selig et al., Cole–Mueller, and the e^9 references—reattachment 0.02–0.04 c late at $\alpha \leq 5°$, at the edge of bubble collapse at 7°. Figures 15–16: surface suites. Table 5: forces. Table 6: bubble stations. Table 7: at L2 lift within 0.5%, separation 0.030–0.037 c ahead of and reattachment 0.033–0.045 c behind the SA-AI stations.

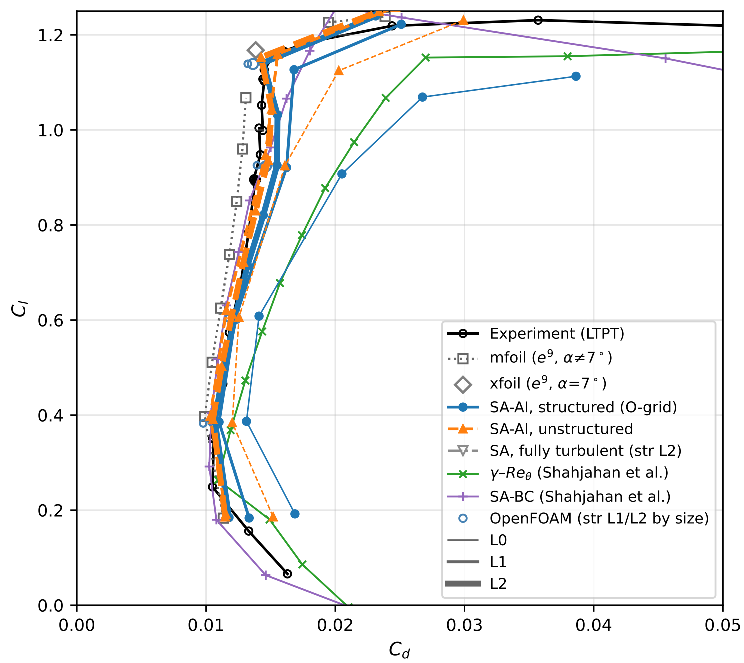


Figure 13: Eppler 387 drag polar, $Re = 2 \times 10^5$; all six grids span $\alpha = -2°$–$8.5°$. Fully turbulent SA ($\chi_\infty = 3$) 66–87% above the SA-AI drag in the bucket, +149% at $\alpha = 7°$; at L2 mesh families agree to $\Delta C_d \leq 6.9 \times 10^{-4}$ over $\alpha = -2°$–$7°$ and 2.7×10^{-4} at 8.5°; OpenFOAM port omits the bypass, worth -6 to -10 counts at $\alpha = 0$–$5°$.

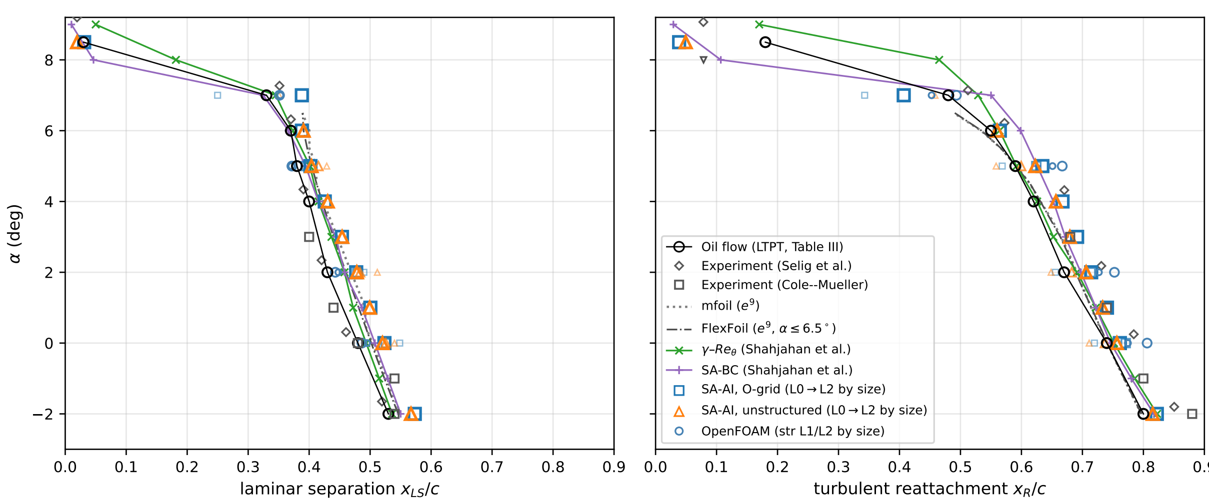


Figure 14: Eppler 387 bubble stations, $Re = 2 \times 10^5$; signed-C_f zero crossings against oil flow (TM-4062 Table III), Selig, Cole–Mueller, e^9. At $\alpha = 7°$ XFOIL $C_l = 1.17$, mfoil $C_l = 0.928$.

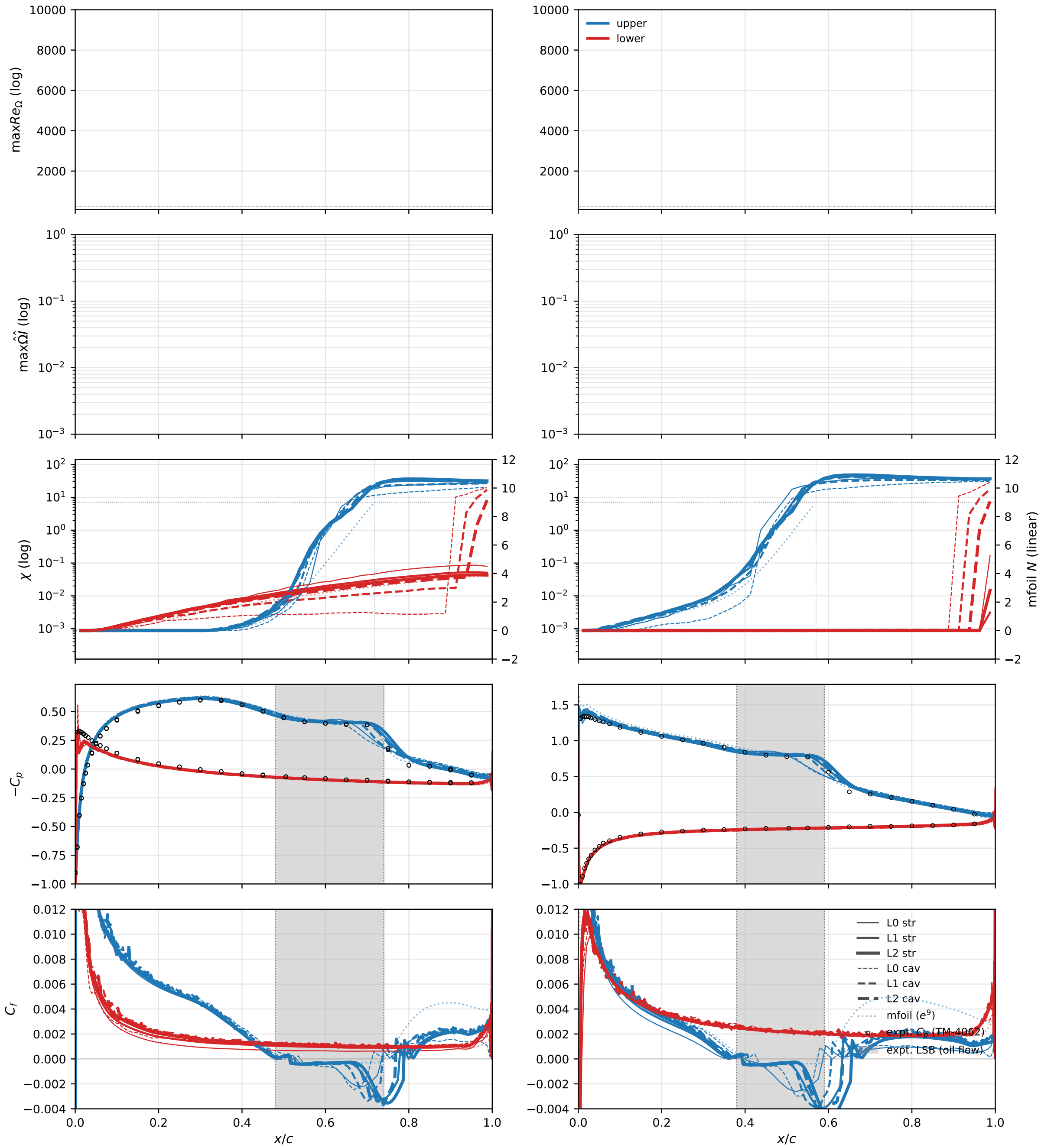


Figure 15: Eppler 387, $\alpha = 0°, 5°$, $Re = 2 \times 10^5$.

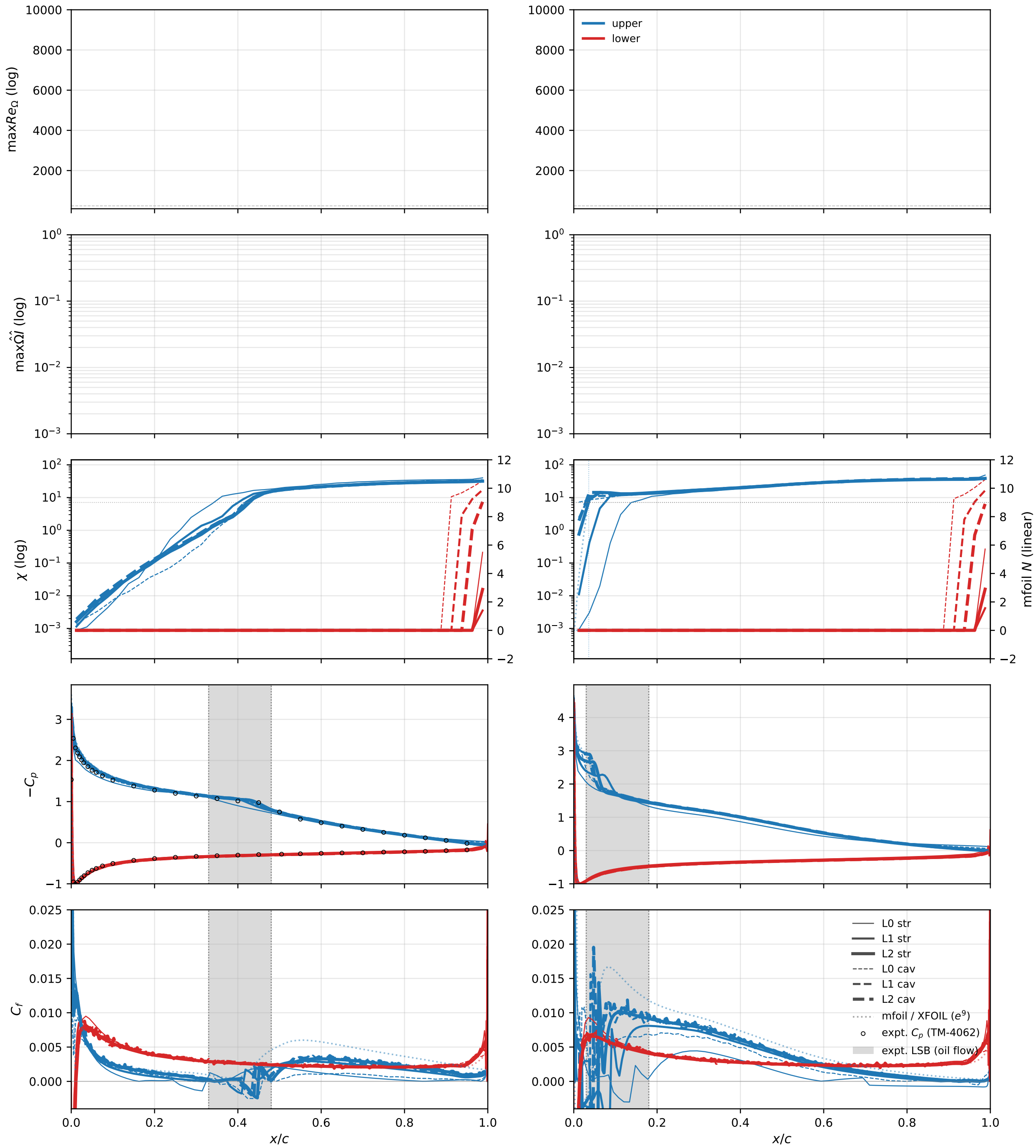


Figure 16: Eppler 387, $\alpha = 7°, 8.5°$, $Re = 2 \times 10^5$.

Table 5: Eppler 387, $Re = 2 \times 10^5$: computed forces (the $-2°$–$8.5°$ extension incidences exist on the finest pair only).

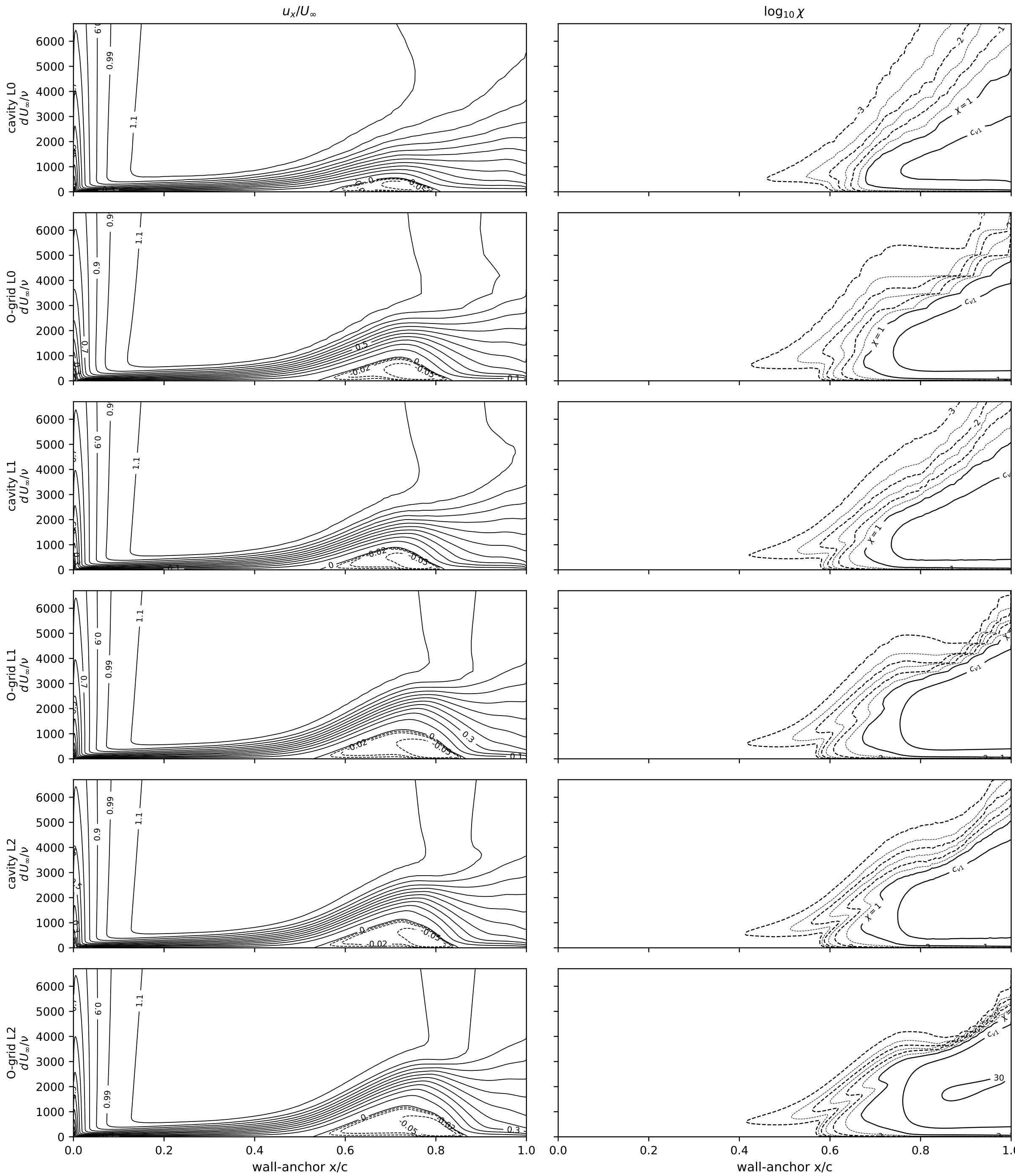
α	grid	structured O-grid C_L	structured O-grid C_D	unstructured cavity C_L	unstructured cavity C_D
−2	L2	0.1752	0.01089	0.1791	0.01080
0	L0	0.3859	0.01292	0.3847	0.01196
0	L1	0.3843	0.01080	0.3969	0.01026
0	L2	0.3844	0.01010	0.3894	0.01030
1	L2	0.4957	0.01078	0.5012	0.01092
2	L0	0.6079	0.01381	0.6061	0.01246
2	L1	0.6048	0.01196	0.6207	0.01152
2	L2	0.6059	0.01154	0.6124	0.01167
3	L2	0.7146	0.01247	0.7218	0.01245
4	L2	0.8218	0.01354	0.8304	0.01341
5	L0	0.9083	0.02007	0.9253	0.01596
5	L1	0.9218	0.01575	0.9465	0.01444
5	L2	0.9274	0.01451	0.9375	0.01428
6	L2	1.0346	0.01453	1.0456	0.01439
7	L0	1.0639	0.02728	1.1241	0.02038
7	L1	1.1262	0.01686	1.1593	0.01557
7	L2	1.1415	0.01431	1.1535	0.01419
8.5	L2	1.2190	0.02310	1.2611	0.02150

Table 6: Eppler 387, $Re = 2 \times 10^5$: computed upper-surface laminar-separation and turbulent-reattachment stations (signed-C_f zero crossings; missing entries transition ahead of separation and have no bubble). The measured oil-flow reattachments (TM-4062 Table III) at $\alpha = 0°/2°/5°$ are $0.74/0.67/0.59\,c$.

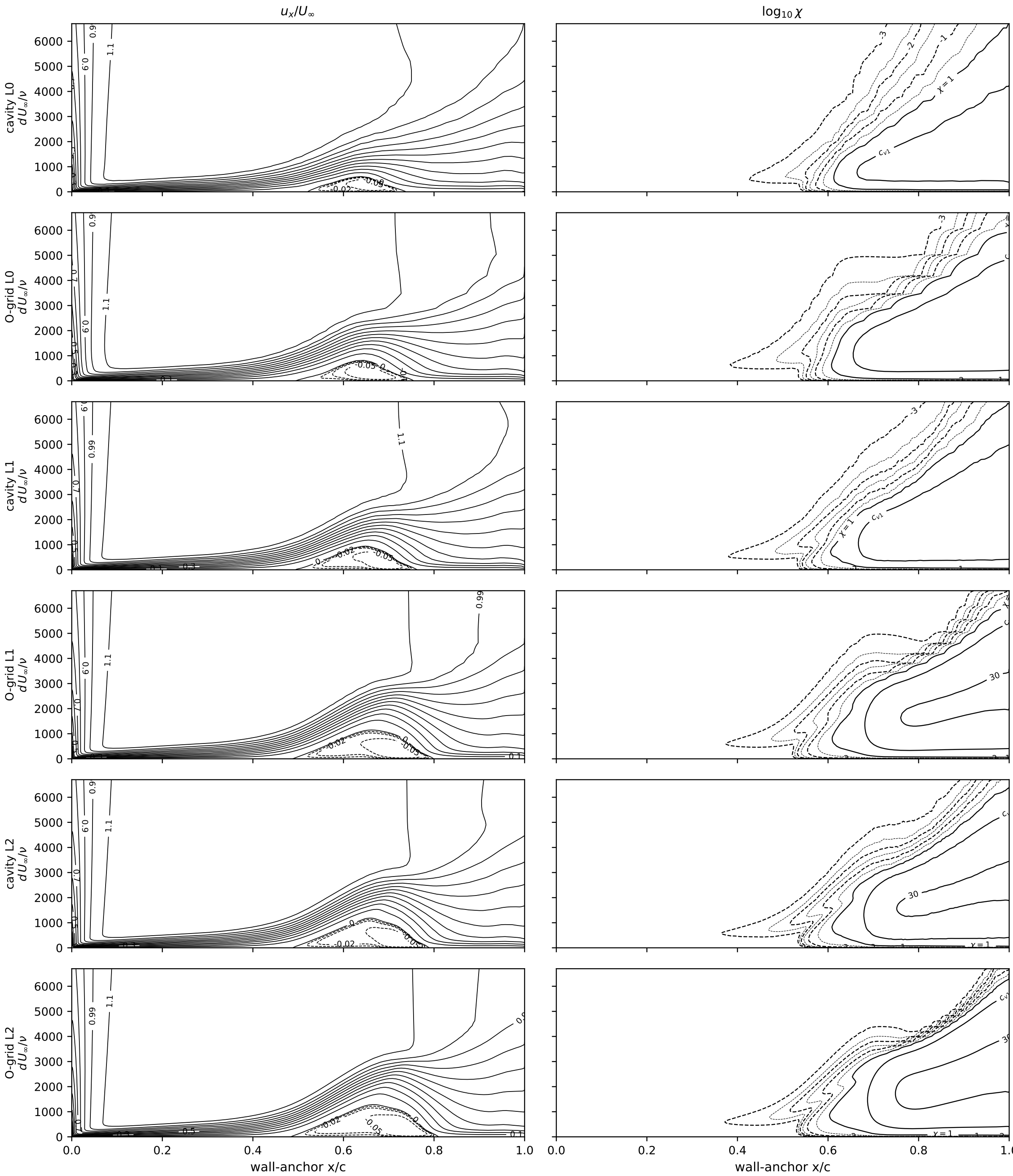
α	grid	structured O-grid x_{LS}/c	structured O-grid x_R/c	unstructured cavity x_{LS}/c	unstructured cavity x_R/c
−2	L2	0.573	0.822	0.567	0.815
0	L0	0.549	0.720	0.540	0.711
0	L1	0.526	0.771	0.526	0.737
0	L2	0.523	0.761	0.520	0.757
1	L2	0.500	0.740	0.499	0.733
2	L0	0.490	0.656	0.512	0.649
2	L1	0.479	0.713	0.483	0.682
2	L2	0.477	0.714	0.478	0.706
3	L2	0.454	0.691	0.454	0.679
4	L2	0.426	0.667	0.430	0.657
5	L0	0.393	0.568	0.429	0.558
5	L1	0.394	0.624	0.415	0.600
5	L2	0.402	0.634	0.404	0.623
6	L2	0.388	0.565	0.391	0.560
7	L0	0.250	0.343	0.351	0.458
7	L1	–	–	–	–
7	L2	0.388	0.407	–	–
8.5	L2	0.031	0.039	0.019	0.049

Table 7: Eppler 387: the independent OpenFOAM implementation on the structured L0, L1 and L2 grids (fronts at the near-wall $\chi = 1$ convention, stations at the signed-C_f zero crossings; the port omits the f_{v1} bypass).

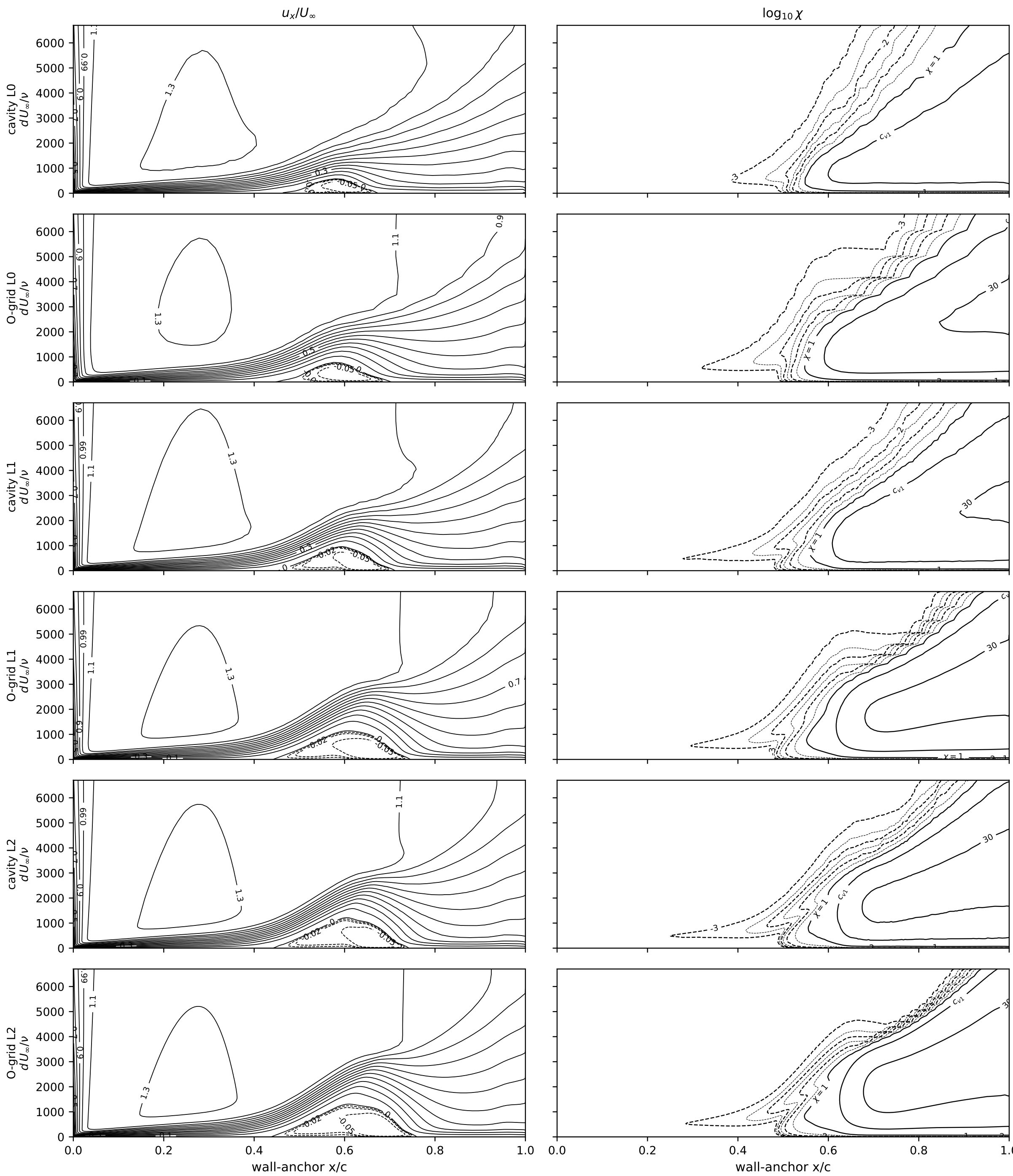
α	grid	C_L	C_D	$x_{\mathrm{tr}}^{\mathrm{up}}$	x_{LS}	x_R
0	L0	0.3887	0.01167	0.638	0.493	0.784
0	L1	0.3826	0.00978	0.605	0.495	0.775
0	L2	0.3857	0.01058	0.621	0.486	0.806
2	L0	0.6133	0.01406	0.568	0.447	0.739
2	L1	0.6037	0.01174	0.543	0.448	0.727
2	L2	0.6039	0.01229	0.564	0.443	0.752
5	L0	0.9292	0.01665	0.457	0.344	0.654
5	L1	0.9256	0.01394	0.468	0.371	0.651
5	L2	0.9234	0.01463	0.487	0.372	0.667
7	L0	1.1173	0.01817	0.139	–	0.201
7	L1	1.1391	0.01323	0.329	0.353	0.453
7	L2	1.1388	0.01363	0.379	0.351	0.493



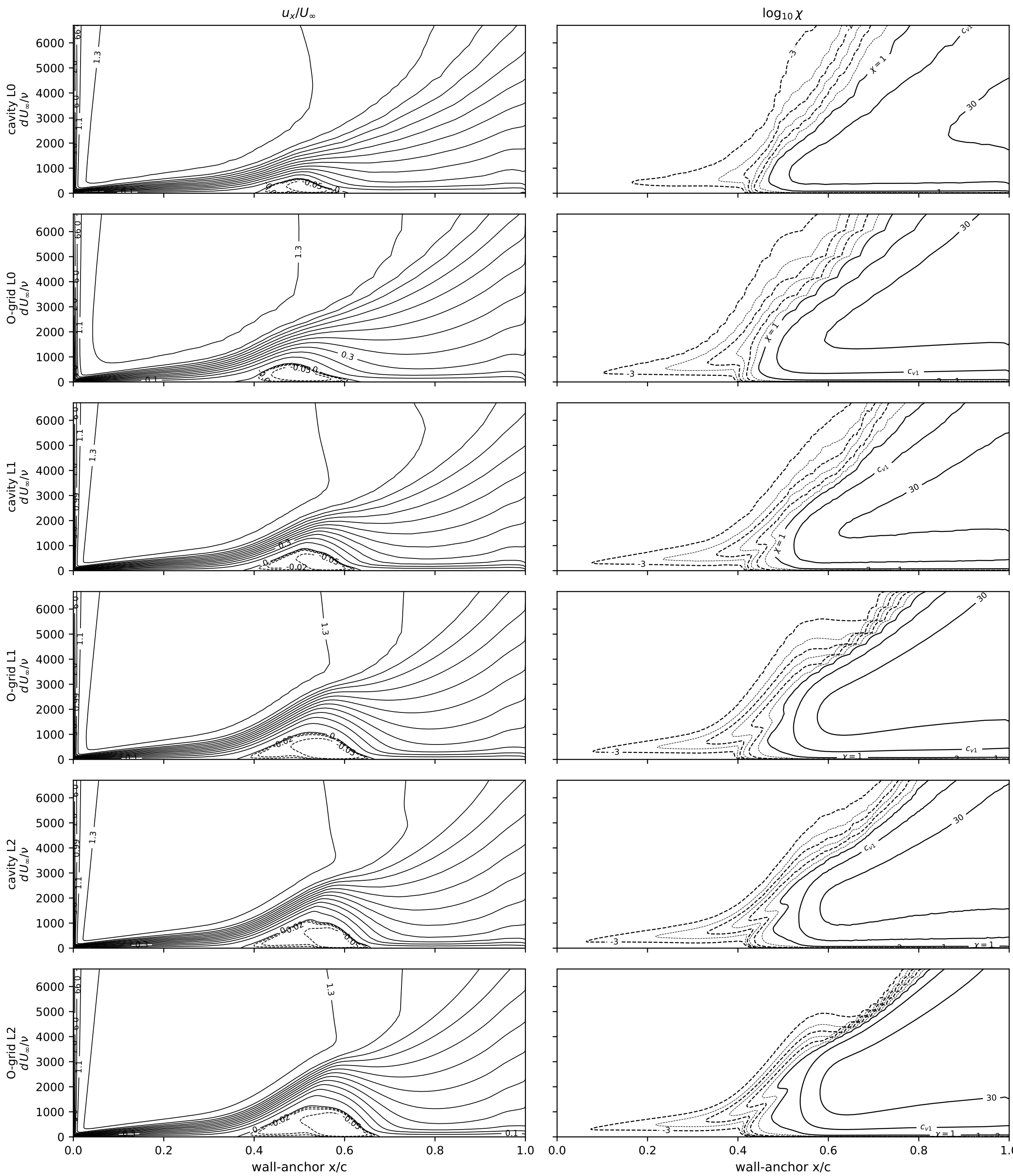
u_x/U_∞
$\log_{10}\chi$
cavity L0
O-grid L0
cavity L1
O-grid L1
cavity L2
O-grid L2
$d\,U_\infty/\nu$
wall-anchor x/c
wall-anchor x/c



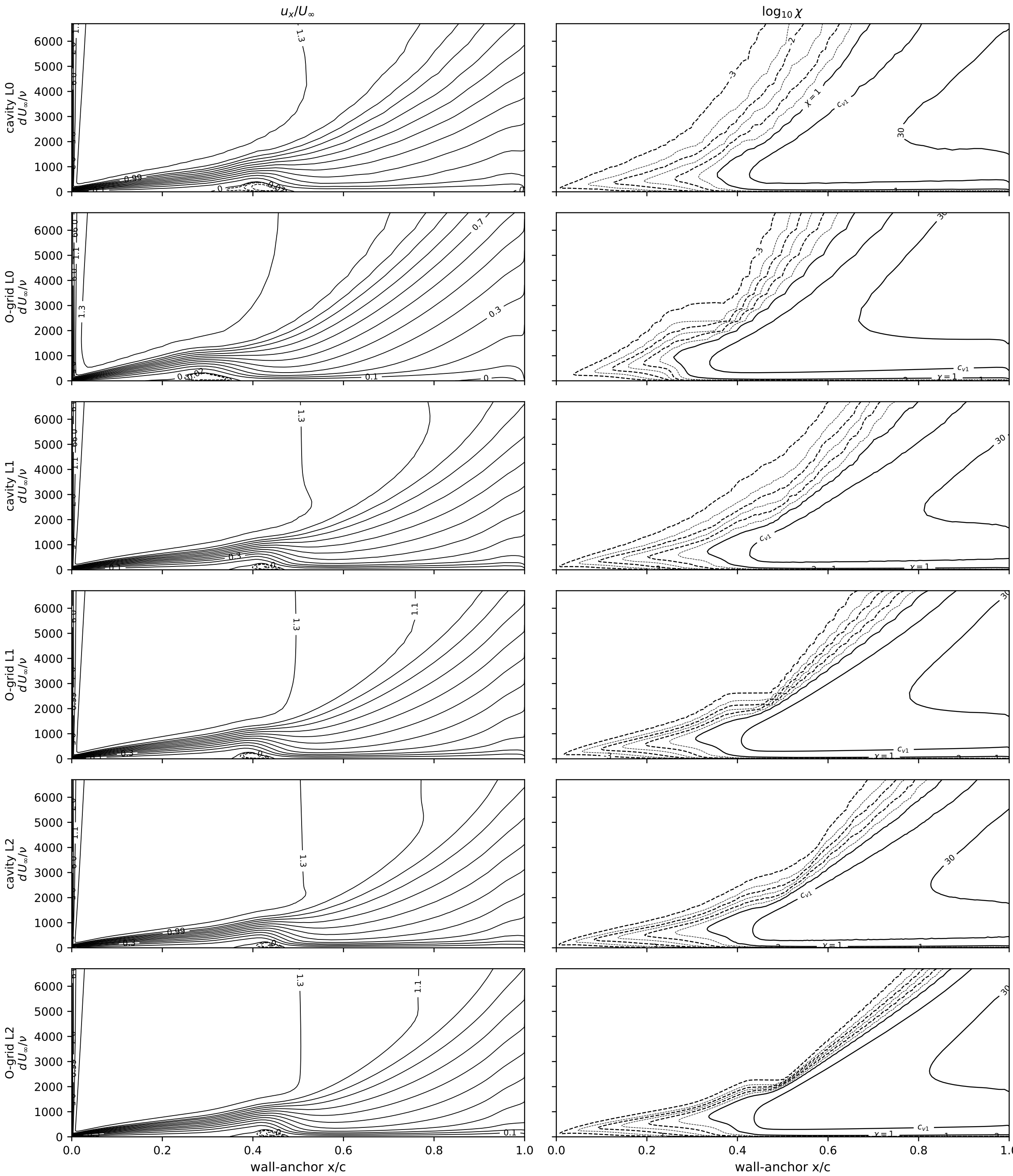
u_x/U_∞
$\log_{10}\chi$
cavity L0
O-grid L0
cavity L1
O-grid L1
cavity L2
O-grid L2
$d\,U_\infty/\nu$
wall-anchor x/c

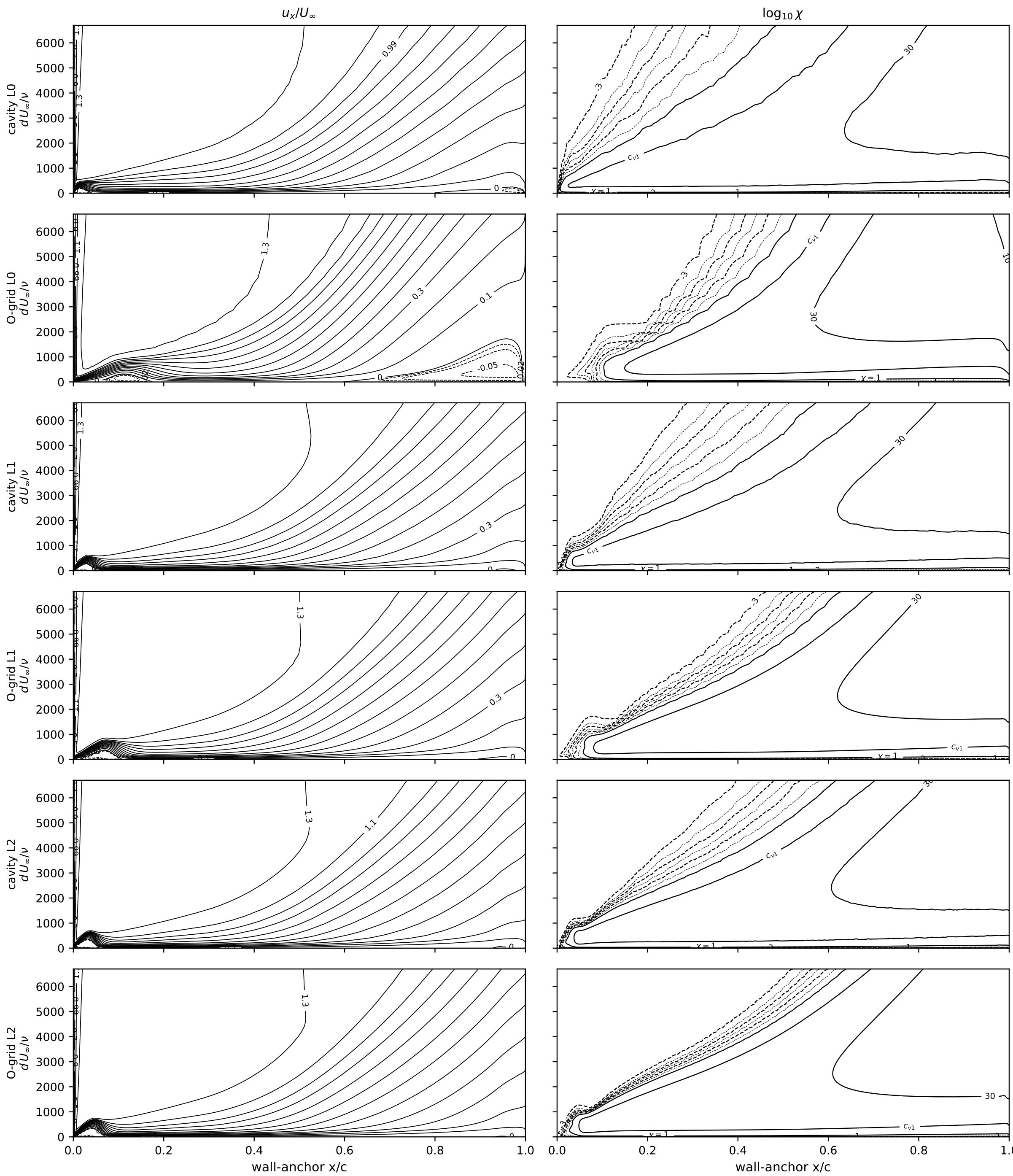


u_x/U_∞
$\log_{10}\chi$
cavity L0 dU_∞/ν
O-grid L0 dU_∞/ν
cavity L1 dU_∞/ν
O-grid L1 dU_∞/ν
cavity L2 dU_∞/ν
O-grid L2 dU_∞/ν
wall-anchor x/c
wall-anchor x/c



u_x/U_∞
$\log_{10}\chi$
cavity L0
O-grid L0
cavity L1
O-grid L1
cavity L2
O-grid L2
$d\,U_\infty/\nu$
wall-anchor x/c


u_x/U_∞
$\log_{10}\chi$
cavity L0
O-grid L0
cavity L1
O-grid L1
cavity L2
O-grid L2
$d\,U_\infty/\nu$
wall-anchor x/c
wall-anchor x/c



2.4 Eppler 387 Reynolds sweep at $\alpha = 5^\circ$

$Re \in \{6\times10^4,\ 10^5,\ 3\times10^5,\ 4.6\times10^5\}$ on the same grids (24 further solutions). Figure 17: the refinement fan closes onto the measurement at 2–4.6×10^5 and opens at 10^5, the model's bursting boundary, one Reynolds step above the measurement's (6×10^4). Figures 18–19: surface suites. Table 8: L2 coefficients.

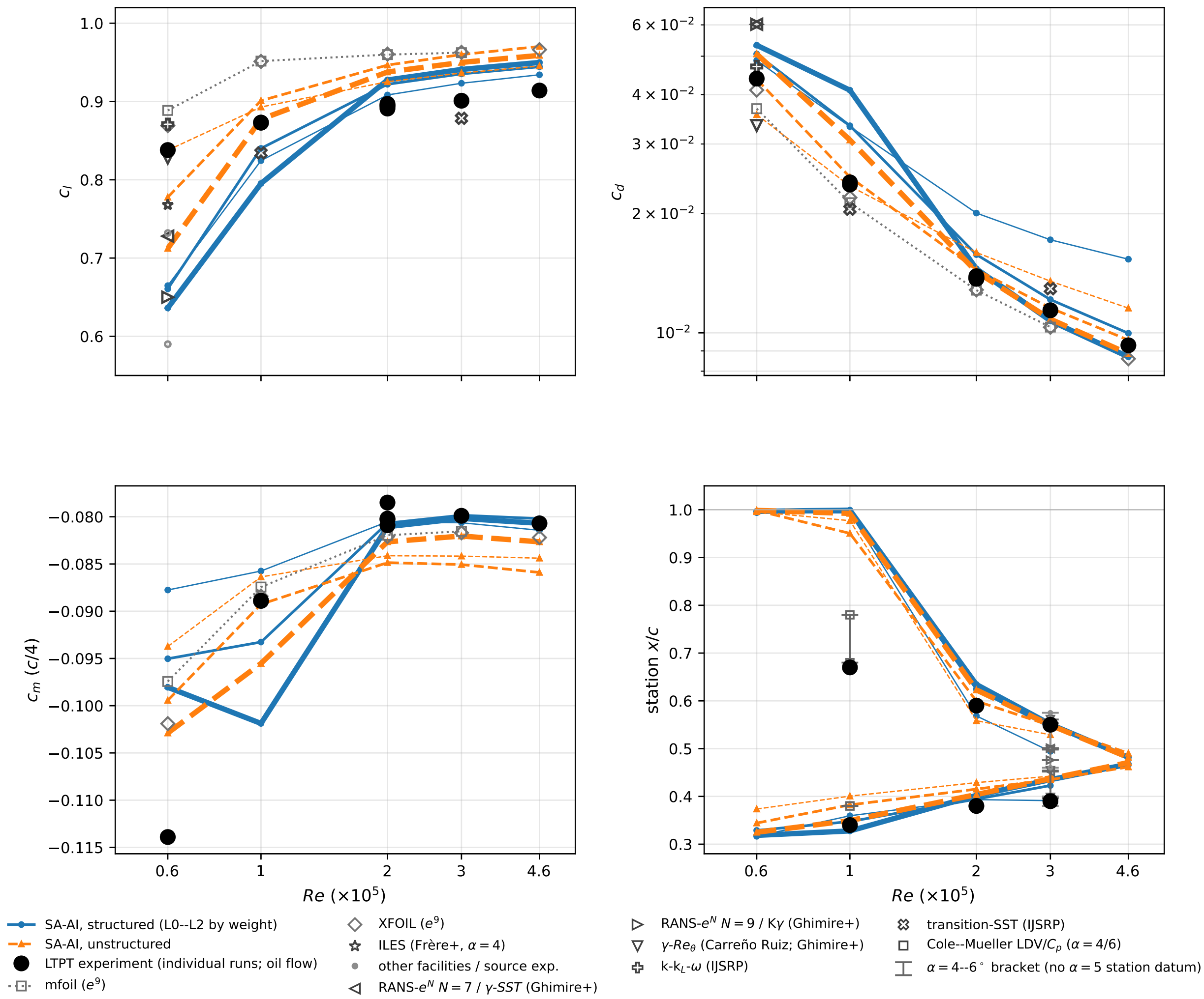


Figure 17: Reynolds-sweep forces and stations, $\alpha = 5°$, $Re = 6 \times 10^4$–4.6×10^5. LTPT runs 1/2/4/1/1 at $0.6/1/2/3/4.6 \times 10^5$.

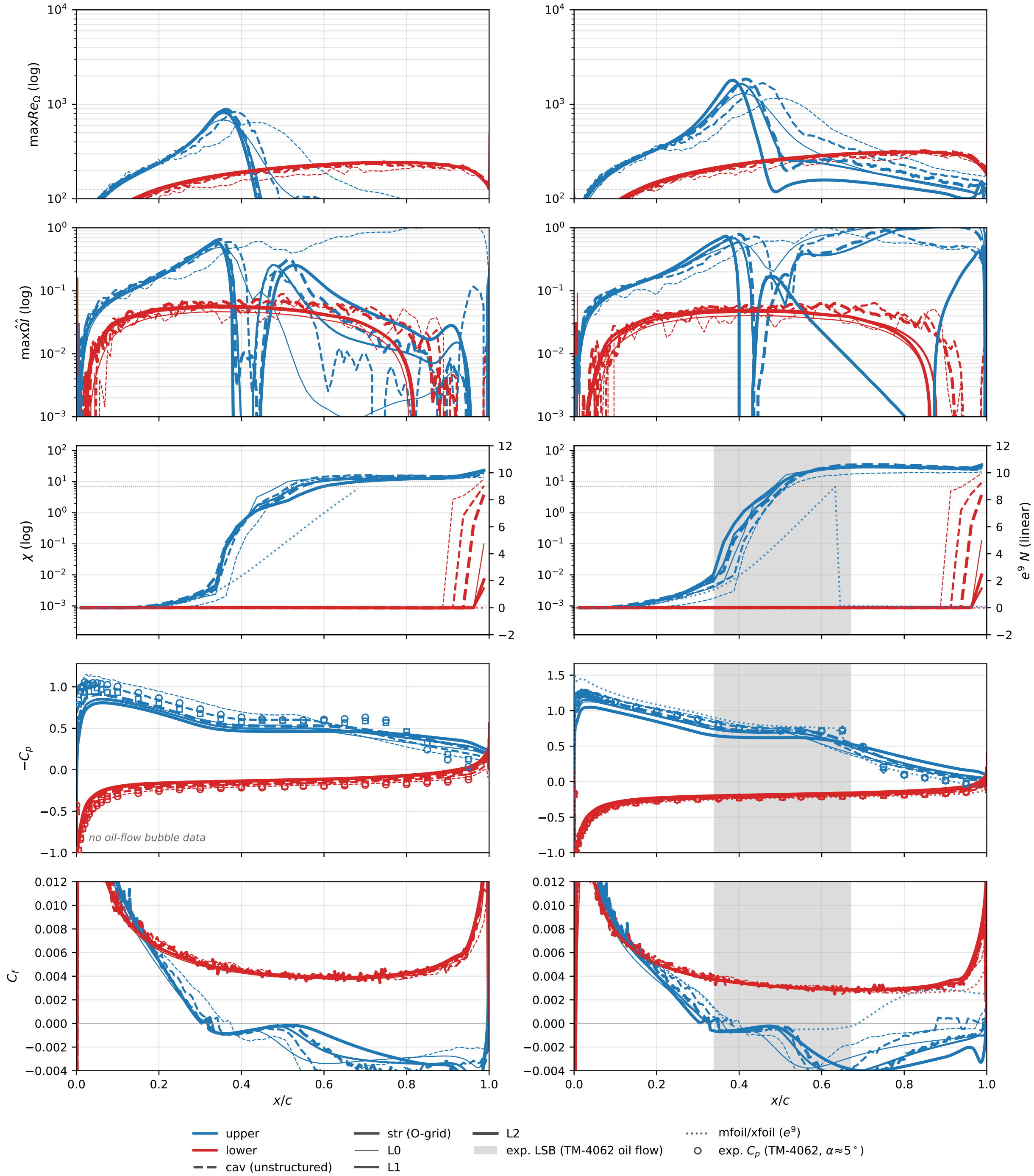


Figure 18: Reynolds sweep, $Re = 6 \times 10^4$, 10^5.

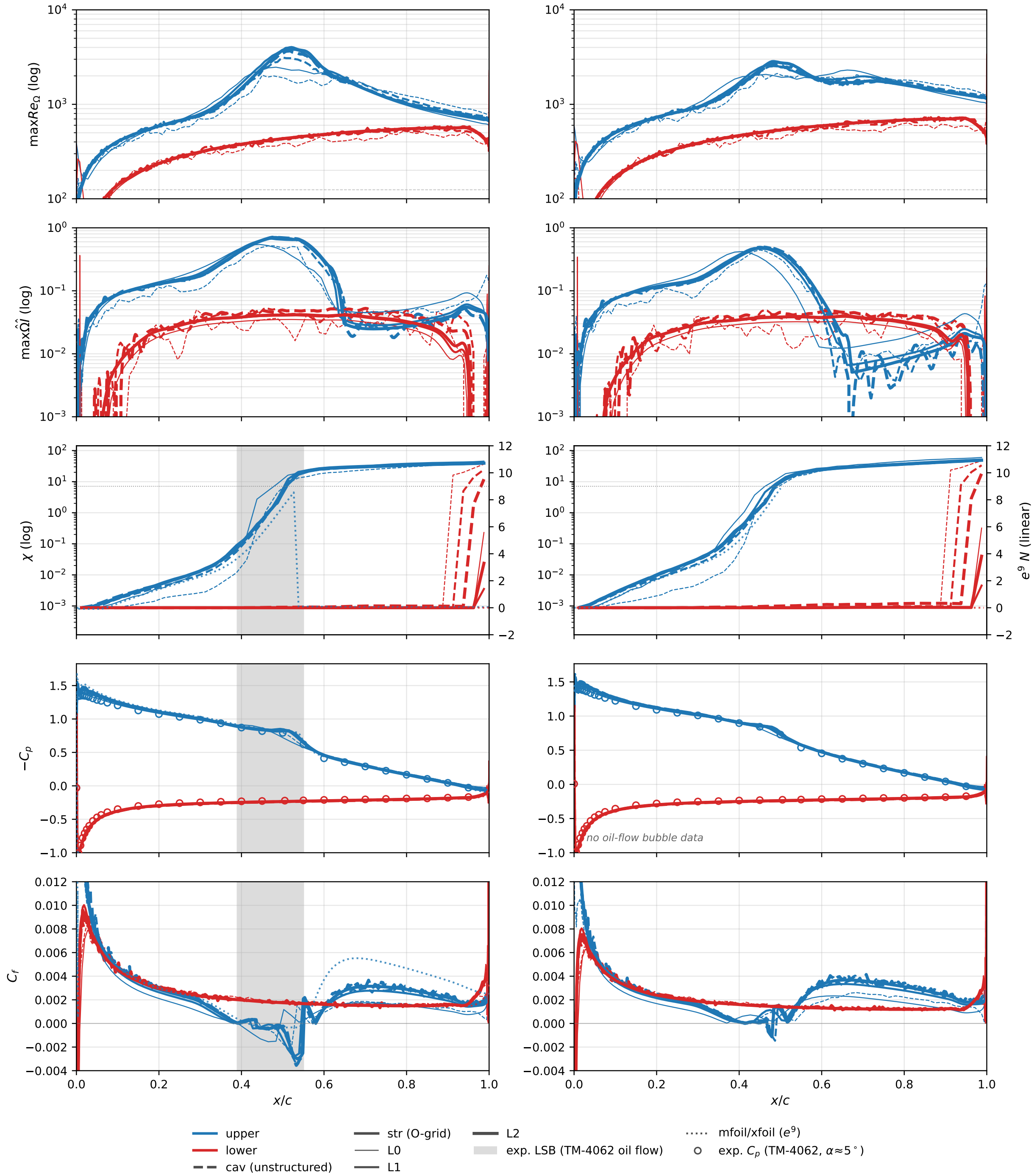
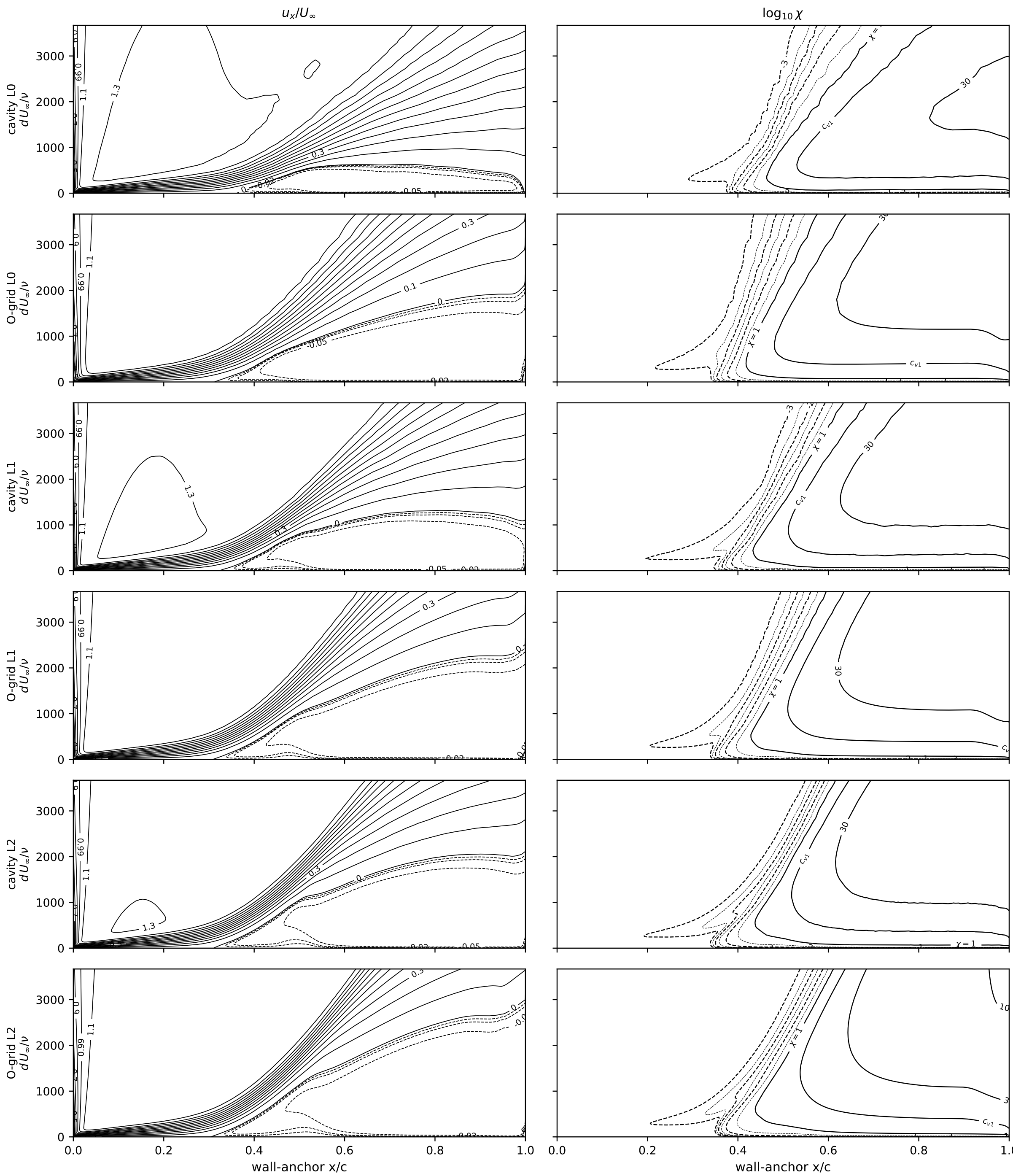


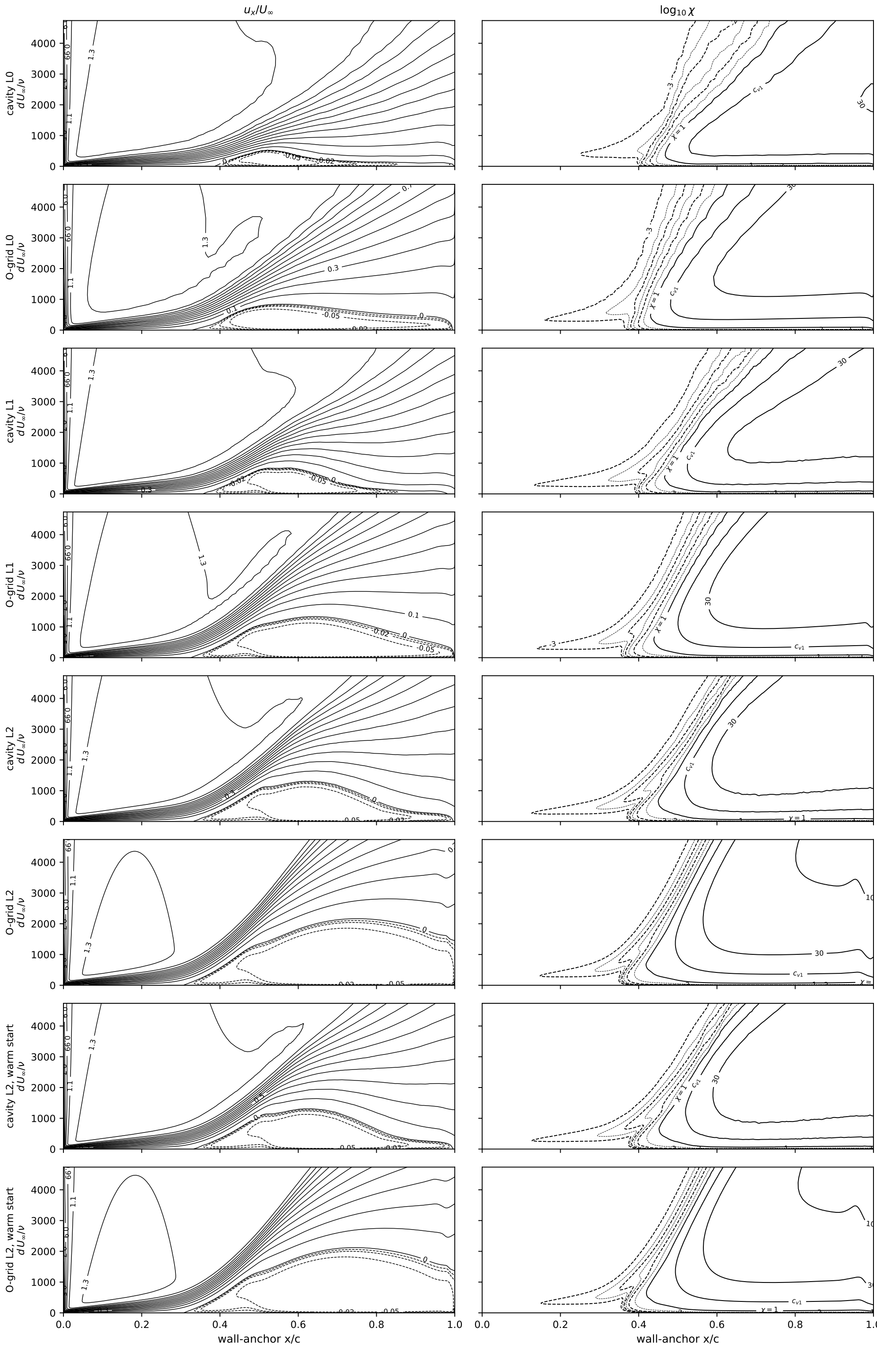
Figure 19: Reynolds sweep, $Re = 3,\ 4.6 \times 10^5$.

Table 8: Eppler 387 at $\alpha = 5°$ across the Reynolds sweep: L2 section coefficients against the e^9 panel reference and the LTPT measurement (read at the tabulated angle nearest 5°). Computed moments transferred to the quarter chord.

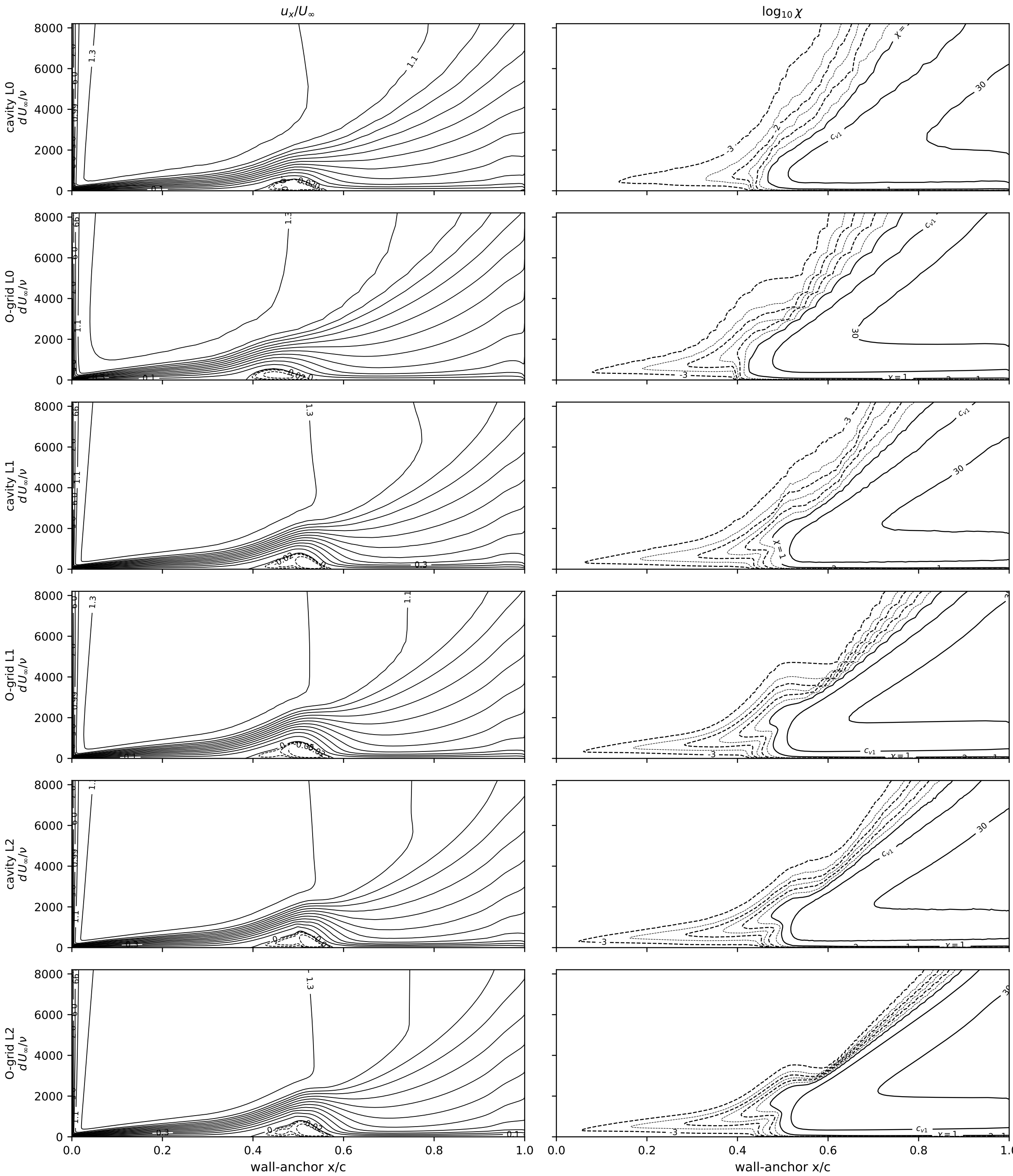
	SA-AI, structured L2			SA-AI, unstructured L2			e^9 panel			Exp (LTPT)		
Re	c_l	c_d	c_m	c_l	c_d	c_m	c_l	c_d	c_m	c_l	c_d	c_m
6×10^4	0.636	0.0533	−0.098	0.712	0.0506	−0.103	0.869‡	0.0411‡	−0.102‡	0.838	0.0439	−0.114
1×10^5	0.795	0.0410	−0.102	0.877	0.0307	−0.096	0.952	0.0213	−0.087	0.873	0.0237	−0.089
2×10^5	0.927	0.0145	−0.081	0.938	0.0143	−0.083	0.960	0.0128	−0.082	0.891	0.0138	−0.081
3×10^5	0.941	0.0107	−0.080	0.950	0.0108	−0.082	0.962	0.0103	−0.082	0.901	0.0114	−0.080
4.6×10^5	0.950	0.0087	−0.081	0.959	0.0089	−0.083	0.966	0.0086	−0.082	0.914	0.0093	−0.081

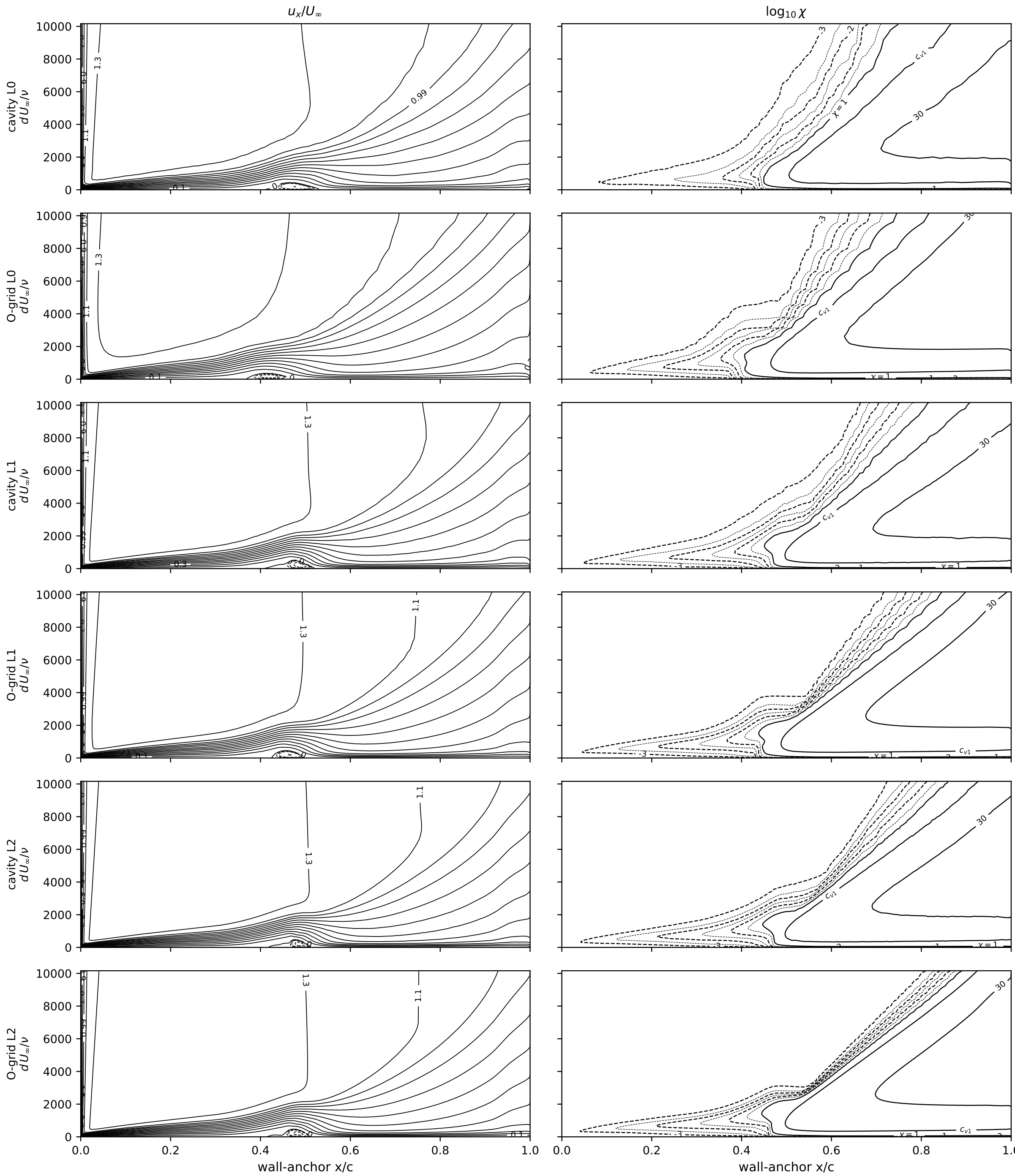
‡XFOIL, whose forces sit closest to the measurement, is the reference at this bistable condition; mfoil gives $c_l = 0.889$, $c_d = 0.0368$, $c_m = -0.097$, and FlexFoil sides with XFOIL ($c_l = 0.868$, $c_d = 0.0406$, $c_m = -0.102$).


u_x/U_∞
$\log_{10}\chi$
cavity L0
O-grid L0
cavity L1
O-grid L1
cavity L2
O-grid L2
$d\,U_\infty/\nu$
wall-anchor x/c
0.0
0.2
0.4
0.6
0.8
1.0
0
1000
2000
3000
1.3
1.1
0.99
0.3
0.1
0
-0.02
-0.05
$\chi = 1$
c_{v1}
30
-3
10



u_x/U_∞
$\log_{10}\chi$
cavity L0 dU_∞/ν
O-grid L0 dU_∞/ν
cavity L1 dU_∞/ν
O-grid L1 dU_∞/ν
cavity L2 dU_∞/ν
O-grid L2 dU_∞/ν
cavity L2, warm start dU_∞/ν
O-grid L2, warm start dU_∞/ν
wall-anchor x/c
wall-anchor x/c


u_x/U_∞
$\log_{10}\chi$
cavity L0
O-grid L0
cavity L1
O-grid L1
cavity L2
O-grid L2
$d\,U_\infty/\nu$
wall-anchor x/c
wall-anchor x/c



2.5 Cylinder drag crisis, $Re_D = 1\text{–}10^{10}$

A steady-branch traverse over ten decades at a single seed ($\chi_\infty = 1.1 \times 10^{-2}$; Mack's map at $Tu = 0.2\%$), swept as one upward continuation ladder on matched quasi-two-dimensional O-grids ($y^+ \lesssim 1$; a large-domain laminar grid for $Re_D \leq 10^3$, a 1200×148 working grid, and 1600×170 fine grids for the supercritical and ultra range), twenty-five points $Re_D = 1\text{–}10^{10}$. Every case lands on the symmetric branch, $|C_L| < 2 \times 10^{-6}$. Figure 20: the crisis falls between $Re_D = 3 \times 10^5$ ($C_d = 0.773$) and 1.78×10^6 (0.228); the supercritical floor settles near $C_d = 0.21$–0.23 (1.78×10^6–10^7) and stays flat to 10^{10}. Figure 21: transition and separation angles. Figure 22: fields across the traverse. Table 9: the up-ladder C_d. Below shedding onset ($Re_D \approx 46$) the steady solution is physical: $C_d = 10.67$ at $Re_D = 1$, 2.797 at 10 (−1.7% vs Dennis & Chang).

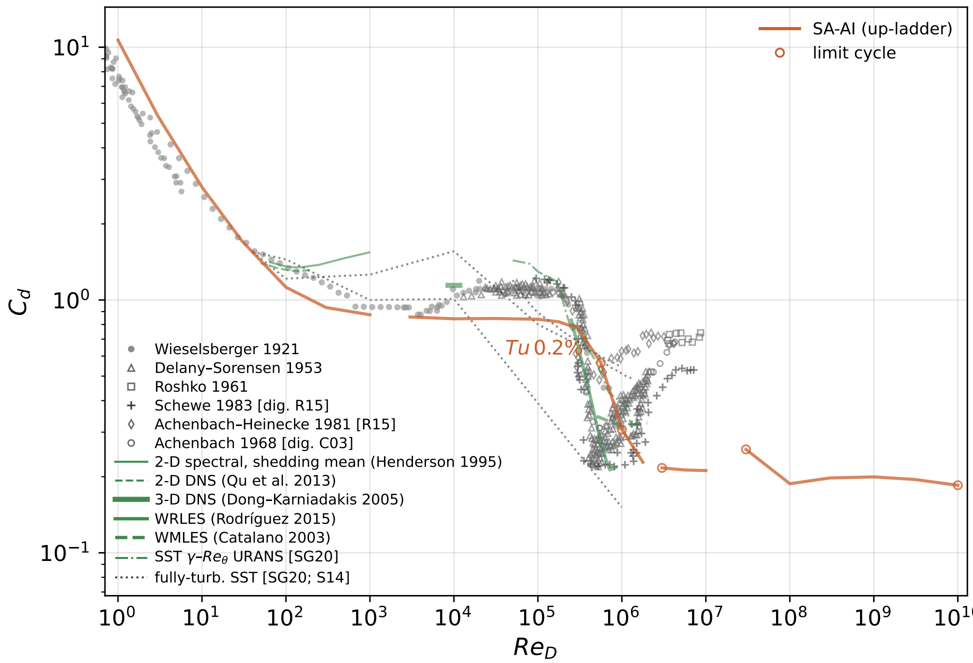


Figure 20: Cylinder drag crisis, $Re_D = 1$–10^{10}; single seed $Tu = 0.2\%$, up-ladder (clean systematic sweep, matched Re_D grid). Symbols experiments, lines computations; open rings mark the cases the steady monitor did not accept.

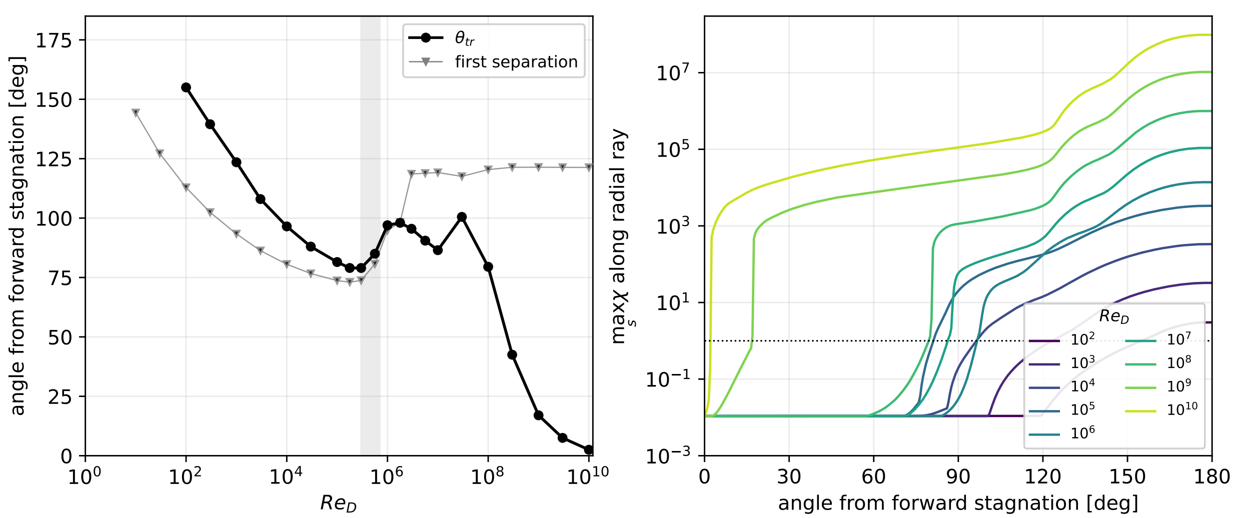


Figure 21: $Tu = 0.2\%$, $Re_D = 1$–10^{10}, up-ladder. (a) transition angle θ_{tr} ($\chi = 1$) and first wall separation vs Re_D (below the crisis θ_{tr} is the wake $\chi = 1$ front); the front marches from $\sim 100°$ to $\sim 2°$ by 10^{10}. (b) $\max_s \chi(\theta)$ per decade 10^2–10^{10}.

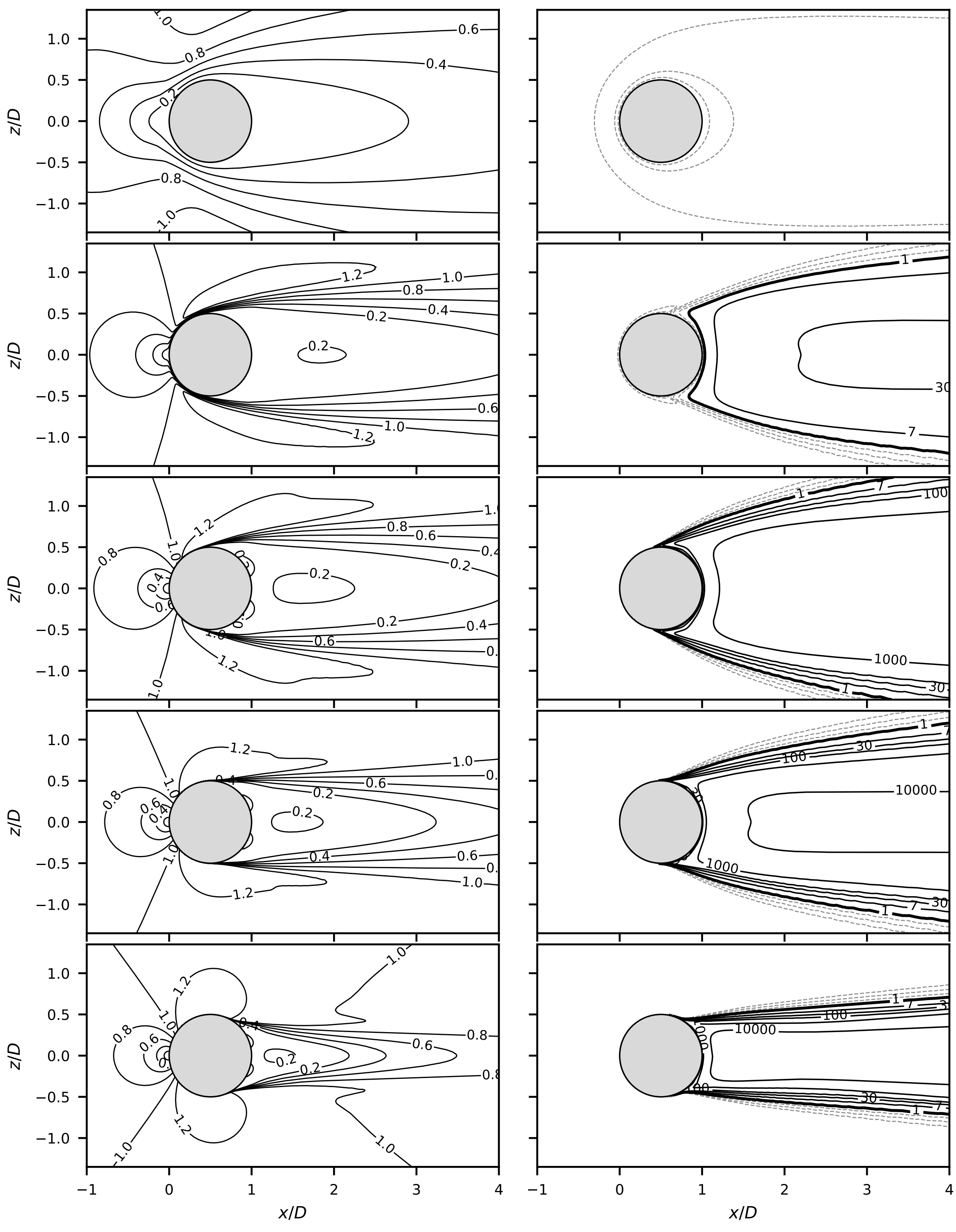


Figure 22: Drag-crisis fields, $Re_D = 10$, 10^3, 10^5, 5×10^5, 2×10^6 (rows, up-ladder $Tu = 0.2\%$); columns $|\mathbf{u}|/U_\infty$ and χ (dashed $\chi < 1$, solid $\chi = 1$, c_{v1}, 30, 10^2–10^4). $C_d = 2.80$, 0.87, 0.84, 0.62, 0.22.

Table 9: Cylinder steady drag-crisis traverse, systematic $Tu = 0.2\%$ up-ladder (C_d median over the tail window, [peak-to-peak] bracketed where the steady monitor did not accept; θ_{tr} the $\chi = 1$ transition angle, θ_{sep} the first separation; $|C_L| < 2 \times 10^{-6}$ throughout).

Re_D	grid	C_d	θ_{tr} (°)	θ_{sep} (°)
1	lowre	10.665	–	–
3	lowre	5.330	–	–
10	lowre	2.797	–	144.3
30	lowre	1.710	–	127.2
100	lowre	1.123	155.0	112.9
300	lowre	0.934	139.5	102.4
1000	lowre	0.874	123.5	93.4
3000	pilot	0.857	108.0	86.3
10^4	pilot	0.843	96.5	80.6
3×10^4	pilot	0.845	88.0	76.7
10^5	pilot	0.839	81.5	73.7
1.78×10^5	pilot	0.820	79.0	73.0
3×10^5	pilot	0.773	79.0	73.8
5.62×10^5	pilot	0.565 [0.075]	85.0	80.7
10^6	pilot	0.307 [0.044]	97.0	94.8
1.78×10^6	pilot	0.228	98.0	98.3
3×10^6	highre	0.217 [0.010]	95.5	118.6
5.62×10^6	highre	0.213	90.5	118.9
10^7	highre	0.212	86.5	119.2
3×10^7	ultra	0.256 [0.171]	100.5	117.6
10^8	ultra	0.188	79.5	120.4
3×10^8	ultra	0.198	42.5	121.4
10^9	ultra	0.200	17.0	121.4
3×10^9	ultra	0.195	7.5	121.3
10^{10}	ultra	0.185 [0.009]	2.5	121.3

2.6 Daedalus wing (three-dimensional deployment)

The Daedalus human-powered-aircraft wing: half-span 17.07 m, root chord 0.917 m, aspect ratio 37.8, DAE-11/21/31 sections; $Re_{\text{root}} = 5 \times 10^5$, $\alpha \in \{4°, 5°, 6°\}$ ($C_L \approx 1.02$–1.23), flight-quiet seed $\chi_\infty = 8.76 \times 10^{-6}$ ($N_{\text{crit}} \approx 13.6$); two mesh families (spanwise-stacked structured O-grid, 0.72–36.7M nodes; unstructured prism/tet/hex, 1.70–54.9M nodes) at three levels, eighteen solutions; the reference is AVL with strip-wise XFOIL section polars at matched local c_l and N_{crit}, trimmed to the RANS lift. Figure 23: the families agree at L2 to 0.35–0.63% in C_L and 0.7–1.7 counts in C_D, and the finest-grid drag runs 15.6/10.3/5.6 counts (7.3/4.4/2.2%) below the matched-lift reference at $\alpha = 4°/5°/6°$. Figure 24: surface maps at $\alpha = 4°$ against the e^N strips—at $\eta = 0.31$, separation within $0.017\,c$ and reattachment within $0.009\,c$ at every incidence, the $\chi = 1$ front sits $\sim 0.09\,c$ ahead of the strips' N_{crit} contour, and the tip vortex trips transition to the leading edge for $\eta \gtrsim 0.992$. Table 10: totals by level ($\alpha = 5°$, $6°$ surface maps and section sheets below).

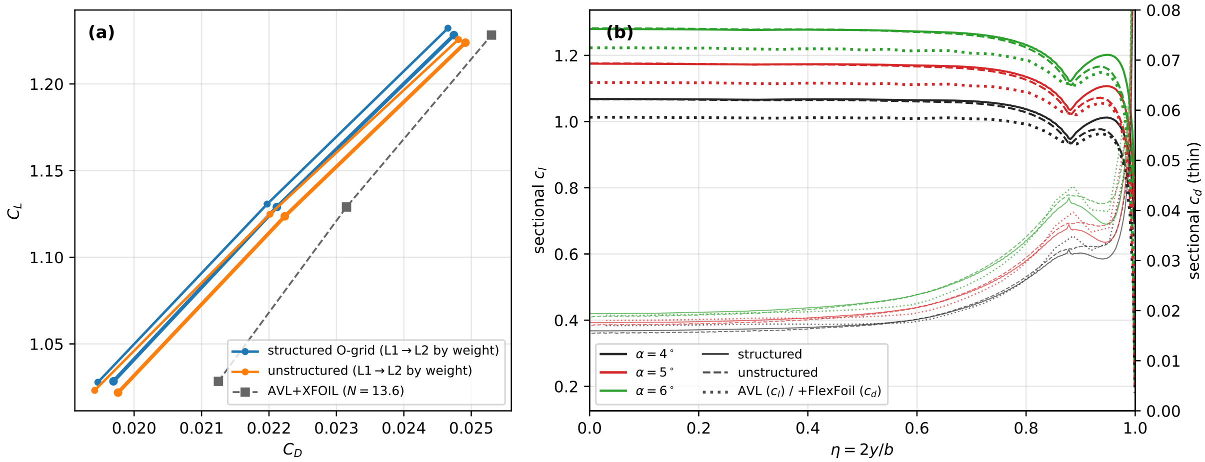


Figure 23: Daedalus polar and sections. Families agree at L1 to 0.46–0.53% in C_L and 0.4–1.6 counts in C_D, at L0 to 0.74–0.97% and 18–21 counts; taper break at $\eta = 0.88$.

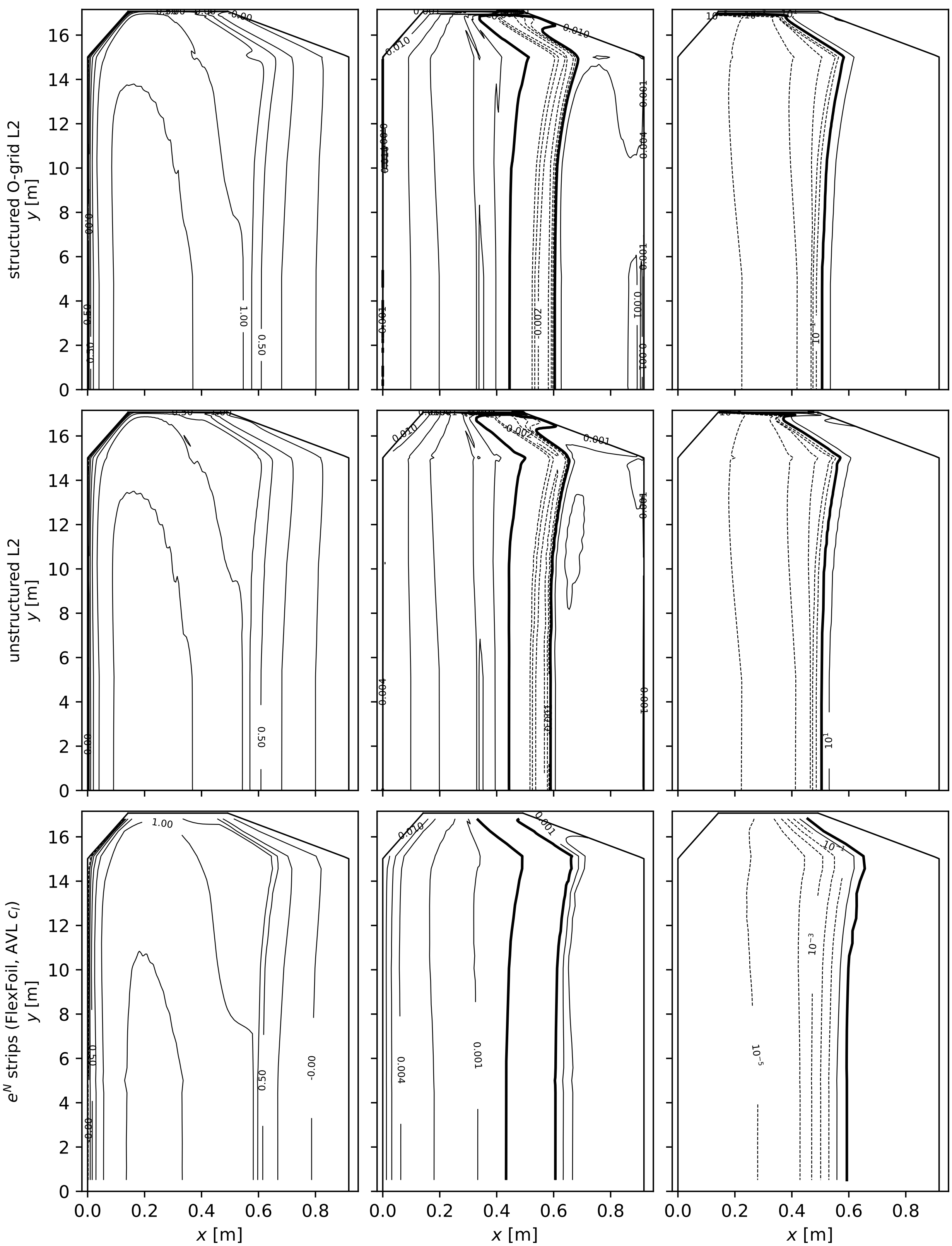
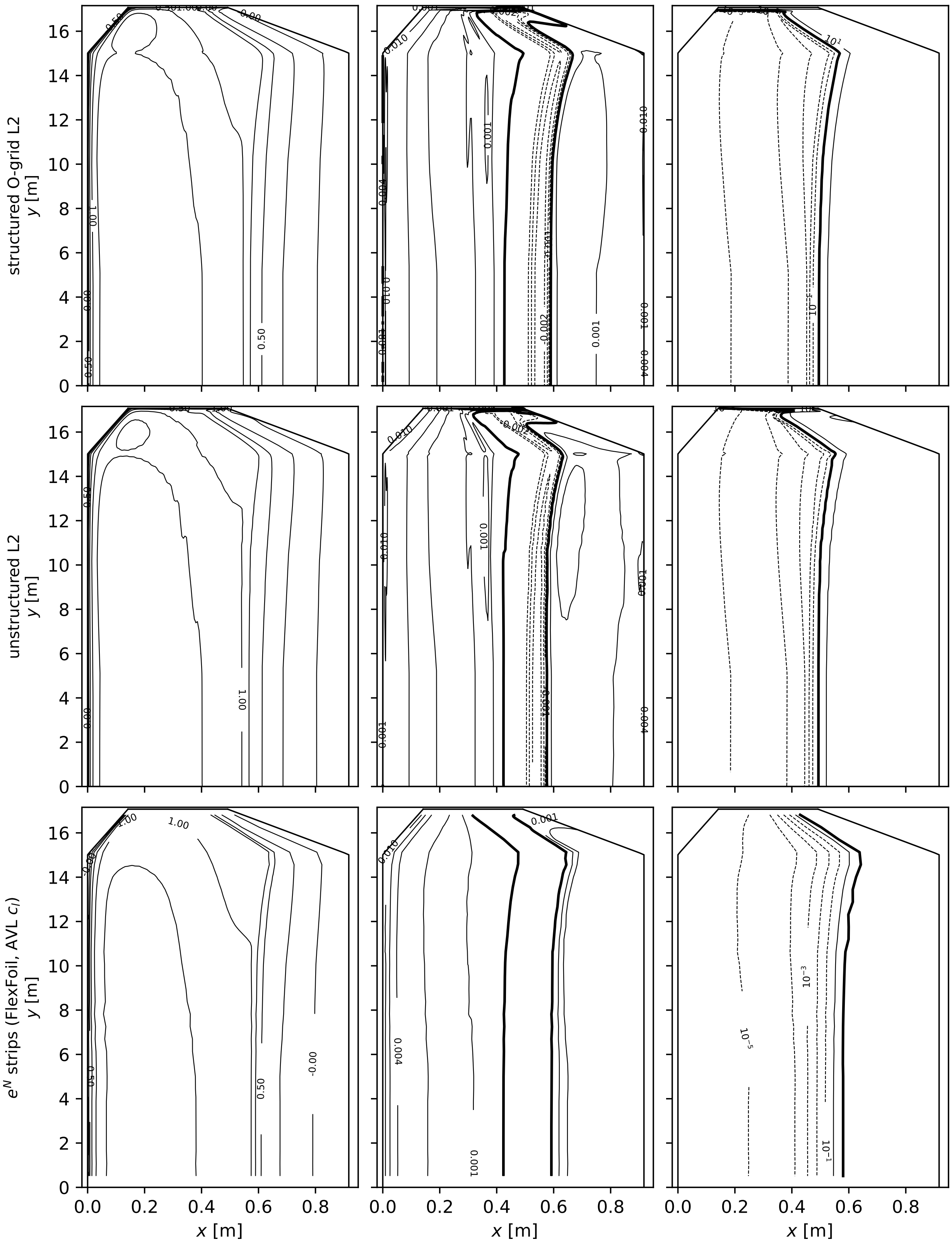


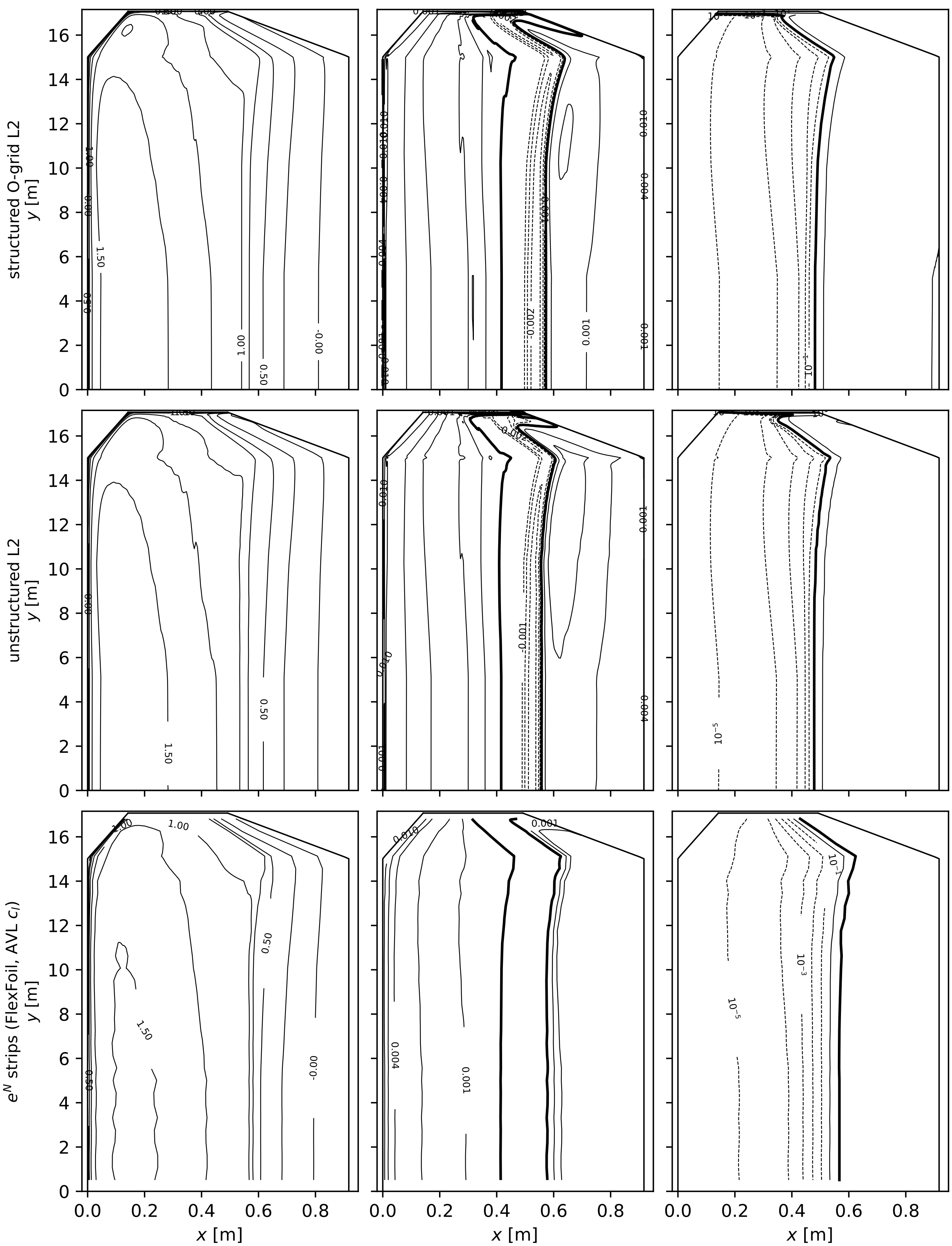
Figure 24: Daedalus surface maps, $\alpha = 4^{\circ}$, $Re_{\mathrm{root}} = 5 \times 10^5$.

Table 10: Daedalus wing totals by grid level: SA-AI on both mesh families against the AVL+XFOIL reference trimmed to the RANS lift at each incidence.

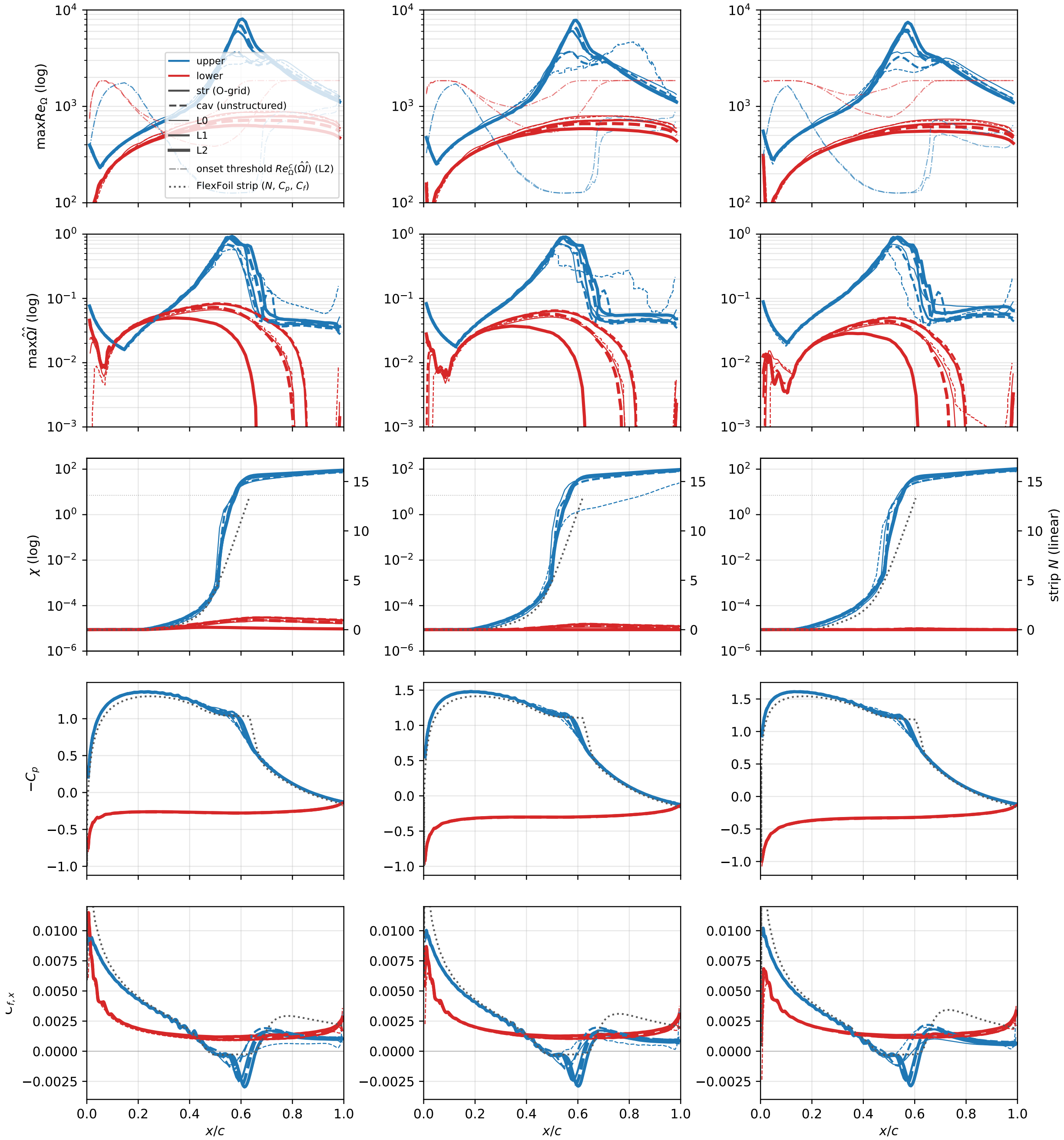
		structured O-grid		unstructured	
α	level	C_L	C_D	C_L	C_D
4°	L0	1.0244	0.01958	1.0156	0.02134
	L1	1.0279	0.01946	1.0232	0.01941
	L2	1.0284	0.01969	1.0219	0.01976
	AVL+XFOIL	1.0284	0.02125		
5°	L0	1.1254	0.02219	1.1145	0.02427
	L1	1.1307	0.02197	1.1248	0.02201
	L2	1.1290	0.02212	1.1236	0.02223
	AVL+XFOIL	1.1290	0.02315		
6°	L0	1.2262	0.02505	1.2171	0.02713
	L1	1.2321	0.02465	1.2256	0.02481
	L2	1.2282	0.02474	1.2238	0.02491
	AVL+XFOIL	1.2282	0.02530		



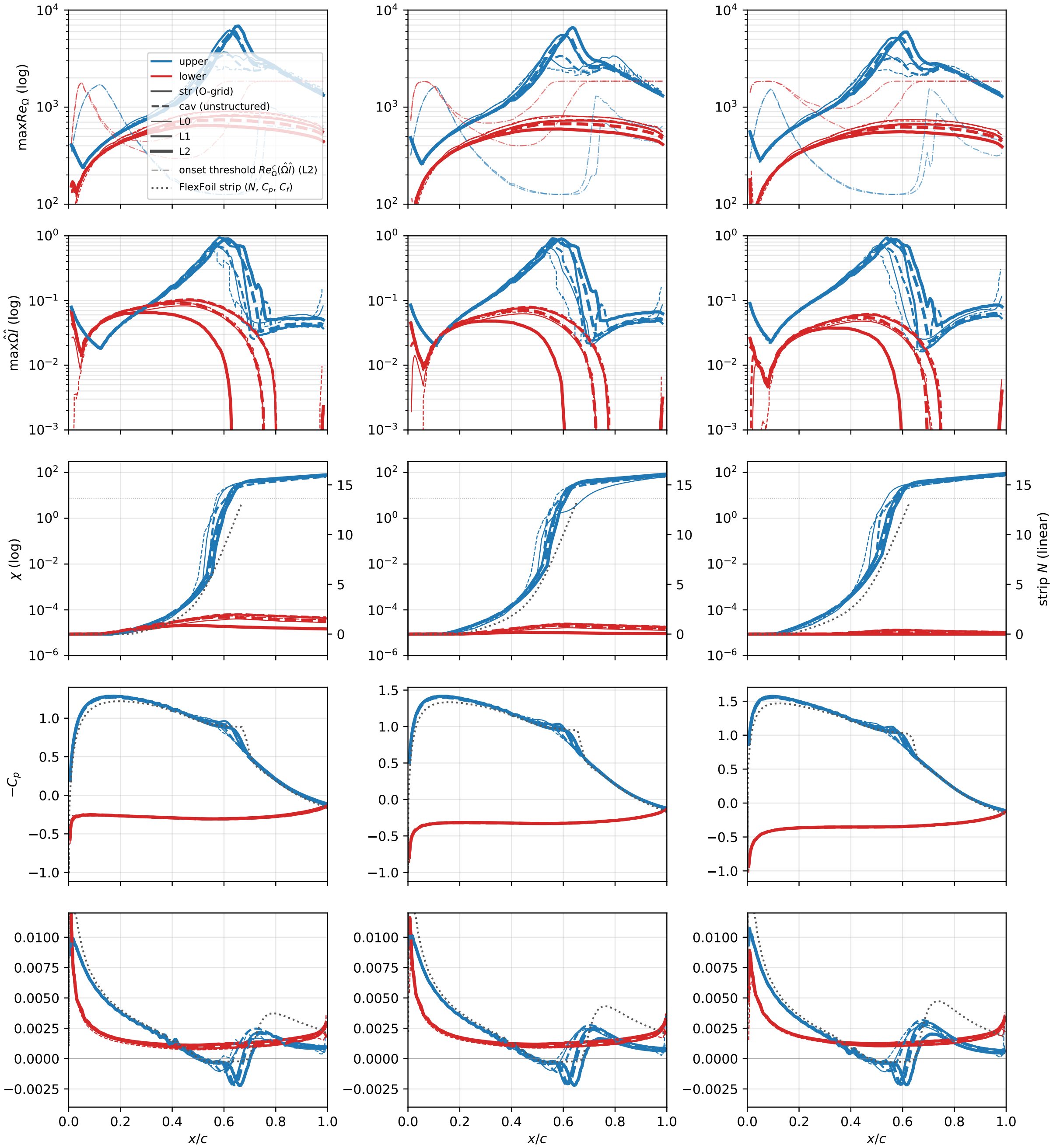
structured O-grid L2
y [m]
unstructured L2
y [m]
e^N strips (FlexFoil, AVL c_l)
y [m]
x [m]
x [m]
x [m]

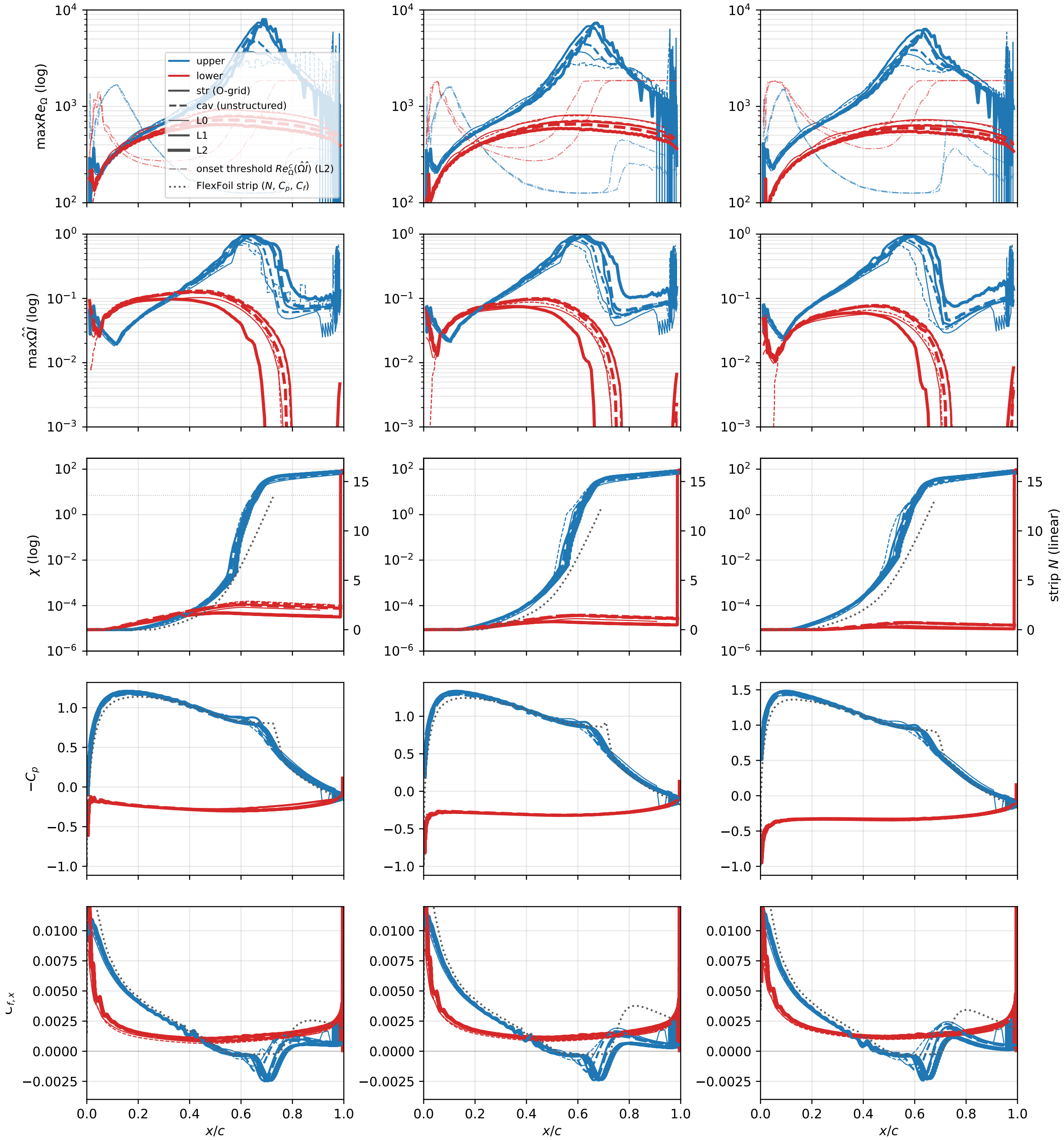


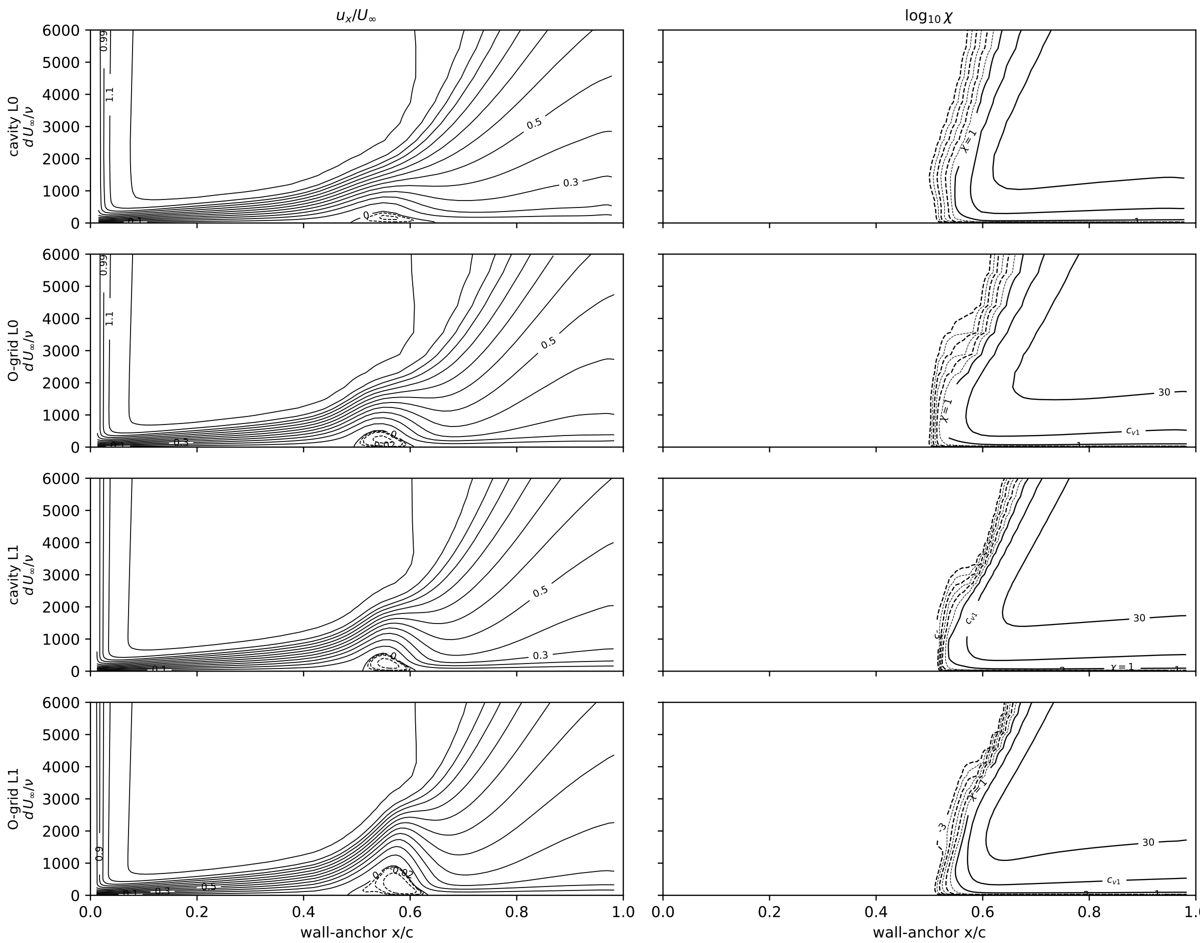
structured O-grid L2
unstructured L2
e^N strips (FlexFoil, AVL c_l)
y [m]
x [m]

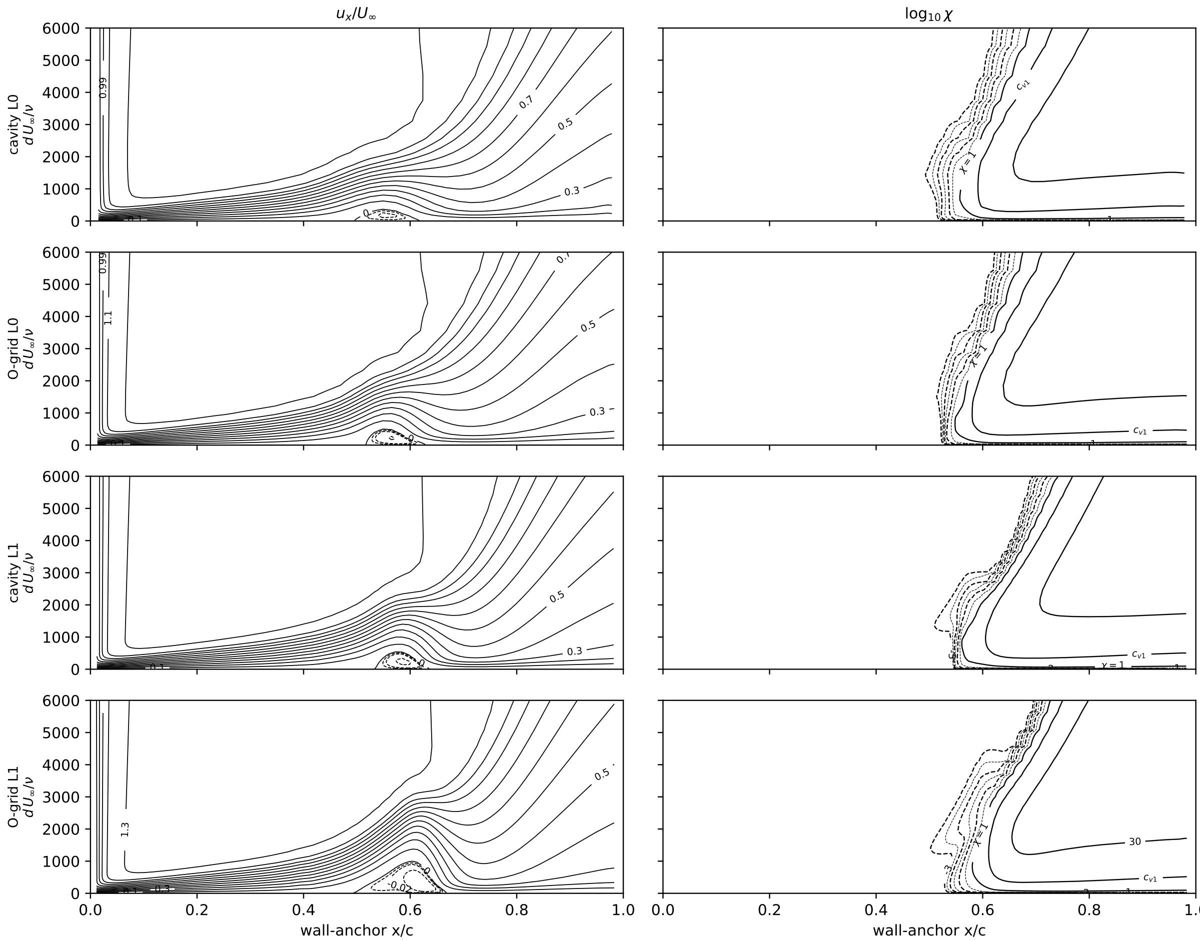


upper
lower
str (O-grid)
cav (unstructured)
L0
L1
L2
onset threshold $Re^c_\Omega(\hat{\Omega}\hat{l})$ (L2)
FlexFoil strip (N, C_p, C_f)
$\max Re_\Omega$ (log)
$\max \hat{\Omega}\hat{l}$ (log)
χ (log)
strip N (linear)
$-C_p$
$C_{f,x}$
x/c

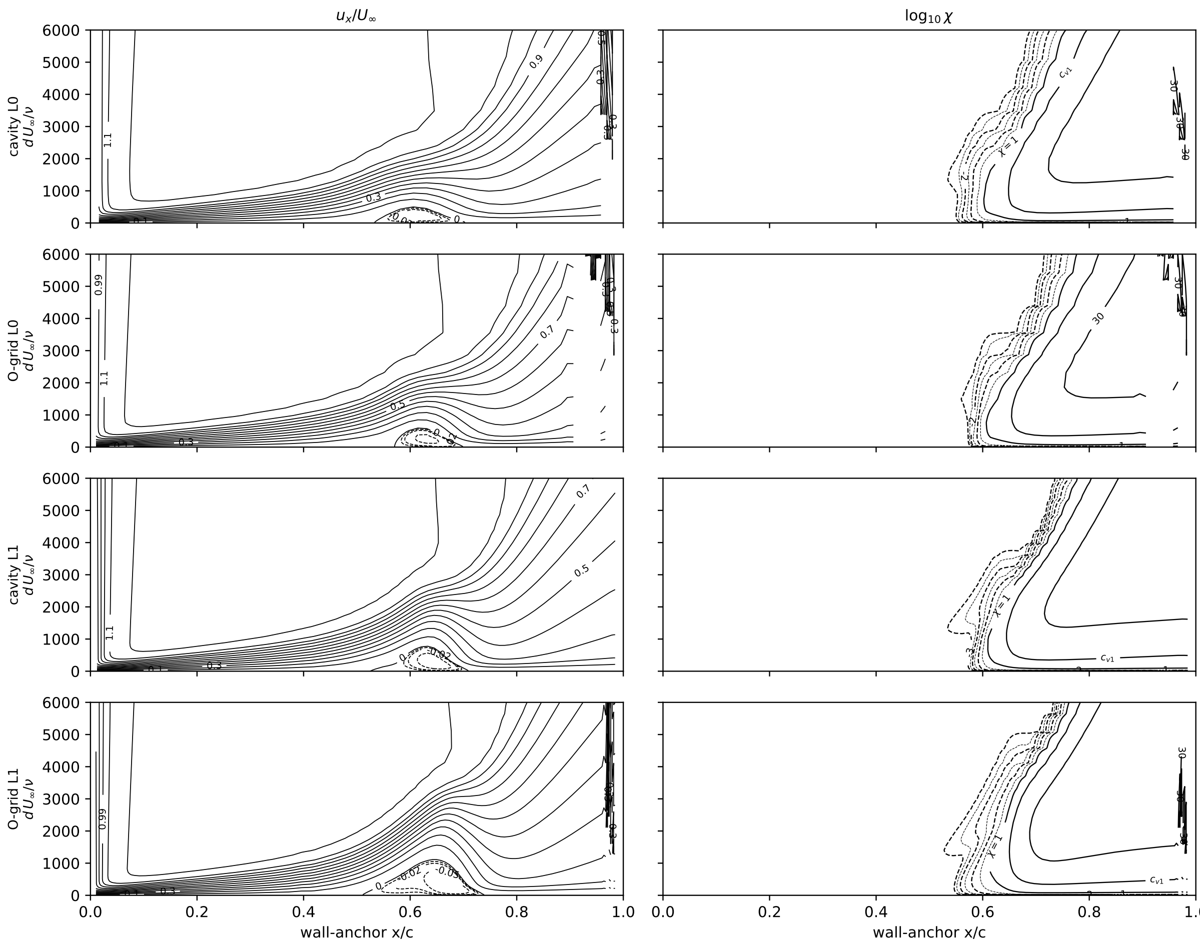

upper
lower
str (O-grid)
cav (unstructured)
L0
L1
L2
onset threshold $Re_{\Omega}^{c}(\hat{\Omega}\hat{l})$ (L2)
FlexFoil strip (N, C_p, C_f)
maxRe_{Ω} (log)
max$\hat{\Omega}\hat{l}$ (log)
χ (log)
strip N (linear)
$-C_p$
$C_{f,x}$
x/c


upper
lower
str (O-grid)
cav (unstructured)
L0
L1
L2
onset threshold $Re_\Omega^c(\hat{\Omega}\hat{l})$ (L2)
FlexFoil strip (N, C_p, C_f)
$\max Re_\Omega$ (log)
$\max \hat{\Omega}\hat{l}$ (log)
χ (log)
strip N (linear)
$-C_p$
$C_{f,x}$
x/c

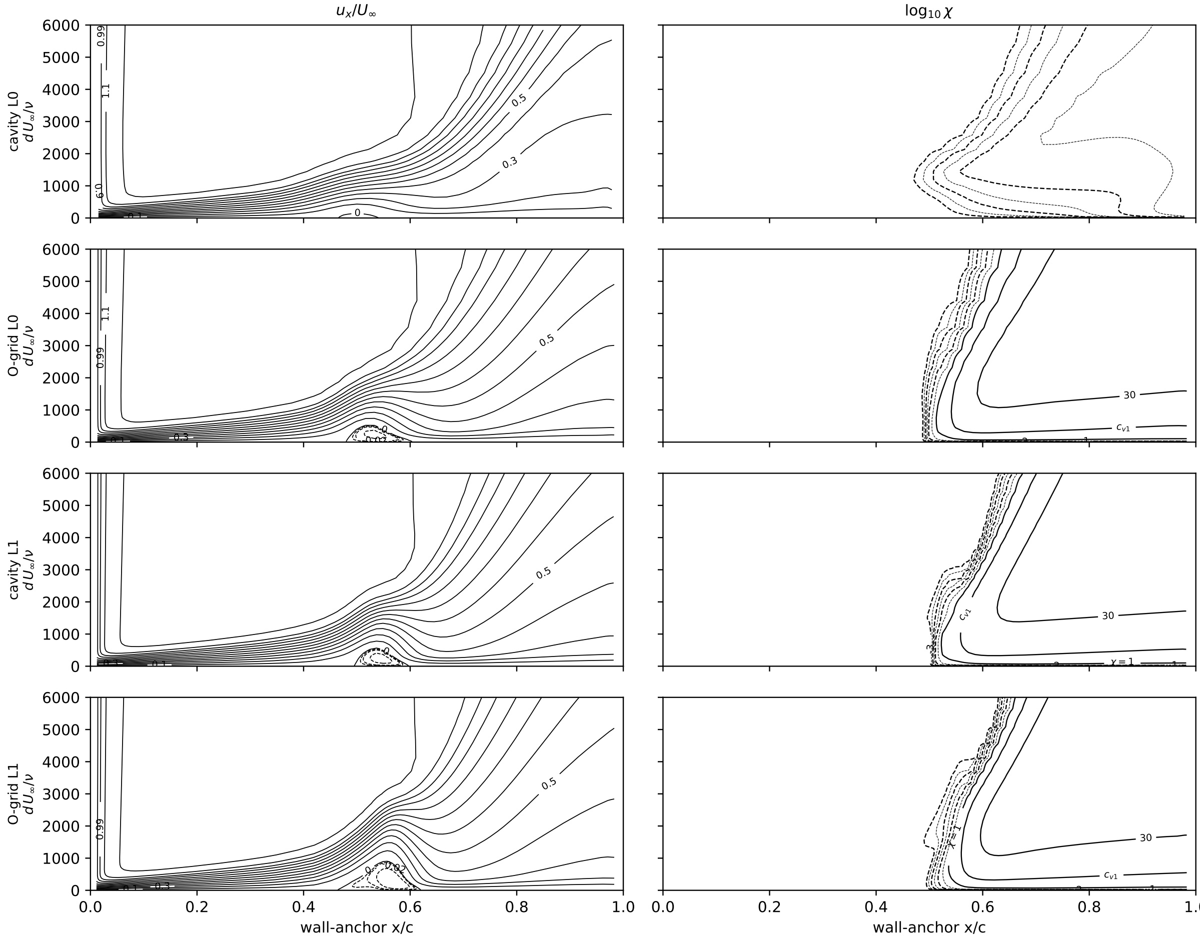

u_x/U_∞
$\log_{10}\chi$
cavity L0
O-grid L0
cavity L1
O-grid L1
dU_∞/ν
wall-anchor x/c
0.5
0.3
1.1
0.99
30
$\chi = 1$
c_{v1}

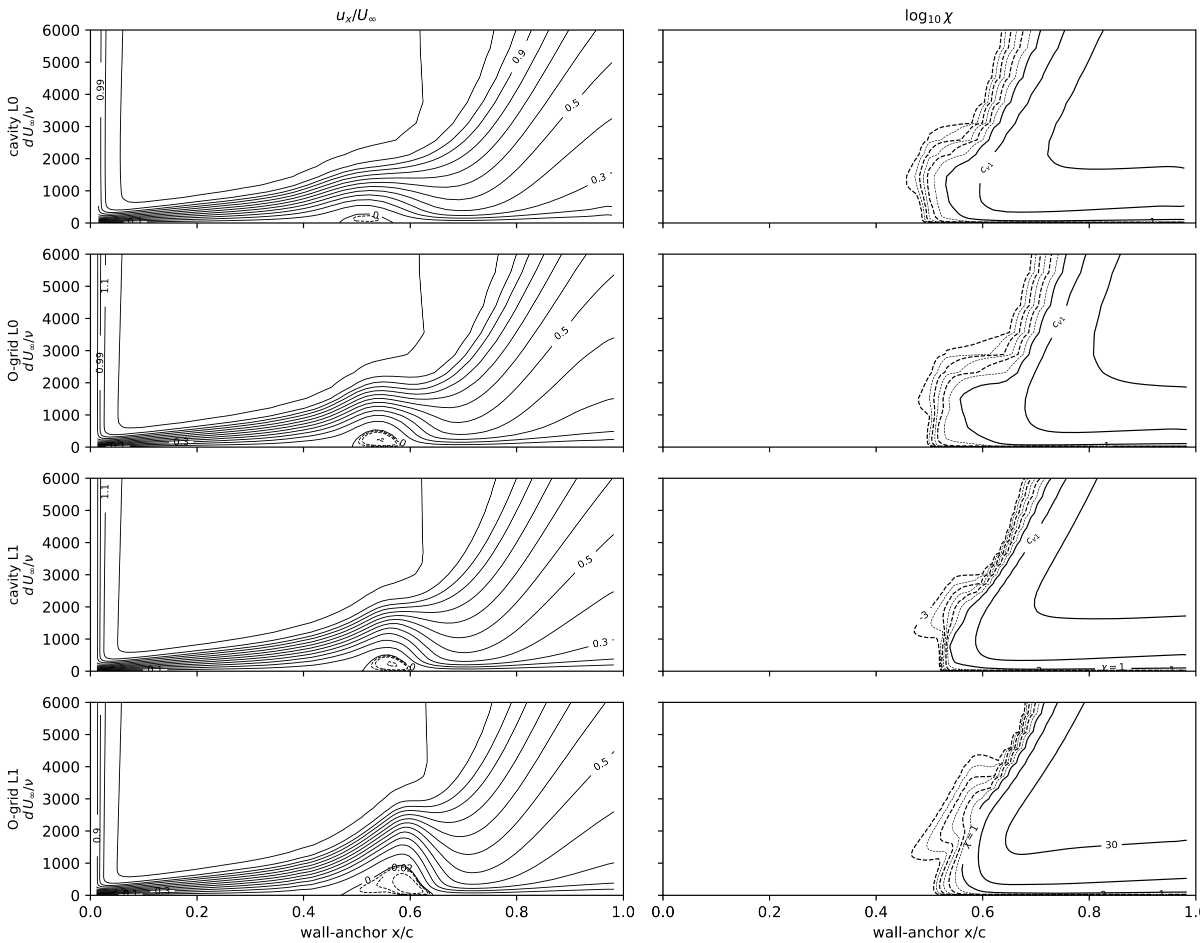


u_x/U_∞
$\log_{10}\chi$
cavity L0 dU_∞/ν
O-grid L0 dU_∞/ν
cavity L1 dU_∞/ν
O-grid L1 dU_∞/ν
wall-anchor x/c
wall-anchor x/c

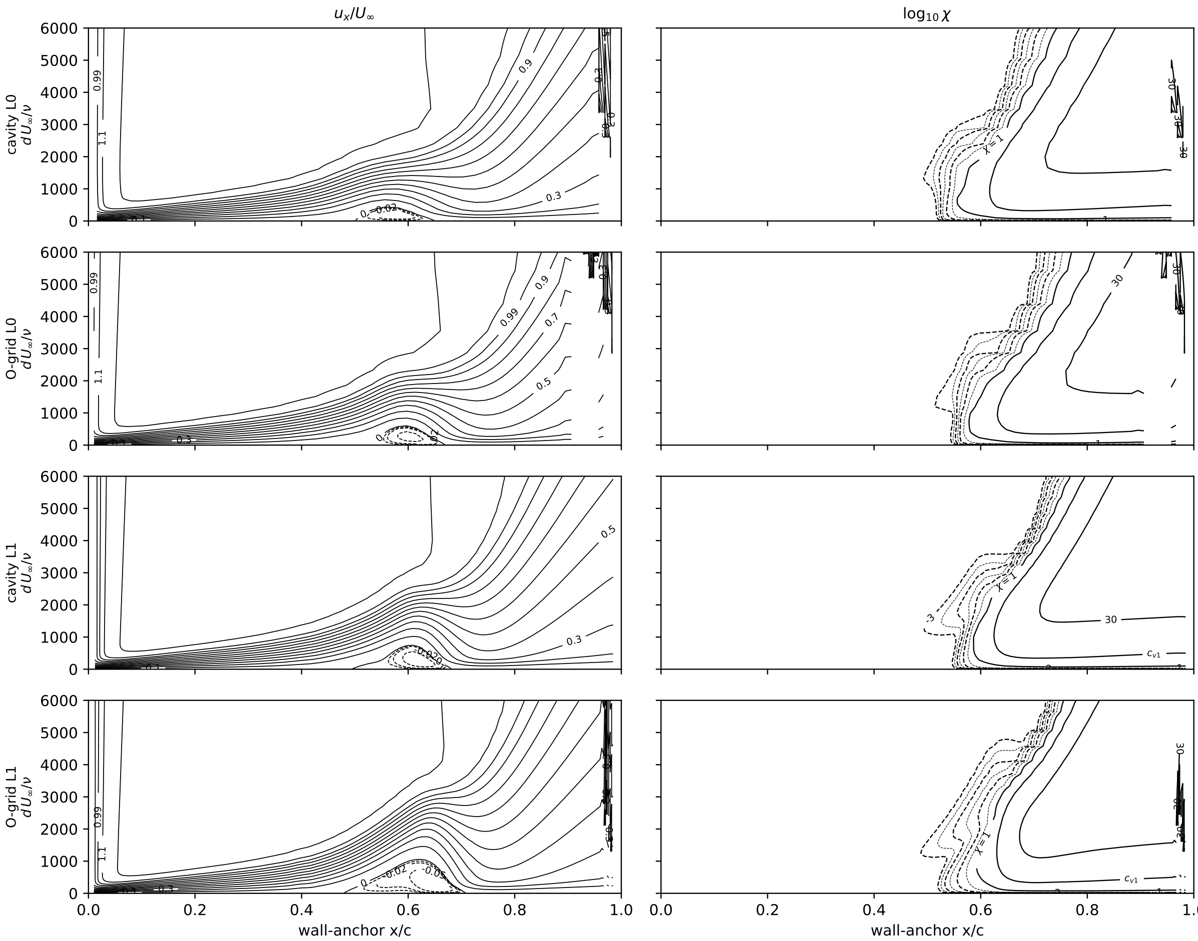


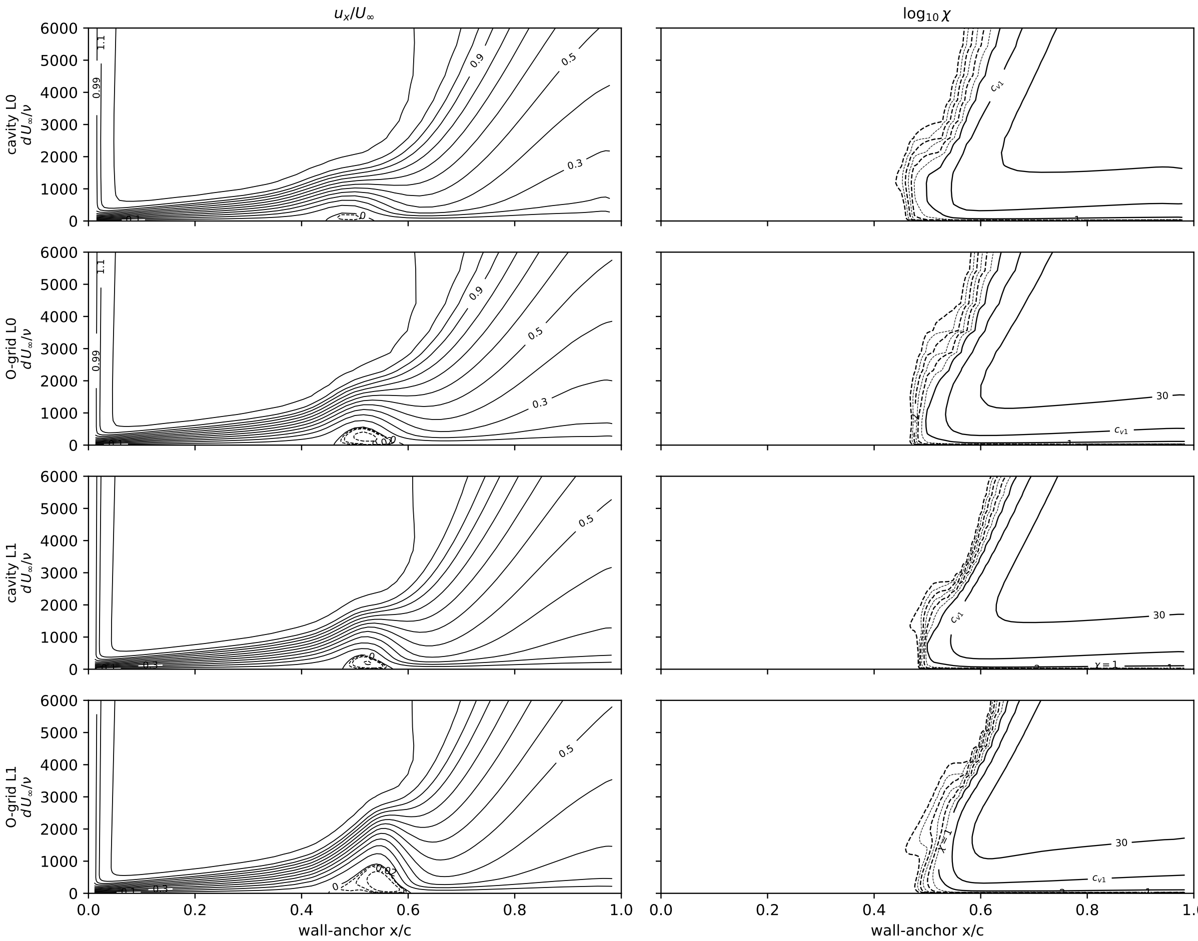
u_x/U_∞
$\log_{10}\chi$
cavity L0 dU_∞/ν
O-grid L0 dU_∞/ν
cavity L1 dU_∞/ν
O-grid L1 dU_∞/ν
wall-anchor x/c
wall-anchor x/c

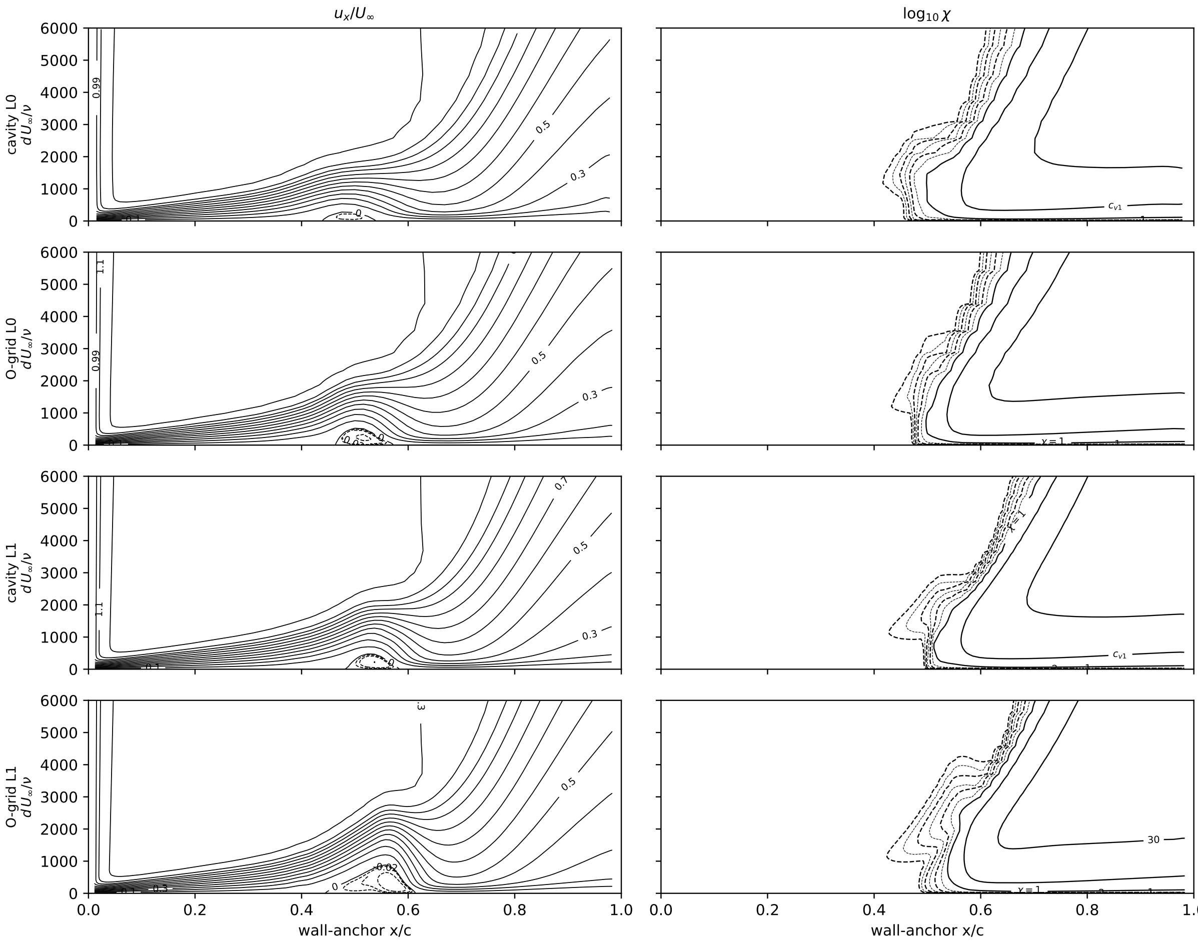

u_x/U_∞
$\log_{10}\chi$
cavity L0
dU_∞/ν
O-grid L0
dU_∞/ν
cavity L1
dU_∞/ν
O-grid L1
dU_∞/ν
wall-anchor x/c
wall-anchor x/c
0.99
1.1
0.5
0.3
30
c_{v1}
$\chi = 1$

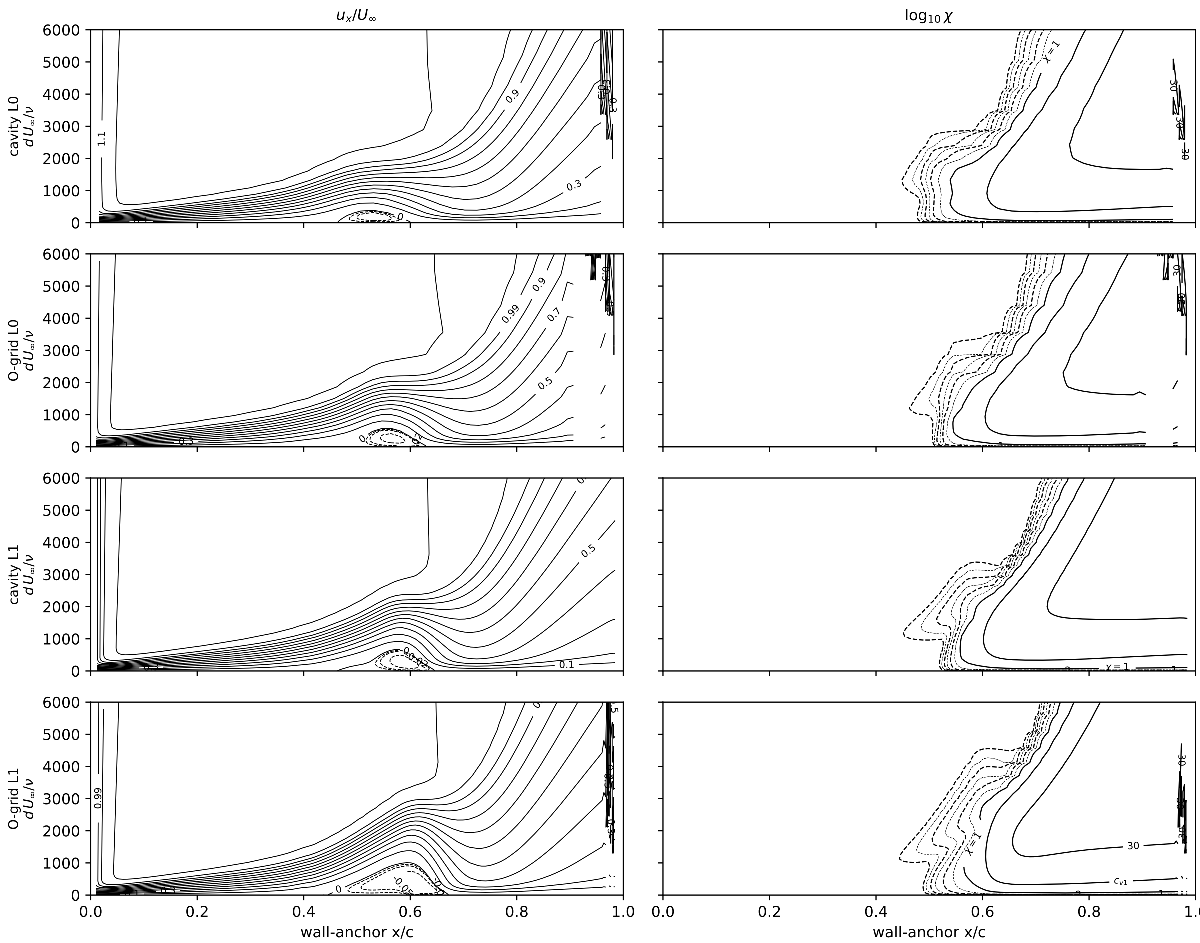


u_x/U_∞
$\log_{10}\chi$
cavity L0
O-grid L0
cavity L1
O-grid L1
dU_∞/ν
wall-anchor x/c
0.99
0.9
0.5
0.3
1.1
0
-0.3
-0.02
c_{v1}
-3
$\chi=1$
30


u_x/U_∞
$\log_{10}\chi$
cavity L0
O-grid L0
cavity L1
O-grid L1
dU_∞/ν
wall-anchor x/c
wall-anchor x/c


u_x/U_∞
$\log_{10}\chi$
cavity L0 dU_∞/ν
O-grid L0 dU_∞/ν
cavity L1 dU_∞/ν
O-grid L1 dU_∞/ν
wall-anchor x/c
wall-anchor x/c


u_x/U_∞
$\log_{10}\chi$
cavity L0
dU_∞/ν
O-grid L0
dU_∞/ν
cavity L1
dU_∞/ν
O-grid L1
dU_∞/ν
wall-anchor x/c
wall-anchor x/c



u_x/U_∞
$\log_{10}\chi$
cavity L0
O-grid L0
cavity L1
O-grid L1
$d U_\infty/\nu$
wall-anchor x/c

2.7 6:1 prolate spheroid, $\alpha = 0$–$10°$

Analytic spheroid ($x^2/A^2 + r^2/B^2 = 1$, $A = L/2$, $B = L/12$, half-model) on the L1 structured O-grid (2.2M nodes, first wall spacing $1.5 \times 10^{-6} L$), campaign constants and kernel unchanged, at the DFVLR/ONERA measured conditions. Mach is the tunnel's: the Göttingen 3×3 m tunnel is atmospheric, so $M = 1.994 \times 10^{-8}\, Re_L$ (calibrated on its two published (Re_L, U_∞) pairs, 7.70×10^6 at 55.0 m/s and 6.54×10^6 at 45.0 m/s, reproduced to 2%), i.e. $M = 0.030$ at $Re_L = 1.52 \times 10^6$ to 0.144 at 7.2×10^6; the pressurised ONERA F1 tunnel does not have M fixed by Re_L, so its condition is run at its Göttingen twin's Mach. 45 000 cold-start pseudo-steps, seed applied in the freestream and the far-field boundary with a cold-start freestream initial condition. Each condition at the facility-calibrated seed (limiting $N_{\mathrm{TS}} = 8.0$ Göttingen, 7.0 ONERA F1) and at the facility-measured hot-wire level (Göttingen 0.33% streamwise, ONERA F1 <0.1%).

Table 11: computed near-wall $\chi = 1$ front minus measured, azimuth by azimuth. $\alpha = 0$, $Re_L = 7.2 \times 10^6$: $+0.077$ and -0.097 at the calibrated and measured seed, uniform to ± 0.001 across seventeen azimuths. $\alpha = 2.5°$: $+0.016$ leeward, $+0.132$ windward, against a measured azimuthal spread of 0.018. $\alpha = 5°$: -0.088 to $+0.246$ at 6.49×10^6. $\alpha = 10°$: rms 0.027 at 1.52×10^6 and the measured seed; at 6.56×10^6 the windward residual is $+0.375$ to $+0.549$. Seed swing in the windward residual between the two facility levels: 0.174 at $\alpha = 0$, 0.181 at 2.5°, 0.239 at 5° (6.49×10^6), 0.012 at 10° (6.56×10^6), 0.000 on the ONERA twin.

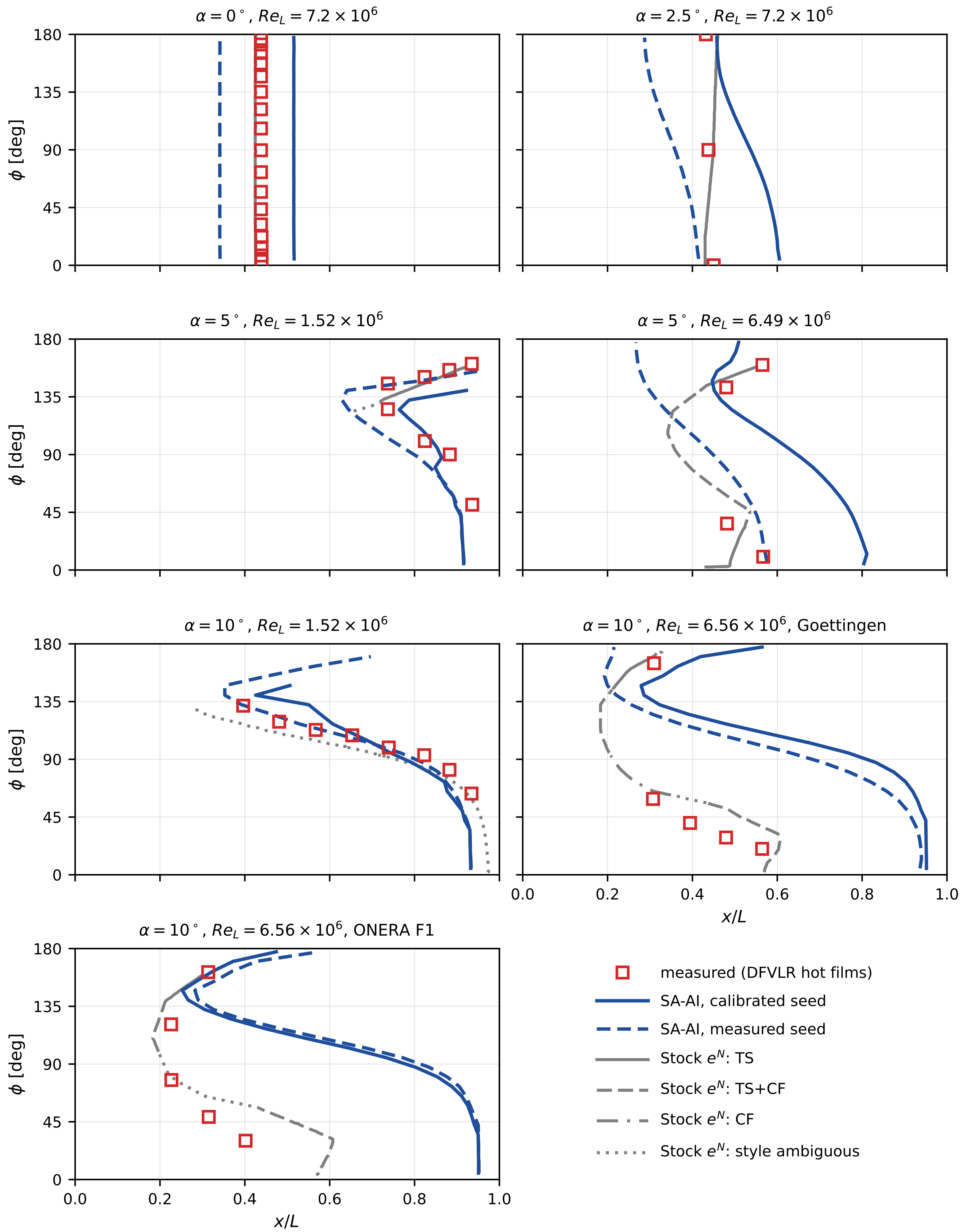


Figure 25: Transition front by azimuth at each measured condition, in the order of Table 11: measurement (open red squares, DFVLR hot films), Stock's two-N-factor e^N computation (grey, line style carrying his printed mechanism coding: TS solid, TS+CF dashed, CF dash-dot, style ambiguous dotted), and SA-AI (blue, solid at the facility-calibrated seed and dashed at the facility-measured hot-wire seed). $\phi = 0$ windward. Mechanism boundaries at $\alpha = 10°$, 6.56×10^6: TS 172–145°, TS+CF 144–95°, CF 93–63°, TS+CF 56–2°. At $\alpha = 0$ Stock's front lies at $x/L = 0.425$, beneath the measured symbols.

Table 11: 6:1 prolate spheroid at the tunnel conditions: computed near-wall $\chi=1$ front minus the measured front, at every azimuth carrying a measurement, on the L1 O-grid at the measured Mach number. Each condition is run at the facility-calibrated seed (the tunnel-calibrated limiting N_{TS}) and at the facility-measured hot-wire level. Δ lee / Δ wind are the residuals at the most leeward and most windward measured azimuth ($\phi=0$ windward). The n column is the number of measured azimuths at which the computed front was found, over the number measured: where they differ the near-wall χ never reaches 1 within $x/L\leq 0.97$, i.e. the computed layer is still laminar at the tail and the residual there is a bound, not a value. The last two columns are the same residual for Stocks own two-N-factor e^N computation, digitized from the printed panels at the same azimuths; at $Re_L=1.52\times10^6$ his computed transition line is the laminar separation line.

α	Re_L [10^6]	tunnel	seed	Tu [%]	n	SA-AI − measured				Stock − measured	
						mean	rms	lee	wind	mean	rms
0°	7.2	Göttingen	measured	0.330	17/17	-0.097	0.097	-0.097	-0.098	-0.013	0.013
0°	7.2	Göttingen	calibrated	0.106	17/17	+0.077	0.077	+0.078	+0.076	-0.013	0.013
2.5°	7.2	Göttingen	calibrated	0.106	3/3	+0.082	0.096	+0.016	+0.132	+0.006	0.021
5°	1.52	Göttingen	calibrated	0.106	4/8	-0.006	0.028	+0.026	-0.040	-0.064	0.064
5°	6.49	Göttingen	calibrated	0.106	4/4	+0.107	0.201	-0.088	+0.246	-0.024	0.051
10°	1.52	Göttingen	measured	0.330	8/8	-0.013	0.027	+0.002	-0.041	-0.091	0.100
10°	1.52	Göttingen	calibrated	0.106	8/8	+0.019	0.082	+0.160	-0.053	-0.091	0.100
10°	6.56	Göttingen	measured	0.330	5/5	+0.365	0.440	-0.105	+0.375	+0.073	0.098
10°	6.56	Göttingen	calibrated	0.106	5/5	+0.421	0.463	+0.068	+0.387	+0.073	0.098
10°	6.56	ONERA F1	calibrated	0.161	5/5	+0.399	0.475	+0.002	+0.549	+0.088	0.134
10°	6.56	ONERA F1	measured	0.100	5/5	+0.420	0.486	+0.051	+0.549	+0.088	0.134

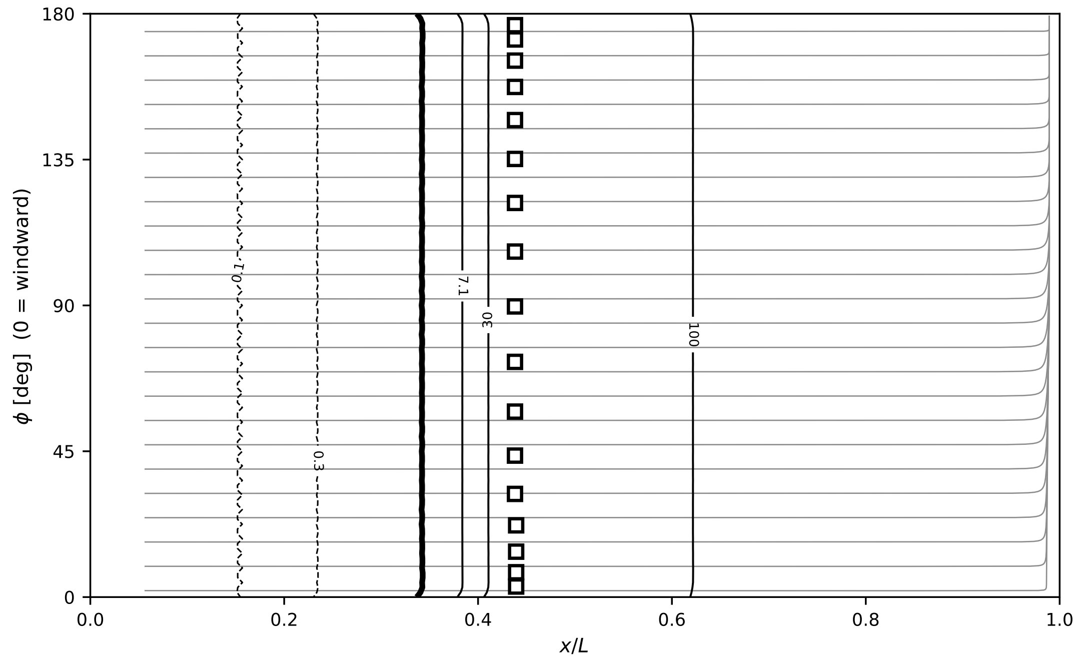


Figure 26: Surface at the measured-seed condition, unrolled into $(x/L, \phi)$, $\phi = 0$ windward; line art. Grey skin-friction lines integrated from the wall shear vector itself – the oil-flow analogue, not the inviscid streamlines of Stock Figs. 14–17, and their convergence is the separation signature – under the $\max_n \chi$ contour family (sub-unity dashed, $\chi = 1$ heavy, $\chi = c_{v1}$ and above solid, labeled), $\max_n \chi$ taken on a wall-normal segment shot from each surface point. White-filled squares: measured DFVLR transition. $\alpha = 0°$, $Re_L = 7.2 \times 10^6$.

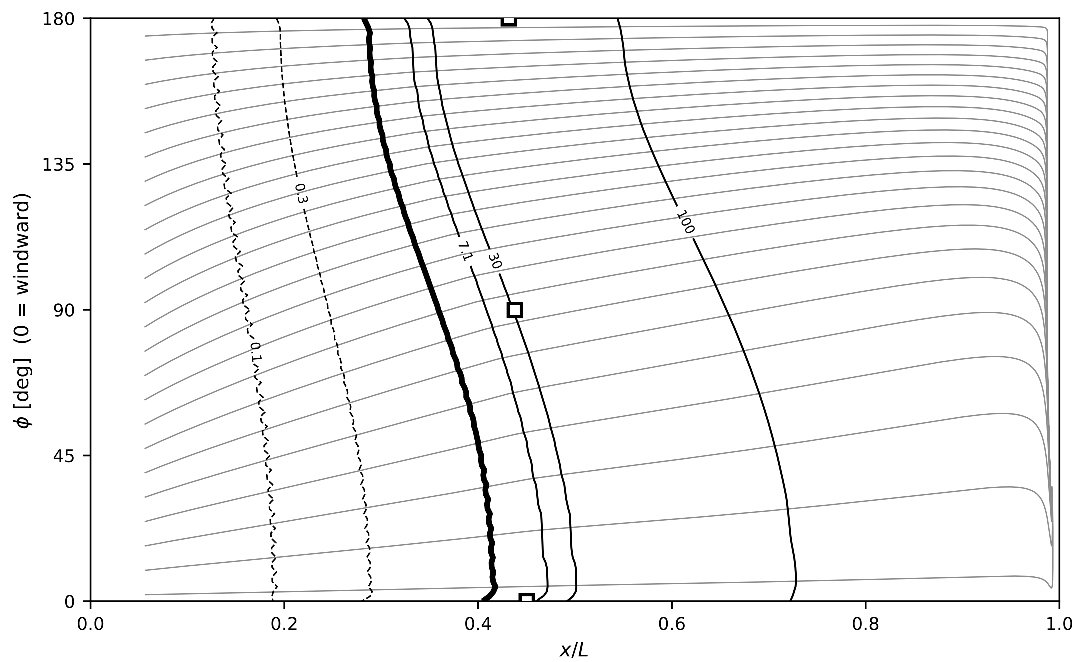


Figure 27: $\alpha = 2.5°$, $Re_L = 7.2 \times 10^6$.

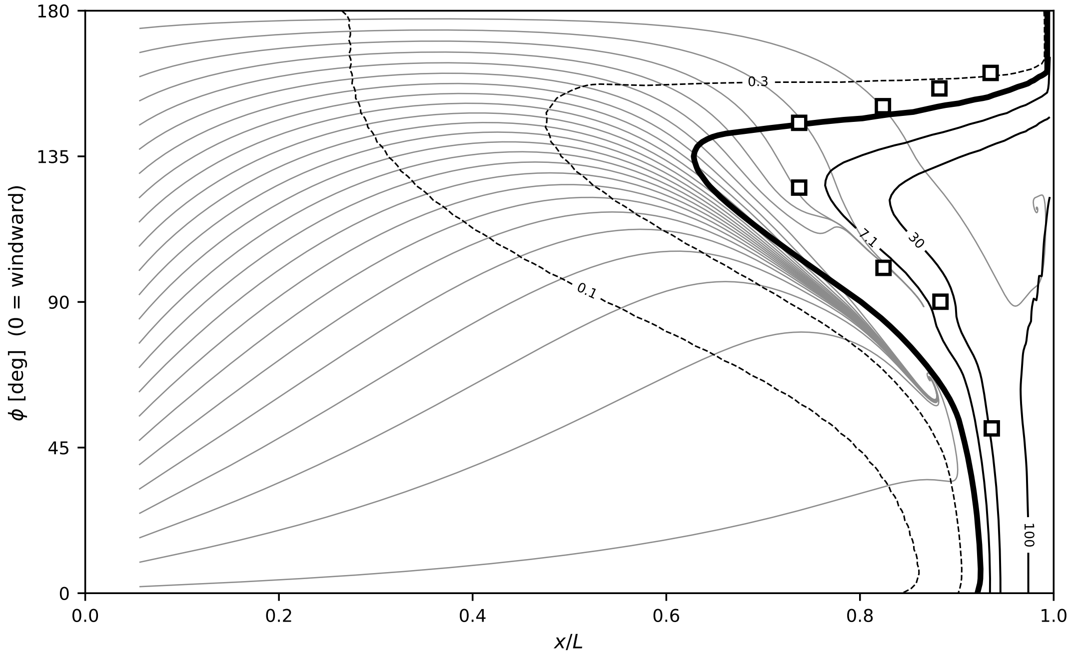


Figure 28: $\alpha = 5°$, $Re_L = 1.52 \times 10^6$.

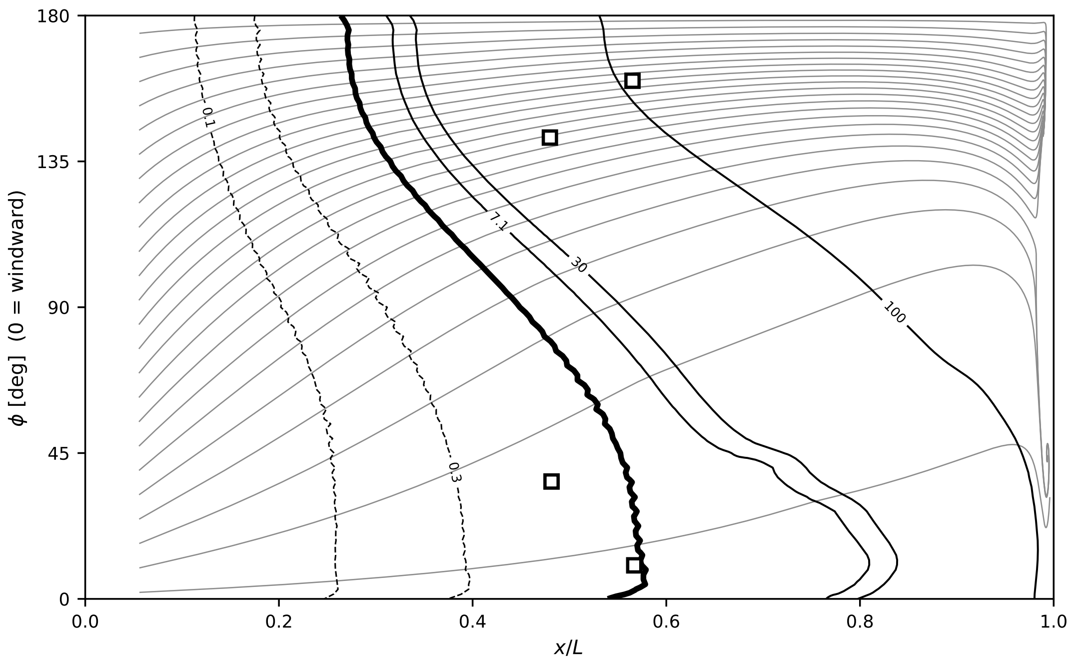


Figure 29: $\alpha = 5°$, $Re_L = 6.49 \times 10^6$.

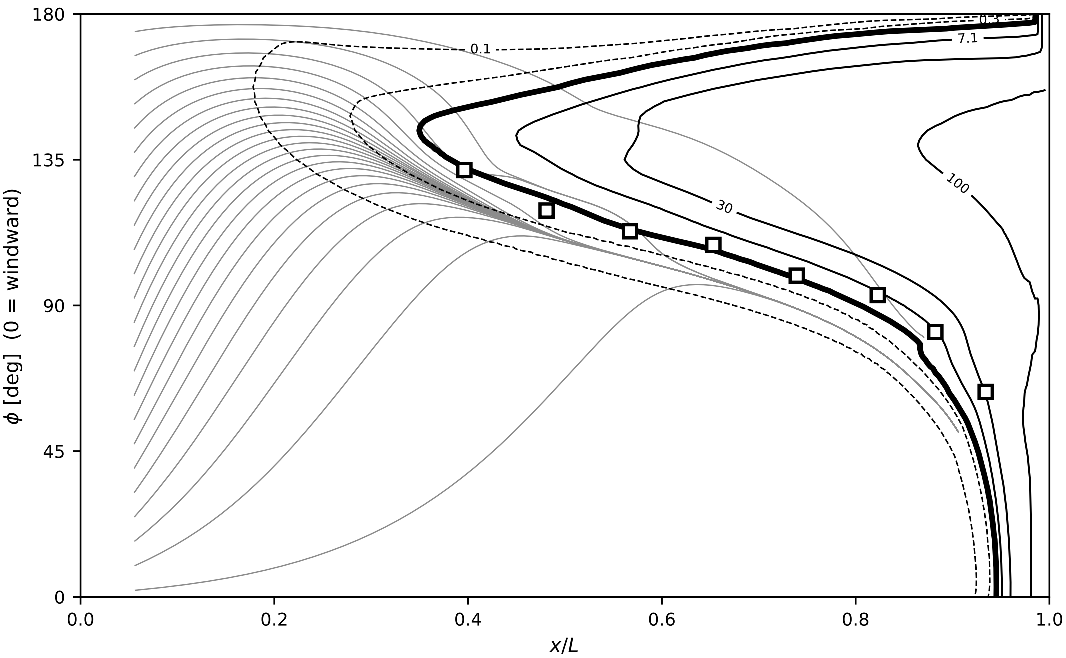


Figure 30: $\alpha = 10°$, $Re_L = 1.52 \times 10^6$.

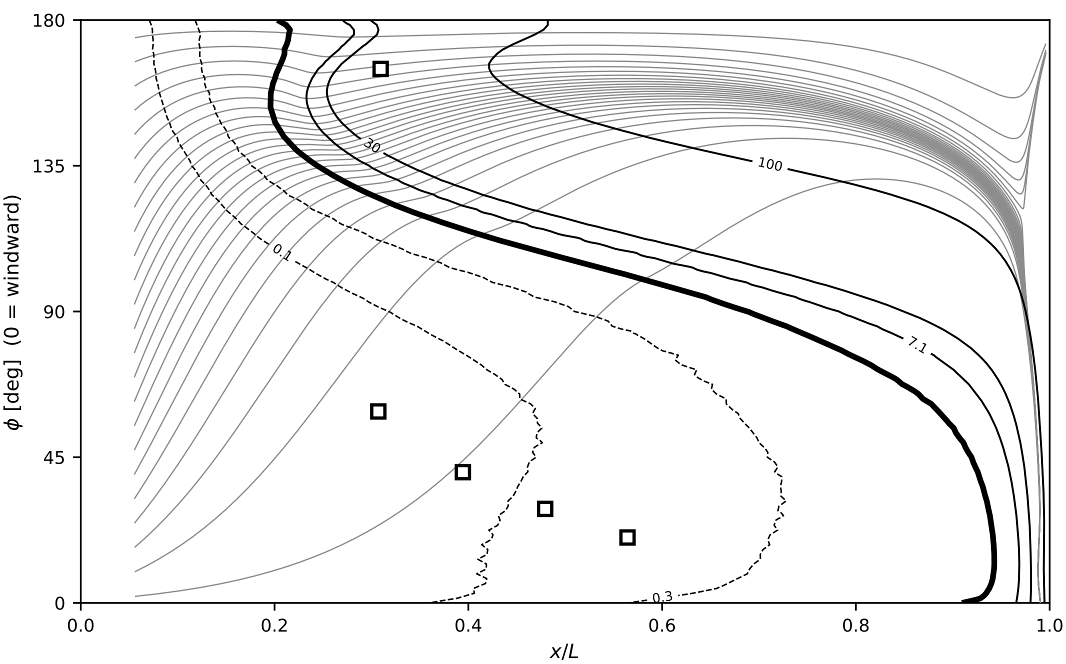


Figure 31: $\alpha = 10°$, $Re_L = 6.56 \times 10^6$, Göttingen.

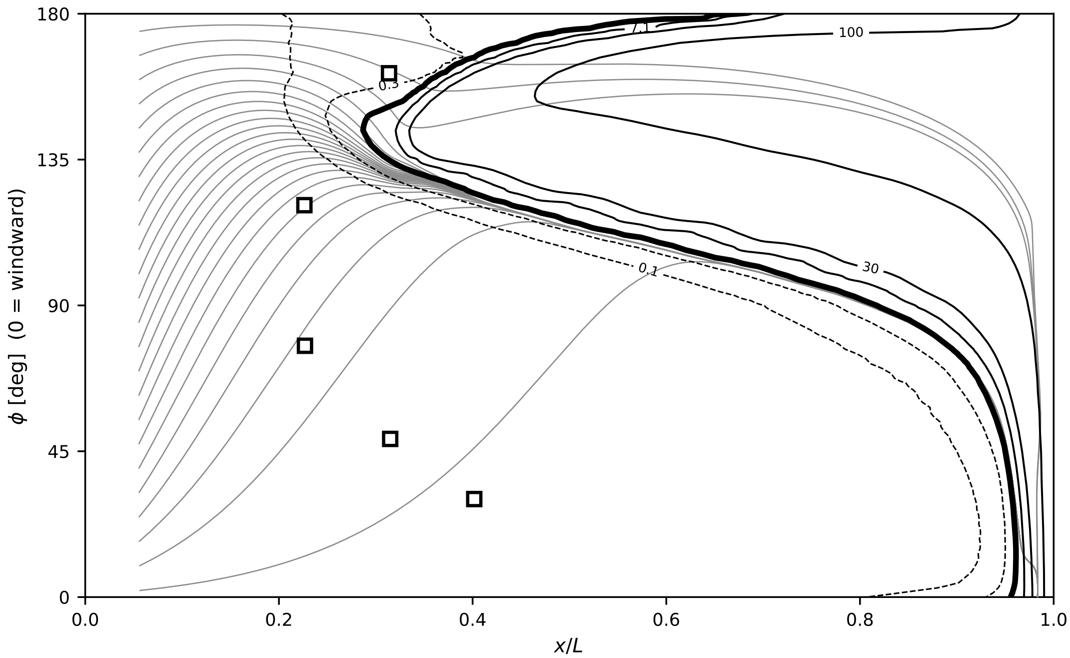


Figure 32: $\alpha = 10^{\circ}$, $Re_L = 6.56 \times 10^6$, ONERA F1.

Table 12: Peak dimensionless source $a\,S$ on Falkner–Skan wedges: the single-source rate, and the two-source form of Eq. (15) at $(a_{\rm visc}, B_{\rm c}) = (0.0276, 130)$.

β	Re_θ	single-source	two-source	ratio
0	500	0.01485	0.01485	1.000
0	1000	0.01485	0.01487	1.001
−0.10	400	0.03082	0.03083	1.000
−0.1988	300	0.09692	0.09764	1.007
+0.10	1000	0.00695	0.00696	1.000
+0.20	2000	0.00281	0.00282	1.004
+0.35	4000	0.00033	0.00405	12.4

A Appendix: the two-source rate — inviscid and viscous branches

The rate of the model is built on the single even coordinate $\hat{\Omega}\hat{I}$, the departure of the profile from its osculating parabola. That coordinate carries both the inviscid–inflectional and the viscous Tollmien–Schlichting content, because the two co-vary monotonically across the Falkner–Skan family it is calibrated on. They separate on exactly one profile—a *parabola*, where $\hat{I} \equiv 0$ identically. Two consequences bound the single-coordinate form. Plane Poiseuille flow is exactly parabolic in wall distance, so the rate returns identically zero at every Reynolds number, while plane Poiseuille is linearly unstable—by a purely viscous mechanism—at $Re_c = 5772$ (Orszag 1971). And on the strongly favorable Falkner–Skan branch the same collapse appears as the rate deficit of Fig. 33: by $\beta = 0.35$ the booked amplification is an order of magnitude below the Drela–Giles envelope, which retains a floor near 5×10^{-3} per unit Re_θ. Note that Blasius itself carries no inflection point, so the canonical Tollmien–Schlichting instability underlying natural-laminar-flow practice is already a viscous instability of a non-inflectional profile: the mechanism the missing branch represents is the main one, not an exotic case.

A two-branch rate separates the mechanisms. Alongside the inflectional coordinate $P_I = \hat{\Omega}\langle\hat{I}\rangle_+$ we form the curvature coordinate

$$P_{\rm c} \;=\; \hat{\Omega}\,\big\langle -\hat{Z}\big\rangle_+, \qquad \hat{Z} = Z/R, \tag{14}$$

positive wherever $u'' < 0$ (filled, non-inflected profiles), clipped to zero at an adverse wall, vanishing like d toward the wall and zero in the free stream. Each branch carries its own rate *and its own onset*, and the two complete *sources*—not the two rates—are combined:

$$P_{\rm AI} = \omega\,\tilde{\nu}\;\operatorname{softmax}_2\Big[a_{\max}\,P_I\;S\big(Re_\Omega/Re_\Omega^{{\rm c},i}\big),\;\; a_{\rm visc}\,P_{\rm c}\;S\big(Re_\Omega/Re_\Omega^{{\rm c},v}\big)\Big], \tag{15}$$

$$Re_\Omega^{{\rm c},i} = k\,\operatorname{softmin}_2\Big(C,\; A + B\,P_I^{-2}\Big), \qquad Re_\Omega^{{\rm c},v} = A + B_{\rm c}\,P_{\rm c}^{-2}, \qquad \operatorname{softmax}_2(x,y) = \sqrt{x^2+y^2}.$$

The inviscid threshold is the model's existing gate unchanged, ceiling included; the viscous branch shares its floor A and adds one coefficient $B_{\rm c}$. Constants: $a_{\max} = 0.19$, $a_{\rm visc} = 0.0276$, $A = 124.6$, $B = 1.424$, $B_{\rm c} = 130$, $C = 1851.2$.

Decoupling the two gates is the substantive change from the form previously explored, which soft-maxed the two *rates* but soft-minimized the two *thresholds* into one shared gate. Kelvin–Helmholtz is inviscid and its threshold is essentially the floor A; Tollmien–Schlichting is viscous and its onset is far higher—on Blasius the two read 358 and 2598. A shared soft-minimum applies the lower of them to *both* rates, so the inviscid branch's low threshold switches on the viscous rate before the viscous mechanism is critical, worth +7.4% on the Blasius anchor. Under Eq. (15) that cross-talk is absent, and the construction reproduces the single-source rate to within 0.7% over the whole calibrated family (Table 12), so $a_{\max}$, k and the airfoil campaign are untouched: the branch adds amplification only where the single-source rate has none. Retaining the ceiling C is what buys that: dropping it, as the earlier form did, sends the $\beta = +0.20$ station to 0.63× its single-source value.

The viscous branch's two constants are anchored as the inviscid branch's are: on canonical linear-stability results rather than on transition cases. $a_{\rm visc}$ follows the favorable Falkner–Skan family against the Drela–Giles

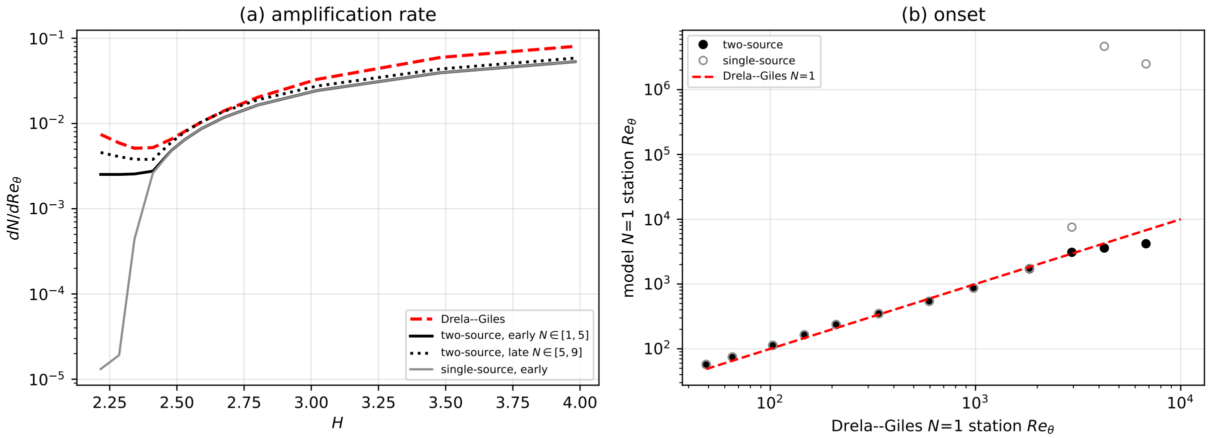


Figure 33: Two-source rate against Drela–Giles across the Falkner–Skan family, $\beta = +1.0$ (stagnation, $H \approx 2.22$) to -0.1988 (incipient separation, $H = 3.98$); analytic march, no CFD. (a) amplification rate as early ($N \in [1, 5]$) and late ($N \in [5, 9]$) secants, with the single-source rate for comparison—the low-H collapse is the deficit the viscous branch removes. (b) the marched $N = 1$ station against the Drela–Giles $N = 1$ station $Re_{\theta 0} + 1/(dN/dRe_\theta)$; the diagonal is exact agreement.

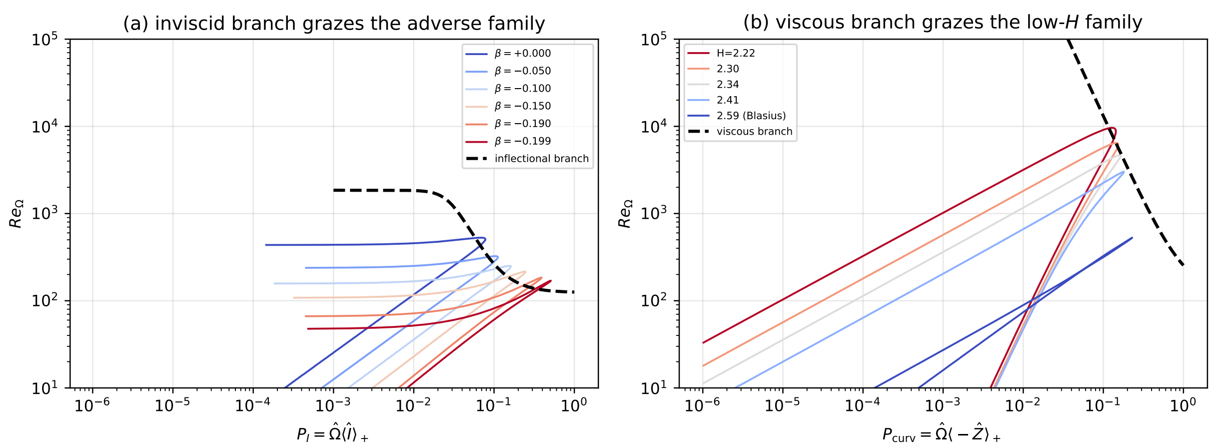


Figure 34: Each onset branch grazes its own family in its own coordinate, each profile evaluated at its Drela–Giles critical Reynolds number $Re_{\theta 0}(H)$. (a) the inflectional branch $A + B/P_I^2$ against the adverse family; the strongly favorable members collapse to $P_I \to 0$ and leave the panel, which is the onset-resolution limit of a single-coordinate gate. (b) the curvature branch $A + B_c/P_c^2$ against exactly those low-H members, in the coordinate P_c of Eq. (14).

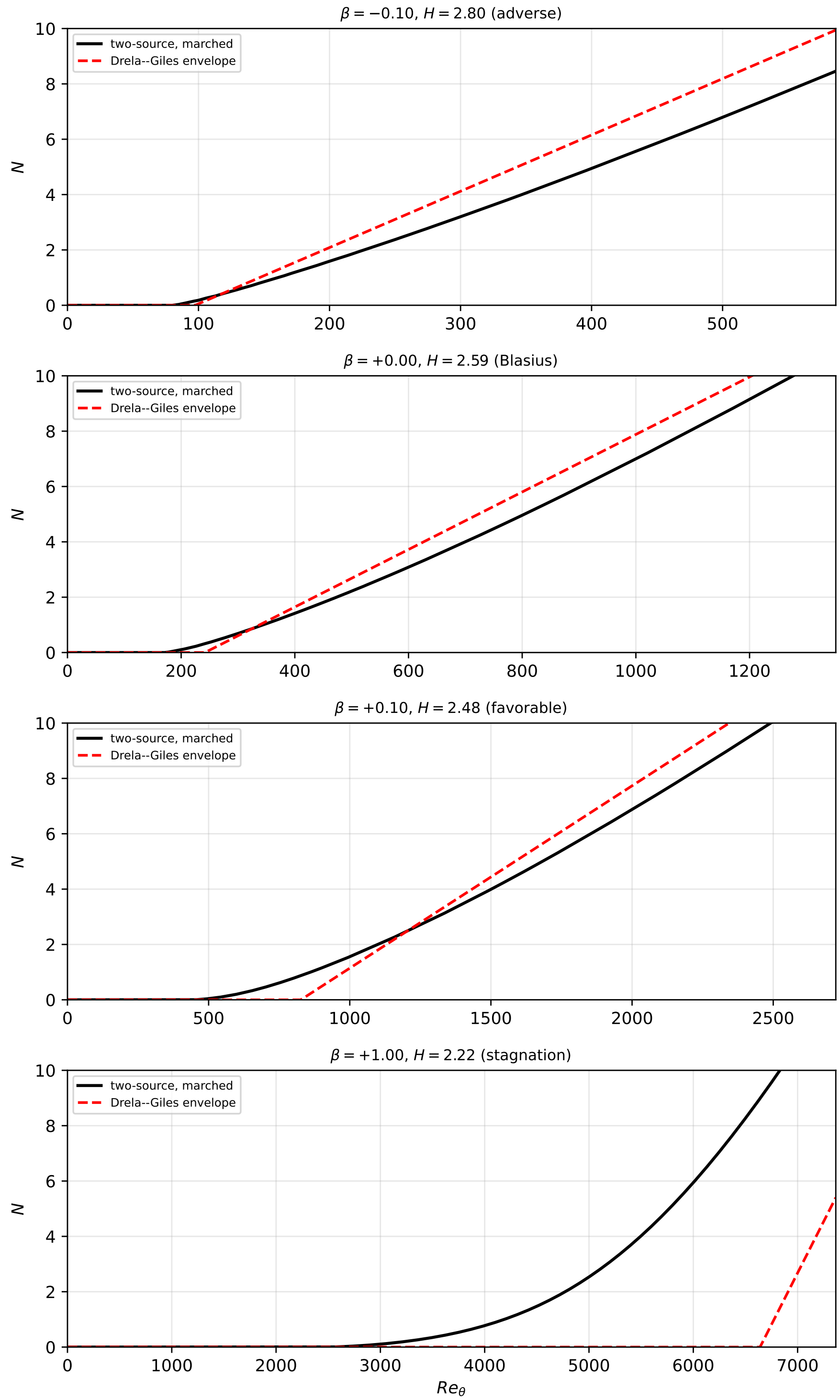


Figure 35: Disturbance transport marched with Eq. (15) on four wedges—adverse, Blasius, favorable, and the stagnation profile ($\beta = +1$) that the single-source kernel cannot ignite at all—against each profile's Drela–Giles envelope. On the stagnation wedge the branch ignites at $Re_\theta \approx 2.5 \times 10^3$ against the correlation's critical station $Re_{\theta 0} = 6.5 \times 10^3$: early, but finite where the single-source rate is identically zero. Same parabolized instrument and same $c_{\nu,\mathrm{ai}} = 1/6$ as the model's own calibration; the viscous branch touches the rate floor and the gate only.

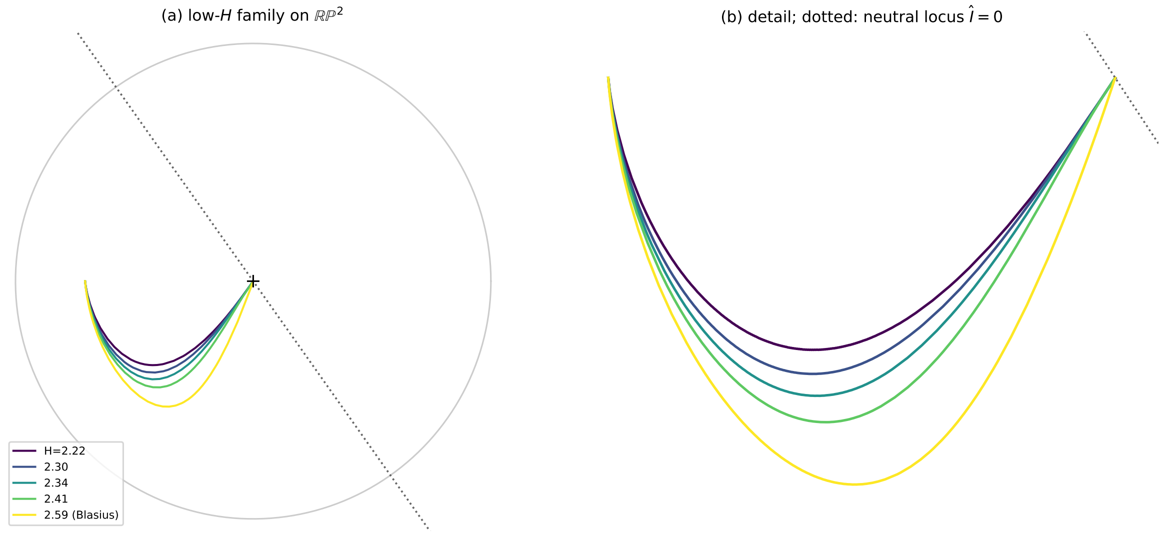


Figure 36: The low-H Falkner–Skan family on the $\mathbb{RP}^2$ indicator sphere (orthographic, curvature vertical), from the stagnation profile $H \approx 2.22$ to Blasius $H = 2.59$, with (b) the near-wall detail. The family bunches against the neutral parabola locus, where $\hat{I}$—and with it P_I—is too small to distinguish members whose Drela–Giles critical Reynolds numbers differ by nearly a factor of two. The curvature coordinate P_c separates them.

Table 13: Plane-channel check: the B_c that would place the neutral point on Orszag's $Re_c = 5772$, and the neutral point actually obtained at the Falkner–Skan-anchored $B_\mathrm{c} = 130$.

a_visc	B_c for $Re_c = 5772$	Re_c at $B_\mathrm{c} = 130$
0.0200	81.0	8095
0.0276	91.0	7409
0.0350	99.9	6916
0.0600	127.9	5834

envelope—itself constructed from Orr–Sommerfeld solutions, and the flow class in which viscous Tollmien–Schlichting amplification is both well characterized and the observed route to transition. Plane Poiseuille then supplies an *independent* check on B_c, exact and with no refitting: on a parabola $P_I \equiv 0$, so the channel constrains the viscous branch with no contamination from the inflectional one. Requiring the frozen-profile eigenvalue of Eq. (15) to change sign at Orszag's $Re_c = 5772$ gives Table 13: at the Falkner–Skan value $a_\mathrm{visc} = 0.0276$ the channel asks for $B_\mathrm{c} = 91$ against the family's 130, and the family's constants place the channel's neutral point at 7409, 1.28× the true value—against a single-source kernel that never destabilizes it at all. Two unrelated flows agree on the same constant to 30%. The channel is a calibration and verification device only, never a validation case: its observed transition is subcritical and finite-amplitude, outside the envelope class this model represents.

Two limits of the construction are worth stating with it. The viscous rate $a_\mathrm{visc} P_\mathrm{c}$ is Reynolds-independent, an assumption inherited from the inviscid branch where it is justified because the mechanism is inviscid; a viscous instability's rate need not be, and the Reynolds dependence of the plane-Poiseuille envelope is the natural test. And the branches are a *decomposition of the rate*, not a claim that each carries one physical mechanism exclusively: the viscous gate on Blasius opens only near $Re_\theta \approx 1190$, so Blasius remains carried by P_I in this parameterization.